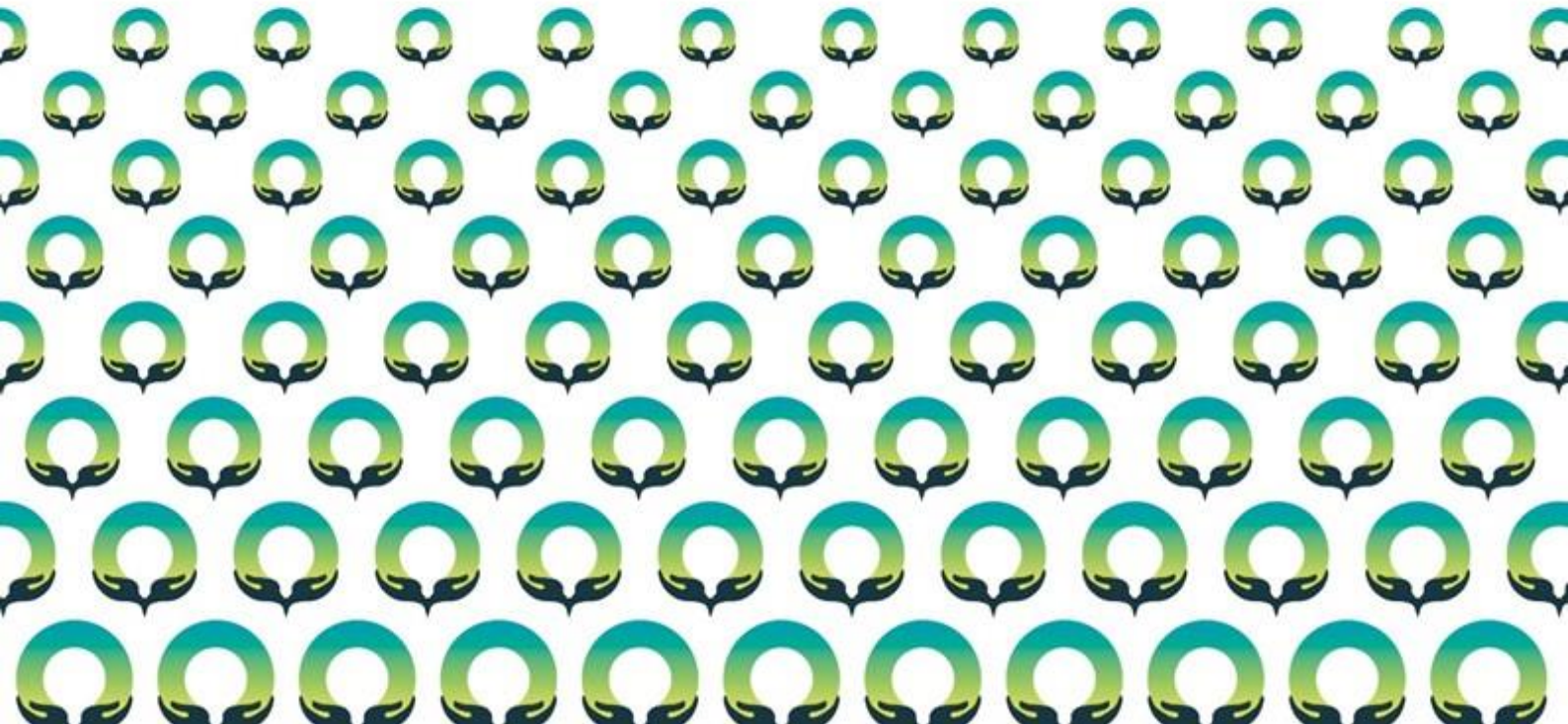
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Hestia

Project Deliverable

D2.2

Participatory design recommendations & user engagement strategy



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Table of Content

List of Figures 8

1. EXECUTIVE SUMMARY 9

1.1 Overview 9

1.2 Overall scope and report summary 9

2. METHODOLOGY FOR DESIGNING AN INCLUSIVE AND PARTICIPATORY DR PLATFORM 15

2.1 Overview 15

2.2 Co-creating the Hestia platform 15

2.3 Opportunities for the development of DR solutions using a combination of co-creation and theories of practice 18

2.4 Participatory and co-creation methods for engaging users in DR solutions 19

2.4.1 Household interactions 19

2.4.2 Participatory workshops in the pilot sites 20

3. HOUSEHOLD INTERACTIONS 22

3.1 Overview 22

3.2 Context and objectives 22

3.3 Time-shifting and everyday life 22

3.3.1 Time-shifting appliances at home 22

3.3.2 Everyday rhythms, sequences and routinised ways of doing 24

3.3.3. Motivations for time-shifting 25

3.4 Household and community dynamics 25

3.4.1 Gender 26

3.4.2 Intergenerational household dynamics 27

3.4.3 Community 27

3.5 Living with technologies 28

3.5.1 Comfort, control, and cleanliness 28

3.5.2 Home (making) and technology 29

HESTIA Holistic dEmand response Services for European residenTIAl communities has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No 957823.

3.5.3 Appreciation and appropriation of energy technologies 30

3.6 Conclusions 34

4. PARTICIPATORY WORKSHOPS' FINDINGS **35**

4.1 Overview 35

4.2 Workshop context 35
4.2.1 Berchidda workshop 35
4.2.2 Voorhout workshop 37

4.3 Workshop engagement process: lessons learned 41
4.3.1 Community matters 41
4.3.2 Timing and venue of workshop is key for engagement 43
4.3.3 Language of facilitation 44
4.3.4 Small groups work better for inclusion 45
4.3.5 Toolkits and serious design games make things easier to engage with 45
4.3.6 Practical planning and follow up 47

4.4 Workshop findings 48
4.4.1 Time-shifting 48
4.4.2 Household and community dynamics 55
4.4.3 Living with technologies 62
4.4.4. Control (of energy technologies) 63
4.4.5. Privacy 64

4.5 Conclusions 65

5. RECOMMENDATIONS AND IMPLICATIONS FOR PARTICIPATORY DR SOLUTIONS **67**

5.1 Overview 67

5.2 Recommendations based on household typologies 67
5.2.1 Families with children 68
5.2.2 Individuals or couples living together without kids 69
5.2.3 Intergenerational families 71
5.2.4 Retirees 72

5.3 Recommendations for the design and technical development of the Hestia platform 74
5.3.1 Design of devices and technology interfaces 74
5.3.2 Frequency of interaction with users and content of interactions 76
5.3.3. Digital literacy and accessibility of users and ways to be inclusive (such as recommendations for voice

HESTIA Holistic dEmand response Services for European residenTIAl communities has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No 957823.

activation or other accessibility issues) 77

5.4 Recommendations for community engagement regarding the Hestia platform 78
5.4.1 Collective vs individual incentives and engagement 78
5.4.2 (Generation and strengthening of) Energy communities 79
5.4.3. Ownership of the energy systems and implications for the design of DR solutions (GRD, DW, AAU) 79
5.4.4. Learning perspective – material vs social learning 80

6. CONCLUSIONS 81

7. REFERENCES 83

8. APPENDICES 85

HESTIA Holistic dEmand response Services for European residenTIAl communities has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No 957823.

List of Figures

Fig.1 Workshop set up in the Wine Museum, Berchidda, Sardinia, Italy Fig.2
Workshop set up in a restaurant in Voorhout, The Netherlands Fig.3 Workshop
set up in a community centre in Camille Claudel, France
Fig.4 The mayor (on the left photo) introducing the Hestia team to the local venue that hosts the Time in Jazz festival that takes place in Berchidda every summer
Fig.5 Welcoming the participants and socialising with them in Voorhout
Fig.6 More senior citizens attended the weekday evening workshops (Berchidda)
Fig.7 More families were engaged in the Camille Claudel workshop, which was held on a Saturday morning Fig.8
Participants in Voorhout playing the energy community game
Fig.9 Participants playing the 'Green Mirror' game in Camille Claudel
Fig.10 Cards presenting different options for householders' everyday energy consumption (ownership of User Studio)
Fig.11 Examples of appliance time-mapping exercise in Berchidda
Fig.12 Participants during the Green Mirror game in Camille Claudel (ownership of Zoe Bonnardot)
Fig.13 Examples of cards showing alternative ways of preparing food in Camille Claudel (ownership of User Studio)
Fig.14 Collated suggestions from participants in Berchidda stating what they consider as an energy community and what is the role of households
Fig.15 Family with kids icon (adapted from Colourbox)
Fig.16 Individuals living together icon (adapted from Colourbox) Fig.17
Intergenerational families icon (adapted from Colourbox) Fig.18
Retirees icon (adapted from Colourbox)
Fig.19 Devices and interfaces icon (adapted from Colourbox) Fig.20
Frequency of interactions icon (adapted from Colourbox)
Fig.21 Digital literacy, accessibility & inclusivity icon (adapted from Colourbox) Fig.22
Community engagement icon (adapted from Colourbox)

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1. EXECUTIVE SUMMARY

1.1 Overview

This deliverable is part of the project Holistic dEmand response Services for European residenTIAl communities (HESTIA), and it reports on the findings of task 2.2.

The report is structured over 5 chapters:

Chapter 1 (Executive summary) presents an overview of the report structure and a summary of the findings. It also outlines abbreviations, acronyms and relevant definitions used in the report.

Chapter 2 (**Methodology for designing an inclusive and participatory DR platform)** presents the methodology and methods used for the task. It begins with an overview of co-creation and its usefulness to Hestia, and then presents the different kinds of data collected, the way these were analysed and an overview of the final recruitment for each interaction.

Chapter 3 (**Household interactions**) presents and discusses the findings of the data collected throughvirtual and real-life household interactions and home tours with participants in two of the pilot sites(Berchidda, Italy and Voorhout, The Netherlands). The chapter outlines the objectives and context for these interactions and proceeds with an analysis of the findings under three themes: Time-shifting and everyday life, Household dynamics and community and Living with technologies. It concludes with important insights and a summary of the findings.

Chapter 4 (**Participatory workshops)** presents and discusses the findings of the three participatory workshops that took place in the pilot sites. It begins by an overview of the context for each workshop.It proceeds by discussing lessons learned during the design and facilitation of these workshops, focusingon issues of engagement. It then presents and discusses the findings, under three themes): Time- shifting and everyday life, Household and community dynamics and Living with technologies. It concludes with important insights into the usefulness of the workshops as methods for engaging communities and by summarizing important issues regarding households' and community energy-related issues.

Chapter 5 (Recommendations and implications for participatory DR solutions), presents recommendations for the Hestia DR solutions, developed using the empirical data from recent interactions. They are presented under three categories: 1. as part of scenarios based on empiricallyidentified household typologies, 2. as recommendations for the design and the technical developmentof the platform and 3. as recommendations for community engagement in relation to the development and maintenance of the platform.

1.2 Overall scope and report summary

This report highlights issues of everyday life of the participating households in the three pilot sites (Berchidda- Italy, Voorhout-The Netherlands and Camille Claudel-France), such as the coordination of household routines, potential for energy flexibility, interactions with energy technologies and ways in which energy communities, current or newly forming, are operating and composing themselves. These issues are identified through the collection of in-depth qualitative data from

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a small sample of participants (householders) in each pilot site of Hestia, carried out through household interactions (real-life and virtual) and through participatory workshops. Therefore results are not to be taken as generalisations but only as context-relevant, empirical observations. Therich qualitative findings should be carefully considered or interpreted within their relevant context, when used toinform further activities in the project, which can relate to the design of technologies or other quantitative representations of households. The following sections presents thematic summaries of the report's findings.

Co-creation

The report presents the importance and relevance of co-creation to the Hestia project, by discussing someempirically identified opportunities and challenges that this approach brings to the project. For example:

- **Co-creation** as the active engagement of participants in the design of DR solutions, has been found to **make the transition processes** (to a smart or more efficient grid) **more meaningful and relevant** for ourparticipants, while it has brought attention to the need for coordination with all stakeholders involved inthe process.
- **Co-creation** is recognised **as a way to assist the empowerment of citizens as consumers**, through collaborative processes (such as their engagement in participatory workshops), where they can co-develop their vision for new services and products.
- **Co-creation can be utilised as a more inclusive innovation opportunity** for the community and for the intermediaries involved. This can take place through the formation of community-driven initiatives and new business models to support the DR solutions developed in the newly created energy communities of the pilot sites.

However, co-creation is also a challenging process, which can threaten the power relationships between stakeholders of projects. Challenges include:

- **Issues of expertise**, which are often encountered during the interactions with participants. While some participants want to be told how to do things 'properly', they also hold tacit or embodied experience of their everyday life at home, which enables them to be the experts of their practices. Some participants challenge the professional expertise by experimenting with different ways of operating technologies of performing energy-related practices. These discrepancies between the technical solutions developed and the lived experience or different perceptions of use from participants can challenge the scheduled activities of partners, or the way in which they have planned to perform a task.
- **Co-creation does not mean co-production**. There is a limit to how much participants can be involved in the design of the Hestia platform. While it is useful to incorporate their experience in the conceptualgeneration of DR solutions, it is also important to find ways to practically incorporate them in the actual production of the solutions. This might bring challenges in the way that the more conceptual insights ofusers are translated and developed into specific technical solutions.
- **Sustaining co-creation for long-term results** is a great challenge. There needs to be careful considerationof what happens after the end of the Hestia project and in what ways communities can sustain local DR initiatives.

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Time-shifting (of appliances and everyday household routines)

The findings also indicate that

- **Household appliances used for cleaning and washing**, such as dishwashers, washing machines and dryers, **were generally regarded as the most meaningful to time-shift**. The availability of appropriate (programmable) appliances and other energy-control technologies (such as accurate smart meters) are considered important tools for participants in this process.
- DR solutions **need to consider the routinised daily rhythms** involved in performing these tasks, for instance consider the sequence of practices in a household, such as if dishwashing is performed after meals and laundry is done when 'the basket is full' or there is an urgent need for specific clothes. Institutional rhythms, such as going to work or school, shape when and how energy is used at home, as the findings reveal. Therefore, the organisation of energy intensive practices performed within the home often reflect broader rhythms, and householders might therefore find it difficult to time-shift energy loads at certain times, because it conflicts with institutional rhythms.
- **Socio-cultural meanings about homemaking**, such as ways and times of cooking and eating together as a family, are challenging the time-shifting of practices such as cooking and dining, which are found to be more 'rigid' than other routines. Reducing or time-shifting the energy required for meals for example need to be tied to social norms and alternative means of satisfying people's dining needs, especially when families or larger households are concerned.
- The **time-shifting** of energy appliances and household practices in the three pilots **is implicated in the gendered division of household labour**, such as the availability of stay-at-home spouses, who are able to perform energy-intensive tasks outside of peak hours. While this availability is useful in terms of practical time-shifting of tasks, it also creates the **challenge of the invisibility of energy-intensive tasks to the rest of the members of the household** who are not present when these take place. There needs to be a better way of monitoring and visualisation of the energy consumption of households and be made visible and available to all members at home.
- Householders are motivated by **financial incentives**, such as lower energy bills, however these **are not the major reason** why they want **to engage in time-shifting**. '**Doing good' for their community and the environment** are considered as high priorities, especially in closely-knit communities. The findings suggest **that energy needs to be considered as more than a financial concern** (energy bill) but also **as a community issue and as a collectively produced and consumed practice of households.**
- There is an **opportunity for time-shifting** of appliances and energy-intensive practices through **the recent adoption of home-working**, due to Covid-19. Being at home for longer, has enabled several participating households to spread energy-intensive practices throughout the day or to amend their irregular routines to respond to local energy loads.

Household and community dynamics

- The findings suggest that **everyday housekeeping is gendered and characterised by divisions of labour**. Most women perform the ordinary, everyday housekeeping, while men, in their majority perform the digital housekeeping and also express the biggest interest in new energy technologies. Most women, as housekeepers or coordinators of everyday routines, are therefore the gatekeepers for introducing new ways of performing everyday practices such as adopting flexibility suggestions
- There in an **identified gap or mis-coordination between the ordinary and the digital housekeeping**. The

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findings highlight the need to bring on board all members of a household in order to introduce and stabilise new practices or alternative performances of existing routines. Therefore, **the coordination of tasks and negotiation between household members is essential in order to make any suggested changes sustainable**. Learning how to adapt existing practices to different purposes is an ongoing learning process, which requires the appropriate skills and meanings to develop in households in order for them to get implemented. **Appropriate intermediation is therefore needed to support this transition**. This can involve 'warm' experts, i.e known or trusted people with similar experiences or professionals who can share tipsand suggestions informally.

- The findings indicate **a diversity of household compositions and dynamics**, ranging from couples or adults living together to intergenerational families under the same roof. **This diversity brings about differences to the in-house dynamics**, as well as to the ways in which these households engage withtheir communities.
- **Communities have been challenged by the long disconnection period, due to COVID-19 restrictions**, especially in newly created ones, such as Voorhout. Participants suggested that their engagement in the Hestia interactions was also a good reason to reconnect with their neighbours and communities and feelpart of a commongoal.Community energy or community ownership models and the role of households are still largely undefined.
- **It is important to have a diversity of perspectives woven into the development of new energy communities**, which represent different ages, genders and roles within the household and the community. Notions of ownership and control of the (newly created) energy communities

Living with technologies

- **Comfort was perceived by participants as more than thermal**. The idea of a comfortable home was disconnected from the technologies within. Meanings of a safe and efficient home were implicated in the understanding of comfort.
- The participants perceived the **sense of control of homely environments as important**, and highlighted**that technological solutions could challenge that.** Several occupants expressed frustration about theirenergy technologies not working and the lack of guidance which they had when moving into the home.
- **Notions of cleanliness,** although differed amongst participants, **were an important factor in shaping ofhow and when energy was used**. Emphasis on practices of showering and cleaning the home were especially present in Berchidda. **Appropriate technologies assist the efficiency and coordination of cleanliness practices** (such as programmable appliances).
- **Ownership of energy technologies**, such as PVS, generally **increased the participants' interest and** engagement with their energy consumption.
- **Participants generally welcome energy notifications and recommendations**. But, **while they appreciatethe comparison with their neighbours, they also do not want to compete with them**. This is an importantaspect to consider, especially if gamification is to be adopted as a way of household engagement indemand response initiatives
- **Most participants preferred mobile phone interfaces** for their energy system
- **Digital literacy is not uniform across and within the pilots**. **Gender, age and stage of life are important factors** to consider **when designing DR interventions and solutions.** Participants need appropriate

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support and guidance, by both familiar people (peers) and professionals throughout the whole process of transition.

Recommendations for the development of DR solutions

Recommendations for the design of the Hestia platform were developed based on the rich qualitative data collected in order to generate more applicable and targeted propositions for action. The recommendations are presented in three sections:

1. **As part of scenarios** based on empirically identified household typologies

As recommendations for the design and the technical development of the platform

2. **As recommendations for community engagement** in relation to the development and maintenance of the platform

- **Four household scenarios were generated**, based on the life stage and household typology encountered in the pilot sites. These are not generalisable to a wider population but only representative of a part of the participating pilot households and they should be used as guidelines only.

    - **Families with children**, present a high and often non-negotiable need for comfort and convenience and their flexibility potential is more restricted compared to other scenarios due to coordination issues in everyday routines
    - **Individuals and couples living together**, perceive comfort and convenience as less important compared to families because of the easier negotiation between the few household members. Therefore they also have a high flexibility potential.
    - **Intergenerational households,** present multiple and more complex notions of comfort and convenience. Their flexibility potential is medium to high, due to the diversity of needs and creating shared meanings and skills for energy efficiency is therefore essential.
    - **Retirees** have well-established notions of comfort, developed throughout life and these notions are not necessarily related to energy technologies. Their flexibility potential is medium to high. There are good opportunities to exploit their loosely routinised everyday life, which provides possibilities of time-shifting practices.

- Design and technical development recommendations:
    - Devices and interfaces need to feel safe and attractive to use and to display in the home
    - Smart phones are preferred and interoperability between devices is desirable
    - The platform should support different levels of digital literacy and be accessible to asmany household members as possible.
    - Generate option for multiple accounts within households
    - Increase the reliability the platform by ensuring that it has been tested in an everydaylife setting outside of the lab.
    - Quantified feedback should be combined with suggestions for alternative ways of performing everyday practices
    - Peer-to-peer support is essential, e.g through warm experts

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- Community and engagement recommendations
    - Incentives should be sensitive to the routinised ways of doing things at home and according to local social norms
    - Involve the whole community in the operation of the DR platform. Participants have an interest in their own energy consumption but also that of their community. Local peer networks are important for sparking the necessary meanings and ideas of why it is important to be energy efficient
    - Experience with other energy technologies and systems, generally expresses a created interest in new modes for shifting or shedding energy consumption levels
    - Notions of community energy ownership should be promoted through the Hestia platform.
    - Tacit knowledge plays an important role in a learning perspective and should be made visible through peer and professionals' interactions
    - Allow regular opportunities for community events, which can support the sharing of tecit knowledge
    - Explicit knowledge should be supported by appropriate intermediation, such as demonstrations and guidance to the platform and how to use it.

1.3 Abbreviations, Acronyms, and definition

PD: Participatory Design

Hestia: Holistic dEmand response Services for European residenTIAl communities

Practice: A practice is a 'nexus of doings and sayings'(Schatzki, 1996). A practice, in the this report is considered as a 'routinized way in which bodies are moved, objects are handled, subjects are treated, things are described and the world is understood.[…]A practice is social, as it is a type of behaving and understanding that appears atdifferent locales and at different points in time and is carried out by different body/ minds' (Reckwitz, 2002,p.250)

Ordinary housekeeping: The term *ordinary housekeeping* is used in this report to indicate the household labour that takes place in homes, such as cleaning, cooking, shopping etc. This does not necessarily involve the handlingof technologies per se, as described in the term below, but in many cases technologies (such as appliances) are involved in the performance of the individual tasks.

Digital housekeeping: Digital housekeeping is a term used to describe the everyday engagements that peoplehave at home with digital technologies, energy systems and media per se or in other words 'the labour required to maintain a networked home'(Kennedy et al., 2015, p.409).

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2. METHODOLOGY FOR DESIGNING AN INCLUSIVE AND PARTICIPATORY DR PLATFORM

2.1 Overview

This chapter outlines the overall methodology for task 2.2 as well as the specific methods used for data collection. These methods, such as the participatory workshops, have previously been explained in principle (in D2.1), so this chapter focuses on the specific empirical characteristics of each pilot workshop, and discusses the kind of data collected and how these were analysed.

The chapter begins by outlining the concept of co-creation, which underscores the overall methodological considerations of the task and the whole of WP2. It explains the benefits and some of the challenges of its practical application in the Hestia project so far as well as in relevance to further planned applications. Following this, the chapter gives a brief evaluation on the intersection of participatory and co-creation methodologies with the theoretical perspective of the sociological theories of practice. This evaluation is based on the empirical experience of the project interactions so far, rather than a mere conceptual one. The implications of combining these two perspectives (participatory methodologies and theories of practice) are translated as actionable practical applications for the design of the Hestia platform in Chapter 5.

2.2 Co-creating the Hestia platform

In the first report of WP2, D2.1, we discussed that the way people use energy at home can be understood as the accumulation of prevalent social norms and as their socio-cultural understandings and doings (Wilhite, 2014). Therefore, understanding the ways in which energy is consumed and sometimes produced, at home, requires a holistic understanding of people's lives and their everyday engagements with the planning, coordination and performances of household practices. Furthermore, the active integration of the households' perspectives and embodied know-how of their everyday routines in the design of appropriate household DR solutions (such as the Hestia platform, can enable the generation of collectively imagined scenarios and everyday routines for the participating households, therefore making it easier to be adopted long-term in their lives. The concept of co- creation refers to this kind of collective activity, which is shared amongst stakeholders of a project, who offer their insights and experiences in order to construct a commonly accepted and meaningful tool or innovation model for their community. Co-creation marks the distinct practice of active involvement of users in the shaping of a product or service, rather than their mere or more passive participation in the process of citizen engagement (Voorberg et al., 2015).

One of Hestia's objectives for T2.2 of WP2, as well as of the Hestia project overall, is to facilitate a dialogue between the technical developers and the prospective users of designed DR solutions in order to allow for the co-creation of these solutions. The reason for this was the opportunity that co-creation brings, as a process, to embed users' needs and practices in the technical solutions, as a way to make them more robust and meaningful to their everyday life. The following section gives a more thorough explanation and background for the concept of co-creation as an active user-engagement process.

Several attempts to implement innovations, particularly of technical solutions, fail, often due to the weak integration of users' perspectives and societal requirements during their making (Gjørtler Elkjær et al., 2021). Furthermore, the development of a technical solution of a given challenge, such as household DR in the Hestia

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case, is often perceived as the responsibility of stakeholders other than the users, such as for example thedevelopers or local authorities (ibid). This separation of the processes of design and use, can lead to the disregardof householders' needs and perspectives, not just in the forming of the solution of an issue but also to thedefinition of the actual problem. This limited or lack of involvement of users, is often reinforced by the view that users such as householders can be difficult to reach and to convince (Itten et al., 2021; Trencher et al., 2014).

In this context, co-creation, as well as co-production, have become relevant concepts to authorities and other organisations that act as stakeholders in socio-technical transitions of cities and households to a more energy- efficient future. The concept of co-creation is currently being used by several disciplines, such as design, innovation, transition studies, planning etc. and it is therefore understood and practised in diverse ways (Puerari et al., 2018). For the purpose the Hestia project, we follow Itten et al's definition of co-creation as a process in which "citizens are involved as co-initiators and/or co-designers of the product or service" (Itten et al., 2020p.22).

The purpose of co-creation in the Hestia project so far has been **to assist participants** (in our case householders)in **collectively understanding (and learning) the purpose of household flexibility** and therefore begin the process of collective construction of specific DR solutions for each pilot site. This collective learning process has variedacross the pilot sites, since these represent very different characteristics in terms of socio-economy, culture,housing,- geography and with regard to capabilities (digital literacy of participants, infrastructure available etc.).

The following sections give a brief empirical review of the benefits and challenges of co-creation as a method to generate collaboration of the stakeholders of Hestia. These insights are empirically derived (but often theoretically supported); from the interactions we have had so far in the project.

2.2.1 Opportunities of co-creation processes in Hestia

- Co-creation can provide **a way 'to make change processes more effective and meaningful'** (Puerari et al.,2018 p.4). This has been obvious through the participatory workshops, where participants were able to share their everyday experiences and express their concerns about living with their energy system at home. The collective discussions enabled participants to contextualise their appreciation of why change needs to happen and in what ways their household can contribute.

- Co-creation can provide **a way to bring attention and coordination to all stakeholders involved** in theprocess of household energy demand and response. It is important for householders for example to 'see', through participatory processes, who is involved in the process of household demand response, andunderstand each stakeholder's responsibility. This helps build transparency in the transition process andassist in the building of trust between the relevant stakeholders. The workshops in Hestia have started the process of introducing users to the range of stakeholders involved and explain their input. However,there needs to be a more coordinated attempt to continue this process across pilots in a more consistent way, allowing users (householders) to build a good appreciation of who is involved and to what extent and to get a clear understanding of the frequency and ways in which households need to interact withthe rest of the stakeholders.

- Co-creation can provide **a way to minimise bias** (of the different groups involved) and allow for a blendingof expertise amongst stakeholders (Itten et al., 2020). It is important to ensure a diverse and fair involvement of different voices of a community in the making of household DR solutions, and therefore bringing people together

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through participatory methods, can allow the 'mixing' of different community groups and perspectives. This has happened to some extent through the Hestia workshops; however, since the community groups of each pilot are quite homogenous in most cases, there has not been a wide variation or striking differences in the characteristics of people participating. However, attention has been put in the diverse ways people perform their everyday housekeeping practices, despite thesimilarities in socio-economic and cultural characteristic. Issues, such as the gendering of everydayhousekeeping or age-related performances of practices, have been highlighted and made more implicit to communities.

- **Co-creation** in projects discussing or introducing technologies **can assist in the collaborative identification of potential challenges** (Saunders, 2018) in relevance to households' everyday live. Thiswas evident in the workshops, through the group discussions of both householders' current experienceswith energy systems (and their relevant interfaces, such as the SWYCS app in Voorhout), and exchangeof opinion based on proposals for the Hestia platform interfaces.

- **Co-creation** can **assist the empowerment of citizens as consumers**. Through collaborative processes (such as the participatory workshops), **citizens can collectively develop their vision and have the opportunity to integrate it in the development of services and products**. In the Hestia case, co-creation processes can assist in the definition and customisation of the platform, as well as in the generation of collaborative peer groups for the long-term maintenance of the DR solutions.

- Co-creation can be utilised **as a more inclusive innovation opportunity** (Gjørtler Elkjær et al., 2021) for the local communities and for the energy providers or intermediaries. Through the collaborations developed throughout the length of the Hestia project (through the workshops and other community interactions), there is an opportunity to inspire or generate community-driven initiatives for new business models to support the DR solutions developed and/or the newly created energy communities in each pilot.

2.2.2 Challenges and limitations of co-creation process for Hestia

- The process of co-creation can sometimes be challenged by the perspective that **not everyone can be a co-creator** as it is a creative practice(Sanders & Stappers, 2008). Issues of expertise are often encountered during the interactions with users, with some participants desiring to be told what is right to do and how to do it by the professionals involved. During the workshops, some participants wanted to know what is the appropriate way to operate things at home with their energy systems for example,in order to benefit themselves, their community and the environment. While there is recognition that thetacit experience of everyday life is valuable, there are still reservations in terms of how to perform certainpractices (involving technologies) which need to be 'confirmed' by the experts. It is important to build confidence to users to trust their actions but also to feel empowered to act in their community without this continuous confirmation from technical or professional experts.

- Co-creation **can threaten power relationships** between stakeholders of projects. Project organisers or coordinators can feel that the process is 'slipping' out of hand, since things do not always move accordingto plan. Furthermore, results of co-creation processes can point to different directions that anticipated, which can cause delays in the programme or change of focus to the project. While there has not been a complete change of focus for Hestia, the fact that users have contributed their insights and desires in issues such as the preferred interface with their energy system (which is mobile phone application),might cause unplanned delays in the delivery of

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the platform, if this was not initially considered. When performed successfully, co-creation equalises the level of control amongst stakeholders, by allocating more decision-making power to the users.

- **Sustaining co-creation** and PD processes **for long-term results** is a great challenge. Questions such aswhat happens when the research project stops and in what ways can communities sustain local DRinitiatives remain relevant and need to be addressed throughout the length of the project. It is importantthat by the end of the Hestia project, an evaluation to be made stating the actual achievements of the co-creation processes in comparison to the expectations set in the first two reports of WP2 (D2.1 and D2.2).

- **Integration of conceptual co-creation with practical co-production**. While it is useful to incorporate users' experience in the conceptual generation of appropriate DR solutions, it is also important to find ways to practically incorporate them in the actual production of the solutions. So, co-creation during the makingof the interfaces of the DR platform (such as the dashboard for example), remains really important as a way to continuously collaborate on the process and improve the service that is being designed. If this link between co-creation (design) and co-production is interrupted or not established, there is a risk thatconceptual or experiential insights are translated in the actual service (or product) in ways not appropriate or acceptable for users.

2.3 Opportunities for the development of DR solutions using a combination of co-creation and theories ofpractice

Co-creation processes are based on the concept of integrating users' perspectives into the development of aproduct or process. However, users are not just individual units, who may or may not act rationally, but also partof social networks, which are characterised by distinct social norms and common understandings. Furthermore, as we have discussed in previous reports (see e.g D2.1 of WP2 and D1.2 of WP1), we need to understand in what ways energy demand is made at home, which implicates the understanding of users' everyday routinised behaviours and household practices. The perspective of the sociological theories of practice, which the Hestia project utilises, directs the focus of the investigation towards people's practices rather than individual behaviours.Theories of practice, which have been discussed in more detail in D2.1 of WP2, focus on the understanding of people's practices and the materials (and technologies) involved, as well as people's competences and understandings of the practice involved.

Therefore, it is important to understand and consider the practices in which users (individuals or collectives) are engaged in, in order to be able to co-identify the challenges they face and co-create their solutions. The combination of these two perspectives (practice theoretical and PD/co-creation) presents opportunities forexpanding the ways in which we research and apply solutions for the transition to more energy efficient households and energy grids. Some of these opportunities, as empirically identified in Hestia so far, include:_

- **The uncovering of the tacit knowledge and experience of users**, through the PD methods, **with an established understanding of users' everyday household practices**. The participatory methods can helpmap and understand the materials, technologies and dynamics of household interactions in greaterdepth, allowing for a better co-creation or application of appropriate solutions for household DR.

- Similarly, the combination of the two perspectives allow for an **in-depth understanding of household consumption patterns.** By taking a practice theoretical perspective, and considering **consumption as 'a moment in almost every practice'** (Warde, 2005, p.137), we realise that consumption (of energy.) takes place while people perform their everyday activities, rather than as a conscious or targeted action. Understanding the patterns of

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these everyday life activities, by engaging people in co-creation processes(such as working with serious games or scenarios), allows us to therefore contextualise the ways inwhich energy consumption can be amended (for example lowered or shifted).

- The combination of the two perspectives can enable the **technology applications/DR solutions developed (such as the Hestia platform) to be collectively produced and understood as part of the wider socio-cultural context** in which they are generated. The technology applications/DR solutions producedcan therefore have a context-based narrative and are appropriate for the distinct needs of a community.

2.4 Participatory and co-creation methods for engaging users in DR solutions

The following two sections give a brief outline of the two methods (household interactions and participatoryworkshops) used, in order to collect data about households' everyday life and energy use at home during the course of Task 2.2. The sections describe the methodological aspects of the methods, such as data collection, evaluation and analysis, rather than the findings through these methods.

2.4.1 Household interactions

Household interactions included in person visits to homes, as well as virtual meetings. The interactions involved semi-structured interviews combined with home tours during which the participants discussed their everyday life at home, including the use of appliances and energy technologies, and the spaces in which they interact wit them. Household interactions are discussed in detail in Chapter 3, however, the next couple of sections present an overview of the data collection process, as well as the evaluation and analysis of this data.

Household interactions were held in two of the three pilot sites (in Berchidda, Italy and in Voorhout, the Netherlands). This was due to coordination issues and delays, which in part happened because of COVID-19, as well as delays in establishing appropriate communication with participants and coordination with the pilot partners. The recommendation is to try and conduct household interactions with the French pilot during theupcoming tasks of WP2, in order to complete the insights developed through the workshops and to go further in depth in some pilot specific practices.

Recruitment

Overall, 17 household interactions were conducted across both pilots (8 in Berchidda and 9 in Voorhout). Participants were recruited through personal contact initiated by the pilot partners in both cases, includingtelephone calls and reminder email messages. Participants reacted positively to this personalised attempt toinvolve them to the project and were happy sharing their experiences of everyday life.

Data collection

The interactions were held partly online, through virtual meetings or through visits at home. The pilot partners designed an outline of the interactions in collaboration with AAU, developing interview guides or topics fordiscussion. A sample of these can be found in Appendices 3 and 8. The aim of the household interactions was toget a more in-depth understanding of householders' everyday engagements with their energy systems, as wellas narratives of coordination and rhythms of housekeeping (including digital housekeeping), alongside people's other everyday life engagements. The interactions were

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informal and lasted about an hour on average. During the process, the researchers collected data through notes, voice recording and sometimes video recording (if thesession was held virtually). In addition, photographs and videos were taken during the interaction, in order to mapdifferent everyday practices and sequences that the householders discussed. Participants were given detailsabout their consent of their personal data collected, as well as an overall GDPR briefing. They had the option to remain anonymous and unidentifiable.

Analysis of collected data

Data from the household interactions was analysed initially by the pilot partners, respectively DuneWorks inVoorhout and GridAbility in Berchidda. A template was then circulated by AAU to the pilot partners, asking them to report on major findings and including their own analytical reflections. The filled templates are attached in appendix 1. In the template, the pilot partners were asked to report on 'raw data' including quotes and observations from the household interactions. Based on this deductive approach, AAU identified three themes (also following the interview guides): time-shifting, living with technology and Household dynamics and community.

2.4.2 Participatory workshops in the pilot sites

Participatory workshops between the Hestia research partners, local community representatives and householders were held-through face-to-face interactions- in all three pilots. Two of the three (Berchidda and Voorhout) workshops were held in September 2021, whereas the third workshop (Camille Claudel) was held in the middle of October.

Recruitment

Berchidda: 30 households were invited but 21 households were present in the workshop, represented by 24 participants in total. Participants were recruited initially through a round of invitations in early September, followed by a reminder the day before the workshop. Some of the participants also took part in the household interviews, and therefore recruited to the workshop through these.

Voorhout: 32 households were invited but 15 households were present in the workshop, represented by 25 participants in total. They were recruited by a personal e-mail, a letter and a postcard reminder.

Camille Claudel: 20-30 households were invited but 4 households were present in the workshop, represented by 11 participants. Participants were recruited through communication in the monthly journal of the Municipality of Palaiseau + CPS's journal, through announcement in the CPS website, through posters in the district's buildings, shops and bus stops, through direct contact on 4th September during the 'Associations Day' (local event), throughface-to-face communications (in front of school, market), and by the social landlords and direct contact with 'building council' and finally through phone contact with respondents from previous surveys inside CPS.

Data collection and evaluation

A mix of ethnographic and participatory design methods were used to collect data in the three participatoryworkshops. The methods for each pilot site are presented in detail below:

Berchidda

The outline for the workshop was developed by GridAbility in collaboration with AAU. The aim was to capture energy consuming practices at home, gendered divisions of household labour and engagement and preference for energy feedback.

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The workshop was conducted at the Wine Museum in Berchidda and lasted for 2,5 hours. Participants were engaged in discussions (with their peers) about their energy consumption at home through a visual tool, representing a 24-hour timeslot. Participants were asked to indicate and discuss at what times of theday they used energy and for which purposes. Data was recorded by video (and audio), and representatives from GirdAbility and AAU made observations. In addition, photographs were taken during the entire workshop session. The participants were given a GDPR briefing and informed about how data would be used and their rights fordata. The workshop was conducted in Italian and was facilitated by Gridability.

Voorhout

The outline for the workshop was developed by DuneWorks in collaboration with AAU. The aim of the workshop was to capture preference for living with energy technologies at home and in the community and additionalidentify scenarios for energy futures. The workshop lasted 2,5 hours, it was conducted in Dutch and facilitatedby DuneWorks. Participants were engaged in a game (developed by DuneWorks), guiding them through different scenarios and asking them to discuss their preferences for their energy community. Data was audio recorded andrepresentatives from DuneWorks and AAU took notes and made observations during the entire workshop. Inaddition, photographs were taken. Participants were briefed about the aim of Hestia and how their data was to be used.

Camille Claudel

The outline for the workshop was developed by EDF, with inputs from CPS, User Studio (local based design agency) and AAU. The aim was to capture the participants' everyday practices at home and their preferences for time-shifting. Participants were, among others, engaged through a game (developed by EDF) were they wereasked to indicate their used of energy in time-slots. Data were recorded through notes and observations, and photographs and videos were taken during the workshop. The workshop was conducted in French and facilitatedby EDF and User Studio. Participants were asked to sign a GDPR briefing in the beginning of the workshop.

Analysis/coding

After the running of the workshops, AAU circulated a template to the pilot partners, in order to fill in the collecteddata, as well as their own reflections on the events. Therefore the analysis of the workshop data was based onan inductive process of identifying different themes that came out of the evaluation of the pilot partners' reports,in combination with AAU's researchers' notes and observations during the workshops. The identified themes were then collated into three categories (time-shifting, household and community dynamics and living withtechnologies), which matched those identified in the household interactions.

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3. HOUSEHOLD INTERACTIONS

3.1 Overview

The following section outlines the findings from the household interactions conducted with the participants in the Hestia pilots in Berchidda and Voorhout.

3.2 Context and objectives

The household interactions were deployed as a method for understanding the participants' everyday practices in-depth. They were initiated as a way of bridging the different engagement initiatives in the Hestia project, fromthe survey to the participatory workshops. The aims were several, and the household interactions also helped support the recruitment of participants for the Hestia project and maintain continuous engagement with theparticipants.

The household interactions were conducted as virtual, with the option of performing a face-to-face if safe and possible. This was due to travel and gatherings' restrictions in the specific pilot-sites. However, as the COVID-19situation gradually improved in the pilot countries, it became possible to conduct some of the interactions as physical meetings, which added an extra layer to the understanding of everyday life at home. The interactionsare thus a mixture of virtual and physical engagement. In total, 17 household interactions were conducted (8 in Berchidda and 9 in Voorhout). Due to low levels of recruitment and lack of time in Camille Claudel, no householdinteractions have been conducted there so far.

Based on the collected data, three overall themes were identified:

1. Time-shifting and everyday life
2. Living with technology
3. Household dynamics and community.

In the following sections, findings from the interactions are presented and discussed under each theme.

3.3 Time-shifting and everyday life

In the following section, findings from the household interactions regarding time-shifting are presented. In DR initiatives, and in the context of Hestia, time-shifting of energy loads is at the core. During the interviews, participants were therefore asked about the preferences for time-shifting everyday appliance, but also how it would relate to their everyday life activities in general.

3.3.1 Time-shifting appliances at home

The role of households in balancing energy production and demand was a theme throughout all of the interactions and the participants reflected on how they could time-shift the use of domestic appliances.

Appliances used for cleaning and washing were generally regarded as the most meaningful to time-shift by the participants. Appliances included dishwashers, dryers, irons and washing machines. Some of the participants, e.g. in Berchidda, already had established a routine of using certain appliances at night. One participant from Berchidda for

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example programmed the washing machine and dishwasher to run during the night to benefit fromfinancial savings. When she woke up at 06:30, she would then empty the dishwasher and get the clothes out of the dryer. However, it was not all the participants who perceived time-shifting of appliances as comfortable, soit is still important to investigate each households preferences for time-shifting.

While the participants generally perceived 'cleanliness-appliances' easy to time-shift, the practices involved in washing clothes and dishes were both routinized and time (and energy) intensive. In Berchidda, washing machines were run on a daily basis (e.g. a female participant from Berchidda who washes between 1-3 times a day), and so were dishwashers. One of the participants from Berchidda, explained how use of the dishwasher was very routinised and always happened in the evening after dinner. Besides being used for cleaning dishes, thedishwasher also played a role in the routines of cleaning the kitchen, stoving away the things that had been usedand freeing up space. Despite the perceive ease in shifting the appliances, the participants found it more difficultto time-shift the practices, asthey were both routinized and time intensive.

The use of washing machines and dishwashers were entangled in several everyday practices performed in the home at certain times and by certain household members. In Italy, many of the female participants were stay-at-home-spouses. They were therefore responsible for doing the domestic chores in the home. For stay-at home spouses, washing and dishwashing was done at certain times of the day (in the morning), as part of a daily routine of 'getting the house ready' for when other households members would return home. In that aspect, the use of everyday appliances was closely linked to the temporal rhythms of the household members not staying at home, such as going to work or school. Some participants expressed how washing, cleaning and the operation ofappliances mainly happened during the weekdays and not in the weekends, as this was considered as leisure time. The use of domestic appliances was also coordinated in time and space, e.g. the use of the washing machine was followed by using the dryer and then an iron. For that reason, some occupants regarded it as easier to time-shift the dryer rather than the washing machine to night hours, as the clothes would then not stay in a wetcondition for many hours.

Many participants explained that their appliances were energy efficient (class A and up) and perceived this as something that benefitted them financially. However, the use of appliances did not reflect a similar focus onenergy efficiency, and programmes of e.g. the washing machine were not selected for their economic or environmental benefits. Instead, participants often used one programme (they do not shift between programmes) and often prefer those that had a shorter length or the once that promised a more in-depth washing.

Dishwashing was for most participants related to the daily rhythms of cooking and eating. The dishwasher wasusually operated in the morning, after breakfast or in the evening after dinner. Operation of the dishwasher (and the washing machine) was also dependent, for some participants, on whether, it was 'full' or not (or the laundry basket being full).

Some of the participants who had their own PVs installed, mentioned that they would pay attention to the production of energy and coordinate the use of appliances accordingly. For instance, this was the case in Voorhout:

'Something we pay attention to now, washing I do during the day and we try to do the dishwasher as much aspossible during the day when it is still daylight'. (Voorhout participant, Female).

Dishwasher, washings machine and dryers were not the only domestic appliances which the participants commented on during the interactions. In Italy, most of the participants highlighted the TV as an importantappliance within the home, which played a central role for gathering the family or creating a certain atmosphere. In Berchidda, a female participant explained how they would spend most of their time in the living room where the TV was placed. Another participant from Berchidda, said that they had a massive presence of television sets in their home, and that they would be

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turned on for 6+ hours each day. TVs were often used more in the evening,and were in some instances also turned on when nobody was watching.

Heating and cooling of the house (and bodies) were mentioned less during the interactions, but some hints were made in Berchidda, where one participant explained that the AC was on all day during the summer. In Voorhout, participants lived in newbuilt houses, equipped with new energy technologies. For some participants, the possibility of (passive or active) cooling was new, but it was perceived as a good thing: 'Cooling is kind of nice, youcan bring the temperature down 3 or 4 degrees' (Voorhout participant, Female). Another participant in Voorhout explained how these new energy technologies actually enabled them to be more efficient with their use of energyat home: 'You know it's a game: we have the stuff [the technology in this house], so now we take [theenvironment/efficiency] into account' (Voorhout participant, Male).

3.3.2 Everyday rhythms, sequences and routinised ways of doing

As it is presented in the previous section, time-shifting domestic appliances is not only about knowing how to doso or having the necessary technologies. The use of domestic appliances is deeply entangled in the activities thatparticipants perform at home. These are often routinised ways of doing and performed without much reflection or 'we do this, because we have always done it like this' attitudes. In the following, focus will be on the routinisedactivities performed at home and how that relates to the challenge of doing time-shifting.

When (and how) energy is used at home, is very much related to institutional rhythms such as going to work or school. While institutional rhythms often occur outside of the home, everyday routines within the home alsoshape and are shaped by these institutional rhythms (e.g. by sharing certain expectations and norms). In Berchidda, a participant gave an example of the coordination between routines performed outside of the home and domestic routines. The female participant would spend about 12 hours outside of the home per day, leavinghome around 8 in the morning and returning 12 hours later. After arriving home from work, she would take a shower, followed by arranging the clothes that needed washing (and had been used for work during the day). Dishwashing was done in the evening, after the dinner, while the mornings were used for doing more 'basic' housekeeping. The more in-depth cleaning/housekeeping tasks were done on Friday nights or in weekends. Thisinvolved changing bedsheets or washings windows. In this way, institutional rhythms influenced which domestic tasks that were performed at which time of the day and the week.

Seasonal variations also influence the coordination and synchronisation of energy consumption at home. The specific season of the year influence how often (and for how long) participants spend time at home. Some participants in Berchidda explained how they would work longer hours during the winter season, and stay athome in the weekends, while in the summer season, they would more often be outside (e.g. going to the beach). For example, a male participant who explained how Saturdays (in the winter) were often used to do a thorough cleaning of the home, while Sundays (mainly in the summer) were spent outside of the home (visiting family).

Another important theme, especially evident in the case of Berchidda, was related to the division of labour at home. Many of the female participants were stay-at-home-spouses, while the male participants would work longer hours outside of the home. A female participant explained how her husband would leave the house around 3:30-4:00 every morning. She would wake up a bit later, around 7-8, prepare breakfast and take the kids toschool. Upon returning, she would start doing housekeeping. Around midday, the husband would return for lunch(which she had prepared). In the afternoon, she would pick up the kids. In the evening, they would stay together in the living room, usually with the TV on.

In another household, extraordinary circumstances, such as illness in the family, would also have an effect on theenergy consumption and the possibility for shifting it. The participant explained how she would do most of the housekeeping, as her husband is ill. In this situation, the shifting of energy loads became less important/meaningful and providing

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care was regarded as more important.

Routinised ways of doing are often performed in sequence. A male participant from Berchidda gave an exampleof a typical morning routine. First thing they would do, was to use the microwave to heat a hot drink, beforestarting the washing machine. He explained that there was not a specific need for washing clothes in the morning,but that it remained part of a sequenced routine of doings. The TV was turned on at lunchtime and remained on until 4-5 in the afternoon. It is switched back on around 20 in the evening and the family would watch the TV until late in the evening. The use of the TV would therefore be in the background throughout the day, only turnedoff during dinner time. Another example of a routinised sequence was given by a participant from Berchidda. Shewould start the day by going for a run, before waking up the children and preparing (and eating) breakfast. For lunch, her husband would return home to eat, while she would be out of the home (eating with friends). Around 6:30 in the evening, they all return home.

A final example comes from a couple living in Berchidda. In their household, the husband is the first one to wakeup, eat breakfast and go to work. The wife would wake up later, eat breakfast and start doing housework. She would first start the washing machine, then vacuum and start the dishwasher. Like many of the other householdsin Berchidda, the family would return for lunch and eat at home. After lunch, the wife would normally do ironing. The TV was on all through the day.

3.3.3. Motivations for time-shifting

During the interactions, several of the participants touched upon their motivation for doing time-shifting. The following section gives a few insights into their beliefs.

The use of technology was generally perceived as a positive motivational aspect, which could help participantsin time-shifting their appliances. A male participant from Berchidda mentioned that he would like to receiveinformation and suggestion on how and when to shift energy consumption and expressed that he was curious towards new technologies in that aspect. An important point regarding the use of technology (to do time-shifting) was that it should not take away control from the participants. The participant highlighted how it was important for him to stay in control and decide whether to intervene or not (based on information).

Regarding financial savings as a motivational factor, there was no clear trend among the participants. Some participants explained that monetary savings was not a motivational factor. They were instead motivated by a care for the environment and the use of novel energy technologies. Other were however very interested in financial savings and had in some cases already installed energy technologies (PV's) with the aim of achieving financial benefits-In Voorhout the questions surrounding financial savings and energy consumption were more suppressed due to the existing tariff structure. At the moment, the households in Voorhout pay a flat tariff of 18 euros a month andthus does not pay much attention to their bills. The participants in Voorhout expressed that they liked the flat rate (P4 and P6), and compared the costs to their previous living arrangement:

'When they told us you'll have an energy bill of 18 euros a month... That's just out of this world! Something you just can't imagine coming from an apartment where you pay about 100 euros a month. You really have to live insuch a concept just to understand and to feel how well it is all done' (Voorhout participant, Male)

3.4 Household and community dynamics

Another theme is how to engage participants in inclusive demand response initiatives is social dynamics. These include household dynamics and extends to the broader community of participants. In the following section,findings from the household interactions are presented regarding who uses energy at home, i.e. how household chores are performed and by

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whom, but also participants' reflections on community and how the broader social dynamics influence energy consumption and ways of shifting such.

3.4.1 Gender

An important dimension of household's dynamics and energy consumption is gender. Most of the participants lived together with a partner of the opposite sex, and especially **in Berchidda the division of labour within thehome followed more traditional lines.** As previously noted, many of the women were stay-at-home-spouses,while the men would often work long hours outside of the home. It was therefore mostly the women who were in charge of doing the housekeeping tasks, such as cleaning and cooking. A female participant from Berchidda described these tasks as a mental burden, but also highlighted that her partner helped out when he could (with laundry, dishwashing and hovering). She considered herself 'educating' in how to do housekeeping. **The division**

of labour within the household was by many, explained as a result of different schedules (the man leaving for work and the women staying at home), rather than something related to more traditional gender roles. This division was therefore not perceived as unfair despite participants acknowledging that their family organization was more traditional. Instead, the participants explained that their different roles complemented each other, and **while the man was perceived as a provider, the woman was seen as a caregiver, looking out for the house andthe kids**.

The women in Berchidda would, given their role in performing housekeeping chores, do most of the cleaning andcooking at home. This would involve using different appliances and thus a use of energy. The ongoing work in thehome would normally be performed every day and follow certain rhythms. **In Berchidda, the men's involvement in housekeeping was more sparse and usually performed on an ad-hoc basis. They would usually take on the role of a 'helper', which occasionally would do the dishes or laundry**. In a Berchidda household, the couple explained that they had an open discussion about the splitting of household chores, despite her acknowledging that she would do most of the work in the home. A similar division was seen in other households from Berchidda. In one, the woman explained that she did all of the housework and that she took great pleasure in doing so, reflecting that this was her 'job' in the family. Contrary, another female participant from Berchidda did not see her role as the main housekeeper as especially valuable, but instead as something that needed to be done (and she was the only one to do so).

In Voorhout, despite not being as visible as in Berchidda, a general observation was that the women tended to speak more in terms of domestic practices. While the organisation of everyday life was less traditional in Voorhout, it still seemed that women played a bigger role in the performance of housekeeping chores.

This was also highlighted by findings, where actual performers of practices (in many cases the women), contesting claims about how much time-shifting could be performed:

F: I used to wait until the night tariff kicked in, but now I do the laundry in one day, everything goes through. I'mnot going to puzzle. That gets me nowhere. So, I'll do 5 or 6 loads in one day.

M: maybe you could take one out [if a request would come, which he initially said was not a problem]? F: Yeah,whatever. I have to juggle so many things already. (Voorhout participants, Couple)

On the contrary, in the interviews, men tended to speak more on technologies and how they work. In Berchidda,it was mostly men who were responsible for installing new energy technologies, as well as using those (e.g.monitor PV production). Likewise, in Voorhout, several on the interviewed women explained that they were not very technical minded, while their husbands remained more technological interested:

'[My husband] handles most of the things. I want to stay informed, but he mainly does the technical stuff. But Ialso want

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to understand some things, because when he is away or something, I need to know something of everything. So sometimes I go into the meter cabinet, because while he'll want to explain it, I have to understand it in my own way as well. So yes, I do know a bit about everything'. (Voorhout participant, Female)

Some of the female participants even feared that their disengagement with the technologies could be a disadvantage as they wouldn't know how to control the home if their partner wasn't there:

'Should my husband die before me and things continue the way they are now, I'll sell the house. I'll leave. It's allthe technical malfunctions. It makes me nervous'. (Voorhout participant, Female)

In both pilots, it was the men who were responsible for most stages in the technological process, from setting itup, doing the continuous monitoring, ring the developers if technical help was needed:

'No, I don't call the developers about malfunctions. My husband always does that. When they're here, I'll talk tothe mechanics, of course'. (Voorhout participant, Female)

3.4.2 Intergenerational household dynamics

A second interesting finding from the interactions was related intergenerational dynamics. In Berchidda, thisextended to households where several generations lived under 'the same roof'. For example, a female participantfrom Berchidda had her mother living in the same building, which meant that both their energy consumption wasincluded in the same energy bill. The different family constellations and intergenerational dynamics can bechallenging in efforts to coordinate everyday practices so that they mirror certain peak periods. During theinteractions, not all the different members of the household were present which meant that certain representations of how energy is used is missing.

This also extended to children. They were often not mentioned in relation to housekeeping, but played an important role how energy was used. In relation to smaller children, care was central, and housekeeping tasks such as laundry, cooking and cleaning were often done in certain sequence to provide care for the children.

Living together under the same roof, also had a certain potential in learning from each other. In one household inBerchidda, one of the two brothers lived in the same building. One of the brothers played an important role in introducing and learning how to use new technologies within the home, and the participant referred to him as having a mentor role, which could lower the adoption barrier for energy technologies

3.4.3 Community

Communities sometimes play an important role in the production and consumption of energy. These are well- know from 'energy communities', where households share certain energy technologies that allow them to produce and store energy. Furthermore, community is a social construct, where people engage with each other. Community is thus more than just living in the same neighbourhood, but it is also about sharing a notion of what it means to live here.

Both Voorhout and Berchidda are relatively small villages, and specifically the pilothouses in Voorhout represent a small enclave of houses (46 units), while Berchidda is much bigger, consisting of several hundred homes. The participants in Voorhout described themselves as a relatively close-knitted neighbourhood, with close relations among the household. In Voorhout, some of the participants e.g. described how they would do things together with their neighbours, such as watching football on a big screen. It was also described how some participants were more engaged in the neighbourhood than others. One Voorhout participant described a 'go-between guy', who would with the occupants on the other street on shared technical issues with the houses.

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During the interviews, the participants in Voorhout were asked about their interest in establishing an energycommunity. The answers were not decisive, and while some participants found the idea of a collective attractive, others were more keen on relying on individual self-consumption. The participants who disapproved of thecollective thoughts, highlighted that it can be tricky to agree with multiple households. Others, and those who approved the neighbourhood idea, saw the community aspect as a strength:

'It should get fun too, to make a transition in our lives, to use more energy in the, in the lower cost period. Howcool would it be to have a game with our neighbours? The winner should be the one who makes the most adjustments to their own life? Wouldn't that be fun?' (Voorhout Participant, Male).

3.5 Living with technologies

A third and important theme in the interactions was how participants perceived and used energy technologies. The participating households were different in many aspects, and also in regard to which technologies they had installed at home. While homes in Voorhout generally had newer energy technologies installed, homes in Berchidda presented a more mixed picture.

3.5.1 Comfort, control, and cleanliness

Throughout the interactions with the participants, comfort, control and cleanliness seemed to influence how the participants perceived and used technology in the home. Among the participants, these notions remained prevalent and of importance to their use of technology, and thereby to the possibility for increasing engagement with DR initiatives.

Comfort was by several participants perceived as more than thermal. A household in Berchidda, described how they liked being at home (and thought of it as comfortable) despite experiencing quite big temperature variations in their home. In the interview, they explained how they only experienced the home as comfortable '15 days a year', but their notion of comfort was instead interwoven in the activities performed in the home, such as gettingtogether in the living room/kitchen. They refer to their home as a nest. Interestingly, after having a child, the household started to think about comfort in a more 'thermal way' and took new measures to ensure a goodindoor environment. The chase of thermal comfort was also reflected in a changing use of appliances, with the household explaining that they had started to use the AC more extensively after having a child.

Comfort was perceived as important for some of the participants, ranking it higher than environmental concernsand costs savings, while others perceived it as less important. To many of the households, comfort was perceived as 'in the background', to habits and routinised ways of doing. A male participant from Berchidda explained that as he perceived comfort as very important, he did not see any problem in time-shifting appliances, as long as the home remained comfortable. In other households, existing comfort problems were prevalent, such as in a household in Berchidda, where half of the house was not used very often due to a lack of comfort. In order to counter this problem, the participants had installed an AC, making it a bit more comfortable.

Cleanliness was important for many of the participants when they performed everyday practices. For some participants, it was important to keep the house as clean as possible, meaning that they would hoover every day and do dishwashing after each meal. Notions of cleanliness was also connected to temporal rhythms of everyday life. A participant from Berchidda explained how the need for cleanliness and related energy consumption was connected to their professional needs (clothes used when working).

Another major theme was control. This includes control of technologies, i.e. knowing how to perform certain **actions using the technology, but also more general control of homely environments.** The latter refers to a notionof being in control. Of specific interest to this report is how the influx of new technologies in the home might reconfigure what

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it means to be in control of the homely environment.

Among many of the participants it was important that technology did not take away control from them. This refers both to the how control is performed, but also how control is perceived. Several participants highlighted that they **did not want to live in a fully automated house, but wanted to decide and intervene themselves.** This puts up specific requirements for the DR technology, with a stronger focus on developing recommendations upon which the participants can act themselves. This notion was explained by a participant from Berchidda: 'I want some spontaneity in life (…) the optimal moment for technology may not be the optimal time for me'.

On the other hand, participants seem to like some of the new control features which technology could bring. A male participant from Berchidda expressed that home automation could be comfortable and enable financial savings. This was especially related to the use of remote control, which by several participants was highlighted as a feature of interest. In these instances, remote control of domestic appliances was perceived as something that increased their sense of control. With remote control it would for instance be possible to turn off devices remotely or monitor problems (such as overconsumption) before the bill arrived.

However, other participants expressed that energy technologies were complicated, and they felt disempowered when faced with new technologies in the home. These participants explained that they lacked the technological competences to control the technologies. A better solution for those was therefore more information about their energy consumption, communicated in a way that was easy to understand (and on a platform they knew).

In Voorhout, there was also a perception that feedback on energy consumption would increase their sense of control over their domestic space, one participant expressing:

'I want to have insight into the house. I simply want to understand what's happening in my own home' (Voorhout participant, Male)

The features of technology were once again highlighted for their possibility to shed light on energy consumption and provide recommendation for how to act upon such. In addition, participants in Voorhout were generally not opposed to having to learn about how to use new technologies in the home. They expressed, as also seen in Berchidda, an interest in learning more about their energy consumption, and energy technologies was here seen as an important tool, which they should know how to master. The participants from Voorhout also expressed that feedback technologies could help them consume energy more 'responsible'.

Expectations and real-life experiences often conflict. In the Voorhout case, many of the participants already had energy technologies installed. Yet, many of them felt frustrated with these and expressed how it did not live up to their expectations. One participant explained:

'I just I just mainly keep my hands off all the equipment here. We simply just don't know how to use it and when to use it'. (Voorhout participant, Male)

Other participants expressed similar thoughts and feared that the new technological installations could be a fire hazard within the home, given extreme temperatures around the battery inverter and in the meter cupboard. Others expressed that the new technical installations in the home had led to more instances, in which technology operated in unexpected ways, and where they were not able to control it.

3.5.2 Home (making) and technology

As detailed in the above sections, living with technology is related to the activities that participants perform at home. Home and the making of home is an evolving process, performed through a myriad of practices.

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To some of the participants, home and the idea of a comfortable home was disconnected from the technologies within and e.g. the ability to maintain a comfortable temperature. For example, this was the case in Berchidda, where a household described their home as a nest and a place they liked to be, despite experiencing it as uncomfortable (too humid) for many days of the year. **For others, technologies play an important role in homemaking.** These people were proud of their home, for instance because they had built the house themselves, and took great pride in making it as comfortable and sustainable as possible. In the case of several participants from Berchidda, the process of homemaking typically involved installing new technologies, and the homemaking process was very much connected to technologies and other building materials (they considered getting an AC and PVs).

Other notions of home were also present, for example a female participant from Berchidda describing the home as a safe-haven: 'I feel safe in this home, it reminds me of my childhood'. In this instance, the notion of home was not (only) related to technologies and material objects, but instead to a feeling and a sentimental attachment to the house.

In Voorhout, the material context is different, and the participants all live in newbuilt houses. During the interview, many of them expressed great delight in their homes:

'We love this place. (…) It's like I'm on permanent vacation, but we're at home'. (Voorhout participant, Female)

The reasons for this positive attitude varied, and while some highlighted the energy-positive design of the homeas an important aspect for their choice of living in this home, other highlighted the location, the spatial layout, the garden and the neighbourhood. In Voorhout, many of the participants were seniors and the home was perceived as something that secured them a safe place of living while getting older, as expressed by a maleparticipant: 'This house gives us peace of mind – we are already enjoying the safety of a future-proof home, whenwe'll be less able [getting older]'.

3.5.3 Appreciation and appropriation of energy technologies

In the following, focus will be on how the participants perceive energy technologies e.g. what they prefer inrelation to communication of energy data. Furthermore, and just as important, extracts on how the participants use energy technologies will be presented.

Awareness

A major theme throughout all interactions was the communication of energy consumption data. Questions were asked on how the participants checked their existing energy consumption and if they were interested in newways of receiving this and, if so, how. **In general, many of the participants did not pay much attention to theirenergy consumption**. Several households would only check their energy consumption when receiving a bill, and they would not pay much attention to how much energy they consume on a day-to-day level. Despite a lack of awareness on energy bills or the amount of energy, participants might be attentive to energy consumption in other ways. They would use know-how, which is not expressed in numerical values (e.g. kilowatt hours, kWh), but instead relates to everyday doings. A participant from Berchidda explained how he remembered certainmonths being more energy intensive by feeling a higher temperature from the heater(s) within the home.

For some participants, a general interest in energy technologies increased their awareness of energy consumption. These household had their own PVs, and paid attention to the production energy. An interest in energy technologies such as PVs, though, did not necessarily generate specific kind of energy expert knowledge,as in the case of a participant from Berchidda, where he knew if the PV would produce electricity by "looking at the sky" (i.e. looking at the weather)..

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Several of the participants already had technologies, which allowed them to see their energy bill on a digitalplatform, but that did not necessarily lead to greater insights on energy consumption. On the contrary, some of the new digital technologies, could lead to confusion for participants and a need for 'translation':

'But don't ask me what all those numbers in the meter cabinet mean. I see things appear there and I think 'that can't be'. The way I think it is, that shouldn't be possible. But it's what I'm seeing' (Voorhout participant, Male).

Preferences for automation and feedback

Most participants perceived remote control and automation as a comfortable technology, that could ease the coordination of everyday tasks as well as generate energy savings. Still, others had no interest in the use of remote control, as they did not see the need for such. In the case of a household from Berchidda, they related this to the fact that they already spent a lot at time at home (and therefore did not need the option of remote control).

Visualisations of energy consumption was perceived as a positive thing by several of the participants. In the interactions, the participants expressed that visual feedback could be helpful in providing an easy overview of how much energy they used and when. **Most participants preferred to have the feedback presented on their smart phone (using an app)**, as they would use the phone frequently throughout the day and it would allow for remote monitoring and control, is possible. Regarding the use of notifications (e.g. recommendations), theanswers from the households were not showing a clear trend. Some participants would find it very useful toreceive suggestions on when to time-shift energy consumption, and they saw this as an important tool forbecoming more flexible in their use of consumption, as explained here by a couple from Voorhout:

M: 'It can also be quite simple: three lights. Green means a great time to consume electricity, orange meanswe'd prefer not, and red only if you have to. You can still do it of course, but it's clear that that's not a goodtime.'

F: 'Yes, I don't need an app to dig into graphs and numbers. But I do want to know how I can make the best useof my house and electricity.'

However, other participants would find it stressful to receive notifications on the phone, and they would prefer to only have visualisations and take their own decisions based on that. The participants who opposed receiving suggestions explained their resistance with the challenge of doing time-shifting based on such suggestions, as this would also involve changing their routinised ways of doing. Receiving suggestions would therefore be perceived as less useful, unless they also adopted new ways of doing. In other words, these participants were generally more sceptical about changing their routines and therefore also more sceptical about receiving notifications.

While feedback on energy consumption was regarded as a helpful feature, it was important to some participants that the information was contextualised and that it explained why a certain suggestion was presented. Providing meaningful insight was seen here as a motivational factor.

Some occupant expressed a preference for energy feedback presented in graphs, others in numbers. There wasno clear trend in the answers regarding whether energy feedback should be presented in graphs or numbers. Forexample, in one household the husband preferred graphical representations, while the partner preferred numbers. One household also expressed interested in comparing energy consumption, perceiving it as a fun wayto learn:

'We used to compare meter readings - almost like a game. It is fun and educational because you cankeep an eye out for any errors or problems'. (Voorhout participant, Male)

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In Voorhout several of the participants had already got experience with energy feedback through an app, from which they could monitor their PV production. Their experiences highlight how expectations and the real-life useof technologies often differ. Several of the participants complained about experiencing technical issues, and theyhad therefore stopped using the app. Others expressed that they did not use the app, because they were not very'technical' or that it did not provide enough insight (had too few features).

Ownership and engagement with energy technologies

In both pilot sites where interactions were conducted, some participants already had experiences with energy technologies. These were either pre-installed in the house, as in the case of Voorhout, or they were installed on the participant's own initiative, which was more the case in Berchidda. Which energy technologies the households had installed therefore also differed, though households in Voorhout were equipped with similar technologies. Inthe following section, findings on ownership and how participants use energy technologies (broadly understood) is in focus.

In Berchidda, several of the participants had PVs on their roofs. These were typically installed on their own initiative, and during the interview the participants expressed their competences regarding how the PV's worked. One male participant from Berchidda, had PVs for 11 years and was interested in exploiting the benefits which this gave him. His vision was to have an electric boiler and a heat pump, so that he could benefit from the electricity that he produced. He, furthermore, expressed an interest in exchanging experiences with other members of the community, who also had PVs on their roof. The participant was also engaged in optimising the use of energy at home, so that it reflected the PV production, and he explained that he was aware of the usage of appliances at home.

A similar thing was expressed by another household from Berchidda, who have had PVs since the summer of 2021. The households were also changing their habits, so that it would reflect the production of energy. As the examples from Berchidda shows, **it seemed that participants who had other energy technologies and initiated the installation themselves, were more often aware and interested in their consumption levels and in some instances also tried to time-shift consumption, so that in followed energy production**. Some households from Berchidda had installed an app on their phones, which allowed them to follow their PV production and some participants who had energy technologies, expressed visions for equipping their house with heat pumps (some had already done so) and energy efficiency measures, such as insulating their home. **These findings suggest that ownership of energy technologies increased engagement with energy – both production and consumption.** Thisshows that while some participants find using energy technologies frustrating and complex leading to possible fewer insights into their own energy consumption, others find them useful and will increase their engagement with energy.

The reasons for installing PVs differed and while some explained that it was due to financial reasons, othersexpressed that care for the environment was their main drive. Contrary to Berchidda, the households in Voorhouthad many energy technologies pre-installed in their homes. **The participants in Voorhout did not express thesame level of knowledge regarding energy technologies in general, and several of the participants expressed that learning to live with these technologies was a learning process,** where they had to figure out how they worked.While some Voorhout participants expressed an interest in learning new things, others expressed frustrationabout a lack if introduction and guidance in how the technologies worked, as e.g. explained by this male participant:

'Living with the home's energy system took a while to adjust. I think it is quite a complex system, for ordinarypeople'.

Participants described how it became a necessity to learn how to use the technologies:

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'Now that we're living here and things aren't going as planned, we're starting to familiarize ourselves more with it. Out of necessity'. (Voorhout participant, Female)

Through their engagement with the technologies, the participants also learned about malfunctions in the technologies. This was often reflected in everyday experiences and not necessarily by looking at a display. One participant from Berchidda explained that they experienced an increase of moisture in their home, leading him to conclude that their heating technologies was not working properly, In another case, the problem was identified through an installer:

'Someone came by from an installation company, to check the optimizers. When he looked in the electric panel he saw that the wires were fried. And the same thing happened a couple doors down. That has us concerned ofcourse. We're worried the house may catch fire you know. These are serious problems'. (Voorhout participant, Female)

Again, several participants expressed frustration about their energy technologies not working and the lack of guidance which they had when moving into the home. They specifically missed more explanations on how to use the technologies: 'We really need some guidance. We feel very lost'. (Voorhout participant, Female)

The lack of competences for knowing how to control and manage the energy technologies, also reflected that many of the participants in Voorhout had no experience with similar technologies. **Many expressed that theywould have preferred guidance from an early point, and that they lacked the interest and competences to engage** and try things out themselves:

'The manufacturer of the house did send over a person and there was an app in which you could only find outhow cool it was in the house. But yes, we definitely need additional training and guidance on how to use everything'. (Voorhout participant, Male)

The frustrations about living in their house, became an issue of communication. Many missed a clear communication link, and found it challenging to get in contact with the 'right people'. These examples show howimportant it is to include intermediaries such as installers, in the guidance and training of participants. This is especially important in newly built homes, where the people who move in, maybe aren't familiar with the energy technologies:

'For every problem, a technician needs to come in. So we always have to call. What if something happens in theweekend? And it's from pillar to the post'. (Voorhout participant, Male)

Competences develop over time and often through experience and exposure. Intermediaries then play an important role in learning participants how to use the technologies, but also in assisting, when technical issuesarise. However, in Voorhout several participants also expressed that they had learned a lot of new things since moving into their home. Some participants felt that they had to learn how to use and fix the technologies themselves, due to the communication problems, and they had therefore undergone a learning curve while living in the home:

'It is a continuous learning process [to use the systems in the right way]'. (Voorhout participant, Male)

'I think I've learned a lot since moving here - But not everything yet'. (Voorhout participant, couple)

'At first I felt what am I supposed to do with all this technology? I don't understand the first thing of it. But nomatter, you ask what you need to know and slowly you make it your own […] I think I've learned a lot since moving here- But not everything yet'. (Voorhout participant, Female)

Other participants did however express, that they had not learned much, and were expressing more concernabout the

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health and security of living in the home. These frustrations also related to the expectations of living inthe home, which was perceived as convenient and comfortable. Several participants expressed an interest inenergy technologies, and the benefit from producing their own energy or storing it in the battery. As thesetechnologies did not work, they expressed a general frustration.

3.6 Conclusions

The findings from the household interactions show that dishwashers, washing machines and dryers are generallyperceived, by the participants, as easier and more meaningful to time-shift (compared to other domestic appliances). However, the findings also show that DR interventions have to consider the routinised daily rhythms involved in doing laundry and dishwashing, for instance is dishwashing often performed after meals and laundry is done when 'the basket is full' or there is an urgent need for specific clothes.

Institutional rhythms, such as going to work or school, very much shape when and how energy is used at home, as the findings reveal. The organisation of energy intensive practices performed within the home often reflect broader rhythms, and householders therefore might find it difficult to time-shift energy loads at certain times, because it conflicts with institutional rhythms.

The participants perceived energy technologies as useful tools, which could help them to time-shift their energy consumption. The participants specifically expressed an interested in technical features such as these of remote control and monitoring of domestic appliances. They did not, however, prefer a full automation of appliances, and wanted to remain in control. The findings showed that financial savings were not perceived as an exclusivemotivational factor for doing time-shifting. In was instead considered (and negotiated) alongside comfort, care for the environment and convenience.

Comfort was, by the participants, not only understood in relation to thermal environments. Notions of comfort were related to ideas of homemaking and the activities performed within. Several participants felt comfortable within their home, despite experiencing uncomfortable thermal environments at certain times. Notions of cleanliness also differed among the participants, but was an important factor in shaping how and when energy was used. Emphasis on practices of showering and cleaning the home were especially present in Berchidda. The participants perceived sense of control of homely environments as important, and highlighted that technological solutions could challenge that.

Energy feedback was perceived as useful by the participants, but they did not have similar preferences for how it should be presented, with some preferring visualizations and others preferring numbers. Most participants preferred to receive energy feedback on an app. The participants especially highlighted the possibilities of monitoring and control from outside of the home as a positive aspect of this solution.

Ownership of other energy technologies, for instance PVS, generally increased the participants' interest and engagement with their energy consumption (and production). These findings also showed that early guidanceand continuous support is key in securing a basis for learning how to use and engage with energy technologies such as the Hestia platform.

Finally, the findings showed that everyday housekeeping is gendered and characterised by divisions of labour. In Berchidda, women performed the majority of the everyday ordinary housekeeping chores, and similar (although to a smaller degree) trends were visible in Voorhout. Male participants performed the majority of the digital housekeeping and expressed the biggest interest in new energy technologies.

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4. PARTICIPATORY WORKSHOPS' FINDINGS

4.1 Overview

This Chapter reports on the organisation, the engagement and findings of the three participatory workshops thattook place during September and October 2021 in the three pilot sites of Hestia.

The chapter begins by reporting on the settings, organisation and general participation information of each pilot.It then presents some key lessons learned in terms of the participants' engagement before and during the workshop and outlines what went well and what was challenging. This is followed by the reporting and analysis of the findings, which are presented under three main categories:

1. Time-shifting (of appliances and household routines)
2. Household and community dynamics
3. Living with technologies

Finally, it concludes with some closing remarks about the contribution of the workshops to the project and with suggestions for follow up interactions that will take place in the next months.

4.2 Workshop context

The workshops were held in the three pilot sites during September and October 2021. The following paragraphs describes the settings for each workshop, as well as practical information on their organisation and recruitment of participants.

4.2.1 Berchidda workshop



Fig.1 Workshop set up in the Wine Museum, Berchidda, Sardinia, Italy

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Workshop organisation

Workshop day of the week and date	Thursday 16th of September 2021
Workshop time	18:30
Workshop duration	2,5 hours
Venue	Berchidda Wine Museum
Incentives/reciprocity offers for participating (if any, e.g dinner)	Dinner
Hestia partner in charge	Gridability
Other Hestia partners present and their roles	AAU: support with workshop set up and materials Sinloc: observer CDLT: support with workshop set up and photographic documentation
Comments/reflections about the overall organisation of the workshop: The workshop organisation was overall very smooth. The mayor pointed us to the Wine Museum director, who provided information and pictures about the space and available equipment ahead of the workshop; she also made available her staff for a preparatory visit the morning of the workshop. The formalisation of the content and structure of the workshop was a collaborative process between GridAbility as pilot leader and AAU as WP2 leader	

Participants and recruitment

Number of participants invited	30 households
Number of participants that turned up	24 participants
Recruitment of participants (how they were invited to the workshop e.g personal invitation, letter, email etc.)	Interviews participants were pre-alerted in July about the date; a round of invitations was done early September, followed by a reminder the day before the workshop

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Gender distribution:	Female: 5	Male :19
Number of households present: 21 households		
Comments/ reflections on participants' recruitment and turn out (e.g representation of age, gender, overall turnout of the specific community, practical issues that affected the running of the workshop and the participants' engagement etc.): We had high participation from the engaged households: most of those who could not attend were not available at that time/day. We had a very good diversity of age groups, ranging from participants in their early 30's to participants over 70. In terms of gender, women were 20% of the participants, half coming with their partner and a half alone. Women's participation in Berchidda will deserve attention during the following phases of the project, since from interviews and workshop it emerged that housekeeping roles are quite gendered. We need to ensure that the project interfaces and devices are not only utilised in the households to "supervise" energy, but also that they become an integral part of energy habits, tools for adapting proactively everyday consumption. Although gender roles are out of the direct scope of the project, we will seek to utilise HESTIA technology components as a way to spark conversations and change around them.		

4.2.2 Voorhout workshop



Fig.2 Workshop set up in a restaurant in Voorhout, The Netherlands

Workshop organisation

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Workshop day of the week and date	Wednesday 22nd of September 2021
Workshop time	19.15
Workshop duration	2,5 hours
Venue	Cheers café & restaurant, Voorhout
Incentives/reciprocity offers for participating (if any, e.g dinner)	Drinks during and after the workshop
Hestia partner in charge	DW
Other Hestia partners present and their roles	i.Leco and 4YEF (observation) AAU (support with workshop set up, time-keeping and materials)
Comments/reflections about the overall organisation of the workshop: It was good to coordinate with the local partners (i.LECO, 4YEF) and stakeholders (Van der Hulst) to make sure that what we were doing wasn't going to make life complicated for anyone involved. In this case, it meant that we had to compromise on what i.Leco had ideally wanted to get out of the workshop, and postpone it to a later intervention.	

Participants and recruitment

Number of participants invited	32 households	
Number of participants that turned up	25	
Recruitment of participants (how they were invited to the workshop e.g personal invitation, letter, email etc.)	Personal e-mail; letter; postcard reminder	
Gender distribution:	Female:10	Male: 15
Number of households present: 15		

Comments/ reflections on participants' recruitment and turn out (e.g representation of age, gender, overall turnout of the specific community, practical issues that affected the running of the workshop and the participants' engagement etc.)

A lot of couples attended, which helped with the gender distribution. Of those who came alone, only one was a woman.

Our major recruitment challenge pertains to the age group: we recruited far more older people than younger ones. We had 4 households representing the younger population (and they were all represented by a single person, rather than a couple). Undoubtedly, there is a time factor here: if you are retired and you don't have to stay home for the kids, you

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are more available for a workshop.

However, there are some additional dynamics as well. As a reminder, the neighbourhood is divided into twomore or less equal groups – the senior citizen homes and the family homes. Each group consists of two streets. On the whole, residents are more closely connected with neighbours on their own street. However, the street-based bonds are stronger on the 'senior' side of the neighbourhood than on the 'family' side.
Moreover, the uncertainties surrounding the smart home energy appliances have been the basis for community building on the senior side, especially for one of the streets. On the family side of things, the connections are less strong and frequent and the energy infrastructure has played a smaller role in streetcommunication.

In other words, the topic of the energy community is much more 'alive' for the senior citizens than for the families.

3.2.3 Camille Claudel workshop



Fig.3 Workshop set up in a community centre in Camille Claudel, France

Workshop organisation

Workshop day of the week and date	Saturday 16thOctober 2021
Workshop time	10:00
Workshop duration	2,5 hours
Venue	City of Palaiseau "Salle de la Rigole" municipal hall
Incentives/reciprocity offers for participating (if any, e.g dinner)	Buffet for lunch at the end of the workshop
Hestia partner in charge	EDF & CPS

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Other Hestia partners present and their roles	User studio (Design company) in charge of workshop animation for EDF City of Palaiseau (overall facilitation and support) AAU (observation, workshop facilitation support and coordination)
Comments/reflections about the overall organisation of the workshop: In spite of some challenges to recruit, the workshop went very well, in a good atmosphere with very involved participants.	

Participants and recruitment

Number of participants invited	20-30 directly contacted during the recruitment process (described below) 13 people confirmed their intention to participate	
Number of participants that turned up	11	
Recruitment of participants (how they were invited to the workshop e.g personal invitation, letter, email etc.)	Several recruitment channels: Communication in the monthly journal of the Municipality of Palaiseau + CPS's journal CPS website posters in the district's buildings, shops and bus stops Direct face-to-face contact on 4th September during the "Associations Day" Face-to-face recruitment (in front of school, market), by the social landlords and direct contact with "building council" Phone contact with respondents from previous surveys within CPS	
Gender distribution:	Female: 6	Male : 5
Number of households present: 4		

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Comments/ reflections on participants' recruitment and turn out (e.g representation of age, gender, overallturnout of the specific community, practical issues that affected the running of the workshop and the participants' engagement etc.):

The workshop involved 11 participants from 20 years old to around 50 years old. There were 3 (highereducation) students (physics studies) among the participants.

Several people who were available for the initial workshop date (on the 28th September), which was postponed due to low recruitment, were unfortunately not available at the second date in which the workshop took place (on the 16th October). Many solicited people were not available on weekends either.However, in the end, we had a good turnout of motivated people with a good gender balance and representation of families with children: some parents came with their children, a table was dedicated tothem and Anthony (CPS) took care of them.

The people present were motivated by ecology. They are also involved in the life of the neighbourhood (associations, co-ownerships etc.). Some of them have been involved previously in the Hestia project throughthe survey questionnaire and some of them were interested in becoming ambassadors for the project and talk about it with their neighbours or acquaintances.

4.3 Workshop engagement process: lessons learned

The workshops, overall, contributed to a better understanding of the context of the three pilots and allowed the researchers to get a first impression of the area, the participants, the buildings and the community as a whole. Because of the situation with COVID-19, the first occasion in which AAU (WP2 leaders) were able to visit the pilots in person and interact with householders and other community partners. The following section outlines some of the empirical experiences achieved from organising and running the workshop, as a team, highlightingthe positive and challenging aspects of participatory events. They can be perceived as an empirical guide for future interactions, within Hestia, or future projects.

4.3.1 Community matters

One of the most important contribution that the workshops made to the Hestia project is the opportunity to not only meet the participants as isolated individuals or households, but together as a community, in the local context. Furthermore, all three pilot workshops were supported by local partners (such as municipalities, developers and local industrial partners) who put a big effort in organising the workshops, sometimes as part of wider community events. In Berchidda for example, the mayor, who played a central role in the organisation and coordination of the workshop, organised a site-seeing event for the Hestia partners, a welcoming-pre- workshop- lunch as well as an after-workshop dinner for the participants. This embracing of the workshop as a part of the local initiatives (extending further than Hestia), gave local citizens a motivation to be part of it, as wellas uniting the citizens toward a common goal (building an energy community).

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Fig.4 The mayor (on the left photo) introducing the Hestia team to the local venue that hosts the Time in Jazz festival that takes place in Berchidda every summer

Similarly, in the other two pilots, workshops were supported and enhanced by local partners, such as the developers of the community homes in Voorhout and the municipality in Camille Claudel.



Fig.5 Welcoming the participants and socialising with them in Voorhout

In Voorhout, the presence of the local developer Van der Hulst and local technical partner 4YEF made a positive contribution to the discussions and activities initiated by organising partner DuneWorks. Despite the fact that they were only observing, participants felt, during the workshop. that their concerns were heard and considered by them, especially in the context of recent technical problems with their home technologies (PV batteries and energy apps). Furthermore, the majority of the participants commented positive on the organisation of the workshop and its effectiveness in (re)connecting the community, as the quote below illustrates:

'I'm glad you [DuneWorks] stepped in, I think you were a great wake-up call, with your interviews, and also that at a certain point everyone is out for themselves and we don't unite, and with you… you have the feeling again that we are [a neighbourhood] and that's nice. That you are not alone'. (Participant in Voorhout, commenting onthe value of the workshop to the community)

Similarly, in Camille Claudel the participants that turned up were also active members of the community, involved in the collective life of their neighbourhood, such as residents/owners associations. Some of them expressed thewish to become ambassadors for the Hestia project and assist in the dissemination of the project actions and objectives to their social networks. The municipality of the city of Palaiseau has also been very supportive in the organisation of the workshop and its overall inclusion in their local sustainability schemes, which involve regular citizen engagement activities and learning sessions. However, participants also highlighted the **disconnection between the material infrastructure of their community**-which is already labelled as an eco-district, **which the socio-cultural aspects being a sustainable community.** In specific,

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they suggested that despite its label, the Camille Claudel district does not emphasise neither the ecology nor the commitment of the citizens or that of their local representatives. So, a more collective vision is required and more opportunities for connecting the individual households. Participants suggested that Hestia could be the connecting link between the technical infrastructure and community action.

Overall, the support and involvement of local community stakeholders is really important for the successful embedment of a project. It is important to mention that at least two of the three pilots are in their majority represented by affluent, middle class citizens, who also own their home. There were other groups, such as tenants living in housing associations in Camille Claudel for example, who have not been easy to reach for the workshop (or that have chosen not to participate), therefore their perspective was not considered in the conversations. It is important to have a good understanding of what groups a community consists of and try to make ways to represent their experiences during co-creation. Therefore, the limitation in our cases, is that despite the great interest and participation in the workshop, there are still specific issues, such as the socio-economic capabilities of different citizen groups and their life commitments, which were not fully integrated in the outcome. A way to respond to this would be to target the follow up interactions of Hestia to households that were not part of the household interactions or workshops.

4.3.2 Timing and venue of workshop is key for engagement

The time and day of the week of organising a workshop is crucial to its success. In the first two pilots (Berchidda and Voorhout), the workshops were scheduled to take place during an evening of a weekday. This was consideredas a good way to attract people after work, while not disrupting their free leisure or family time. While the attendance was good in both these workshops, it was obvious that family households with children were nearly absent, or represented by only one member of the household. This was due to the difficulty of coordinating dailycommitments (such as school and work) with household routines (cooking, dining, putting children to bed).

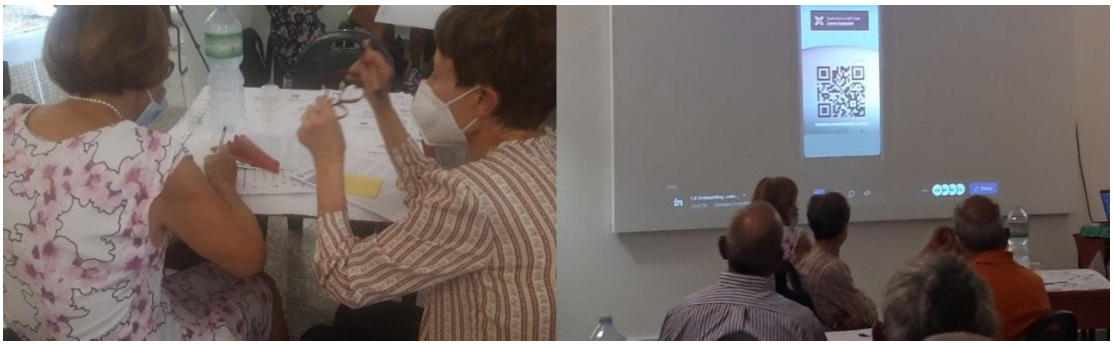


Fig.6 More senior citizens attended the weekday evening workshops (Berchidda)

The third workshop, in Camille Claudel, took this consideration on board (after a poor indication for attendance for a weekday workshop) and was organised on a Saturday morning. This allowed for families in the area toattend. Families were encouraged to bring their children, some of which did, and the organising partners (EDF and CPS) made provisions and entertainment for this children that turned up.

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Fig.7 More families were engaged in the Camille Claudel workshop, which was held on a Saturday morning

In all three pilots, the venue chosen was a centrally located community place or landmark (the Wine museum in Berhidda, a restaurant in Voorhout and a community centre in Camille Claudel), which made it convenient, familiar and pleasant for participants to be there. Furthermore, the inclusion of a social activity, such as lunch, dinner or drinks helped make the workshops a social get-together for each community, who all appreciated this-especiallyduring challenging COVID-19 times-and commended the organising team for it.

Overall, it is important to consider the requirements of the local community and make efforts to accommodate issues such as times of interactions, location and availability in order to include as many different members of the community as possible.

4.3.3 Language of facilitation

All three workshops were facilitated in the local pilot language (Italian, Dutch and French). This was essential, in order to respond the local community and in order to capture culturally-specific insights on people's everyday experiences with energy at home.

However, the initial planning and some of the thematic content of the workshops were done in English, andcoordinated by AAU, as a non-local Hestia partner. This involved translations and appropriate interpretations of the concepts for each local community. Furthermore, it meant that AAU, as coordinators of the task, could not facilitate the workshops but instead had to rely on the local pilot Hestia partners(and sometime rely also on other local non-Hestia partners) in order to run these. Researchers from AAU attended all three workshops and contributed to the practical running, however, it was very challenging to follow the flow and get directly involved with participants during the event because of the lack of language skills.

The language issue presents a limitation for the analysis of data collected during the workshops. Efforts were made to de-brief all three events with the local partners straight after the event, as well as have ongoing discussions about the flow of events, about participants' perceptions as well as cultural interpretations of things said and performed.

Therefore, lack of knowledge of the local language by researchers, can have implications for the analysis and socio-cultural interpretation of data and needs to be considered carefully, both in the conception of activities andwell as their translation and interpretation. One way that we have tried to mitigate this during the past months, was through ongoing communication

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with the local pilot partners, in order to get as much understanding and explanation of the local context and meaning as possible.

4.3.4 Small groups work better for inclusion

Despite the fact that all workshops remained relatively small in size (the largest cohort being 25 participants), it was quite effective to split participants into smaller groups during the different activities.

The pilot workshops used this technique in order to introduce variety in the agenda (each smaller group doing a different activity) or to purposefully divide members of the same household in order to evaluate their perspectives. Furthermore, smaller groups allowed for a better distribution of genders and age groups.

However, not all pilots used purposefully divided groups. There were also occasions, such as in Berchidda, where participants choose their table, and in most cases this involved members of their household or neighbours that they were friendly with. This version enabled participants to feel more familiar and comfortable with discussions and therefore contribute more personal information.

In all cases, small groups allowed for the inclusion of more 'voices' to the collective, some of which might have been more challenging to include in the larger configuration.

4.3.5 Toolkits and serious design games make things easier to engage with

The pilot workshops used different toolkits and probes in order to communicate the ideas behind the objectives, and to generate collective action amongst the participants.

For example, in Voorhout, the participants were split into different groups and were presented with two different activities. One group was given energy application prototypes and the other was presented with a 'serious game', developed by DuneWorks, about the building of an energy community. The energy community values game seemed like it was a good tool for community-building – allowing people to talk about who they were to each other and what role they wanted or felt ready to adopt in their community.









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Fig.8 Participants in Voorhout playing the energy community game

There were seven archetypal energy communities that participants could achieve, each of which representeddifferent priorities, such as financial, environment or community/social. Participants had to move forward as a group through different steps, which meant that a lot of debate and negotiations needed to take place in make decisions. This was a great exercise to initially understand what community means for each participant andhousehold, as well as open conversations about priorities and possible futures in their local context. The findingsof this activity is further analysed in section 4.4.

On the other hand, in Camille Claudel, participants were presented with two different probes and toolkits. The first was the 'Green mirror' game (see developed by Zoe Bonnardot (a PhD student at EDF), which engaged participants in the understanding and exploring of the way they use their household appliances and their everyday energy-related practices at home. The concept of the game was to allow people to share their experiences in the group, to encourage discussions about their daily energy consumption and to try to makeparticipants 'envision' themselves in a future energy scenario. Participants responded positively, and this first game was considered as a 'warming up' for the activities that followed. The downside of the game was that therewas not sufficient time for all players to express their ideas and to contribute to the group. Some participants were a little frustrated by this, however, the game showed a powerful potential for figuring energy consumption by reflecting on a, actual day of the participants' lives (the day before the workshop) and for thinking quickly aboutessential questions, such as the value of energy, the relationship between production and consumption of energy and the effect of energy storage, amongst others. Participants were engaged in lively conversations and advised each other about different issues such as the selling of stored energy. Furthermore, participants added to the game's capacity by suggesting the addition of smart phone technologies. For instance, one participant suggested that the game designers should have added more 'smartphone cards' to show all the moments when the participants charge their smartphone during the day.



Fig.9 Participants playing the 'Green Mirror' game in Camille Claudel

The second set of design probes in Camille Claudel, was a series of action cards, each of which represented the following suggestions/ scenarios:

"Lowering your heating setpoint by 1°C", "Heating outside of peak consumption (6-8am and 6-9 pm)", "Doing your washing when it's sunny" and "Preparing meals when it's sunny", as seen in the image below.

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Fig.10 Cards presenting different options for householders' everyday energy consumption (ownership of User Studio)

During this activity the participants were asked to fill in a card associated with a possible energy efficiency or flexibility action, and explain why it would be easy or not easy to perform (i.e fill in the reasons why it would be easy or not what would be the conditions, and detail their family profile (for example if they are solo, couple, family with children, retired (solo or couple)). The objective of the game was to develop insights into the flexibilityof everyday energy-related household routines and understand the feasibility of the proposals, to understand the reason behind these insights, to identify levers and to develop reflections for the next steps.

The game, which was facilitated by User Studio, was fun and well organised and highlighted important issues to do with household dynamics (how coordination of practices happens amongst household members), issues to do with the gendering of household labour, as well as the identification of recent changes in household routines brought about by COVID-19 restrictions (such as working from home). Full analysis of findings from this activity is presented in section 4.4.

Overall, the adoption of toolkits and scenario-oriented design games enabled participants to engage in activitiesin a fun and playful way and share some of their experiences, concerns and success stories with others in an enjoyable way. Furthermore, it provided a good way to explore issues of flexibility of everyday, energy-related practices at home, while generating some common visions for the near future of energy communities.

4.3.6 Practical planning and follow up

The planning and coordination of the workshops were challenging tasks, which involved a series of partners,some of which did not belong to the Hestia consortium (such as the Berchidda municipality and the Voorhout developer). Furthermore, AAU as coordinators of task 2.2 and engagement leaders, were responsible for coordinating and organising the workshops but not facilitating them. This was due to multiple reasons, such as accessibility to the local pilots and local language skills. Therefore, there has been a period of ongoing communication and negotiation with the pilot partners in order to identify the appropriate local objectives and practicalities as well as organise how the facilitation would take place. This process involved virtual workshops amongst AAU and the pilot partners, in order to brainstorm and develop the content of each workshop, as wellas its practical running.

While the workshops were organised and performed in a very smooth and efficient way in the end, it is importantto put an emphasis on the time and effort required for the practical preparation of activities, the coordination between participating partners as well as last minute hurdles that came up. For example, in the case of Camille Claudel, the workshop had to be postponed a few days before it was originally planned, since recruitment was not so successful. This generated a few

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challenges related with travelling,, notifying participants and re- advertising the event. Good communication between partners and participants as well as between partners is crucial for the successful implementation of such events.

Furthermore, it is important to consider in what ways the engagement and interest of participants can be maintained for a longer term. This continuous engagement along with the considerations of how to move beyond the end of a specific project is termed the infrastructuring of participatory design (Bjögvinsson et al., 2012; Karasti, 2014).

Therefore, important questions are how we deal with participation after the planned project interactions, in our case, how we continue to engage with our communities , for the rest of the project, but also in what ways we recruit these communities to the ongoing engagement with their energy communities. In order to do so, we need to have a good understanding of the existing organisation and context of each community, to understand the technical and socio-cultural capabilities of each site, as well as recognise that not all communities have the same starting point or willingness to contribute to the project.

4.4 Workshop findings

4.4.1 Time-shifting

Time-shifting has been explored in the participatory workshops in two different ways:

1. In relation to the time-shifting of the use of household appliances
2. In regard to the potential for flexibility or time-shifting of households everyday routines (i.e repetitive everyday practices).

The following discuss some of the relevant findings and provide an analysis of the interactions.

Time-shifting of appliances

Time-shifting of appliances was discussed more extensively in two out of the three workshops (Berchidda and Camille Claudel), in order to start the conversation amongst the householders about their everyday consumption.

In Berchidda, participants were asked to place their household daily routines on a 24h schedule in relation to when they use 3-4 recurring appliances. Then, they were asked to draw their group's daily energy load, by lookingat the information they had drawn on their sheets and with the assistance of a sheet with the KWh consumption of most common household appliances.

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Fig.11 Examples of appliance time-mapping exercise in Berchidda

The group discussions revealed that the majority of people are using a lot of energy-intensive appliances and perform energy-intensive practices during the morning. This has mainly two reasons:

1. The first relates to the regular daily routines undertaken by stay-at-home householders (allof them women in this case), who tend to run appliances such as washing machines andvacuum cleaners in the morning as routine housekeeping. Most people in the present households indicated a preference(and availability) to do these chores in the morning.

2. The second reason people perform intensive energy routines in the morning (such as washing clothes, dishwashing and charging) is to follow their PV's energy availability (while at the same time monitoring their consumption). However, all households in this groupsuggested that they are able to do so since at least one member of the household is at homein the morning on a regular basis. They also suggested that they have gradually learned howmuch energy every practice requires (such as washing clothes), which allowed them tochoose the time when they can perform it and therefore make a saving in cost but also use available renewable energy. These households also stressed the importance of having abattery for storing the energy from the PVs in order to then have better operation and choice of how to use the energy they have produced.

In Berchidda there were **more distinct gender roles in the performance of everyday housekeeping.** Women, whooften are stay-at-home spouses or retired parents in intergenerational households, took the majority of the loadin planning, organising and performing the housekeeping tasks. The men in the group were less aware of the daily routines, especially related to the use of appliances, however, they were more involved in the monitoring ofenergy consumption (through meters) and energy bills, as well as in the installation and maintenance of energy technologies such as PVs and batteries. **This discrepancy between the performance of the household tasks andthe monitoring of the consumed energy can create misalignment in the smooth coordination between the digital**

housekeeping of energy systems and the ordinary everyday housekeeping. When new technologies and interfaces enter the home, it is important that they are adopted and 'domesticated' by all members of the household, especially those who perform the regular daily housekeeping. In the case of Berchidda, where the majority of the housekeeping is performed by women and a large percentage of them are senior citizens, it is important to develop an inclusive process of introducing the energy interfaces (dashboards, meters, PV batteries)in order to make connections of how they related and intersect with the everyday practices.

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Most households in Berchidda expressed the perception that energy-rated smart appliances, such as programmable washing machines and dishwashers are essential in this transition to energy flexibility for their community. A few participants also suggested that not only their appliances need to be efficient but that they need to run them in off-peak hours. This was a possibility in the majority of cases in Berchidda, since there is availability of a household member to do so on a daily basis. Therefore, the time-shifting of appliances depends on a well-coordinated household routine, in addition to availability to perform certain practices out of peak times.

Furthermore, an interesting observation is the perception that women held in the group, of being more energy intensive themselves as the main users of energy appliances, as the quote below suggests:

> 'Maybe it's because I am a woman, but my everyday consumption is much higher than the one of my table companions…'(implicit: because she is the main user of housekeeping appliances)

While **men had a more accurate idea of the actual energy consumption based on meter readings and energy bills, they had a vague idea of what this consumption exactly involves, in terms of the household practices**. On the other hand, **most women had a very accurate perception of the exact household practices of their everyday life, but a less accurate or misleading picture of how energy intensive that is**. And while most recognised the more energy-intensive appliances, such as tumble dryers, and tried to use them less, they also lacked a thorough understanding of how practices could be spread in order to help with the household's flexibility. Their **efforts to assist flexibility regarded the coordination of co-related practices** (for example combining cooking with dining and dishwashing) **rather than with decisions based on the energy intensity of appliances and the best time to use them**. This is a good insight into households' capacity for flexibility and time-shifting which suggests that **efforts to spread appliances used throughout the day should consider both the practices that are co-related to** them, as well as the person(s) who coordinates their use.

In Camille Claudel, participants, through a design game (as described in section 4.3.3), allocated their use of household appliances in a 24hr period, using the day before the workshop as reference. The game aimed to bring awareness of participants to their real energy-related practices and to make them reflect on their use and coordination with other activities.



Fig.12 Participants during the Green Mirror game in Camille Claudel (ownership of Zoe Bonnardot)

Participants were very engaged in the process, and it seemed that it helped them visualise their overall energy load, as the quote below suggests and therefore helped them to identify the more energy intense practices:

> 'You buy, you have the money in your hand" (talking about the game […] 'it's concrete, you 'see' your electricity'
> (Participant in Camille Claudel)

Through discussion and exchanges with others during the game, they started to get impressions of how a future energy

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community could work for them and what it could involve. Participants were motivated by the concept of selling their energy and used it as a way to think about their individual consumption and potential time shifts. The**visualisation of energy**, its association with specific practices (rather than just appliances) and its relevance to the community energy are all-important insights from this pilot. These findings illustrate that **energy needs to be considered as more than a number** (meter reading) **or financial concern** (energy bill) **but also as a community issue and as a collectively produced and consumed practice of households.**

Finally, through a design game with cards, facilitated by User Studio, it became obvious that the time-shifting of household appliances, such as washing machines, is often challenging for working professionals and families, who have higher demands and multiple expectations of their members. However, with the recent introduction of remote working, due to COVID-19 restrictions, many participants are now comfortable with the idea that byworking from home, they can run energy intensive appliances, such as washing machines and dryers throughout the day and spread across the week. Therefore, **the newly developed conditions, due to COVID-19, have in manycases enabled energy flexibility and the time-shifting of appliances.** This is not however the case for the majorityof participants, across the pilots, and should not be taken as a generalisation of the wider population. However, it presents an **opportunity which could be exploited in order to introduce new ways of performing energy-demanding practices, for particular households that are able to respond to it.**

Overall, based on the evidence we had from the workshops, we can suggest that **the provision of pricing or other relevant information or notifications for the time-shifting of appliances could be accompanied with suggestions for the coordination of associated household practices**, **and with possible alternative, less energy intensive waysof performing a practice**. Furthermore, the design of energy technologies and their interfaces need to take into consideration of **who performs the energy-related practices** and the overall **context and composition of households**. Therefore, **interventions should be sensitive to different age groups, be more gender-considerate and respond to each household typology in the given socio-cultural context.** Finally, **local social norms and** conventions, including recent work-life amendments, such as working from home, could present opportunitiesfor time-shifting and for introducing flexibility and efficiency in energy-demanding households.

Time-shifting of everyday household routines

Berchida

Our participating households were of different composition (such as families with young children, retired couples, intergenerational families), different size and housing typology. However, despite their disparities in characteristics, there were a few common threads amongst them in terms of the organisation and temporality of everyday practices.

Households consume energy in order to maintain the comfort, cleanliness and safety of their homes and to care for others. Practices of care and cleanliness were particularly distinct in Berchidda, where participants suggested that keeping a clean, healthy home is a big priority for them. It is important to consider that the data were collected during the time of COVID-19, during which cleanliness and hygiene were made even more crucial to households that before. The coordination of routines of cleaning and washing were undertaken primarily by women, who in their majority stayed at home, (as a stay-at-home spouse or as a retiree). These routines were performed mostly in the morning and were associated with an essential daily house maintenance. These are routines which seem to keep households 'locked-in to existing patterns of normality'(Skjølsvold et al., 2017,
p. 3). Furthermore, it was implied that the performance of these everyday routines, define the role (as parents and carers) of these participants, as well as construct and maintain their identities within the home and the community.

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Households that had already installed PVs, suggested that they have amended the timing of their practices to respond to the available energy during the day. Examples given were the charging of devices, doing the laundry and dishwashing. However, these households were also relying on a member of their households always being at home in the mornings. Therefore, the shifting of energy related routines, such as doing the laundry, cleaning,

dishwashing and charging of devices, is interconnected with the availability of householders to be at home and commit to these tasks. However, as we have suggested in the previous section, the women (who in most cases performed these routines) were not always the digital housekeepers, which presents a disconnection between
the ordinary housekeeping and the digital one. While they undertook the daily physical tasks, they did not so
much engage with monitoring the energy that these require.

Furthermore, these practices were then invisible to the rest of the household members who are not regularly present at home when they take place and are dependent on one way of performance (that of the person in charge, usually the stay-at-home spouse). Therefore, there is an issue with the perception and understanding of how energy demand is generated at home, from the perspective of the whole household, which might create challenges in the uptake of more efficient practices, in which all members can engage. If the main housekeeper also drops one of these practices (for example miss to run the dishwasher in the morning), the consequence is that other daily routines might suffer (delay or not get done) or performed at peak times. Finally, these housekeeping practices and their coordination rely on the main housekeeper's tacit way of performing and

coordinating them, which could be problematic if not shared, as there could be a break-down if the main housekeeper is unable to maintain them. It is important to state that most stay-at-home spouses did not consider their housekeeping load as special and suggested that these were standard (i.e expected by local norms) practices of care for their household.

Finally, in the case of Berchidda, the control and management of the everyday housekeeping is an important part of the role of the participants (mostly women) who perform them. The removal or reduction of control, through

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automated services for example, could upset the rhythms and feeling of being in control for these participants. It is important to address in what ways people can maintain the sense of being in control and of that of having a choice of doing things in their homes.

Camille Claudel

The workshop explored the possibilities of time-shifting four different practices or suggestions for introducing flexibility:

- Lowering your heating setpoint by 1°C
- Heating outside of peak consumption (6-8h/18-21h)
- Doing your washing when it's sunny
- Preparing meals when it's sunny

The participants were given cards which were associated with a possible energy efficiency or flexibility action, and were asked to explain why it would be easy or not to perform these(e.g suggesting why, what would be the condition, and detail their family profile (solo, couple, family with children, retired (solo or couple). At the end of this exercise, the facilitators (User Studio) provided suggestions for alternative ways of performing some of these practices, as a household or as a community (e.g as a building block of flats). The following sections discuss someof the challenging issues that were brought about by participants, in terms of adopting the suggested practices, as well as some opportunities and positive points for consideration. Overall there was a mixed picture in terms of the different ways in which routines can be time-shifted; these seem to be depending on issues such as the social norms of the context in which participants belong to, the kind of household they live in as well as the way in which they coordinate and negotiate everyday practices with their household members.

Perceived challenges

On one hand, participants expressed several of the challenges they face **because of the daily practical needs of their household** (such as doing the laundry for a family and accessing common kitchen facilities for students). Such **practices were important to be prioritised in order to coordinate with all household members' routines andtime commitments**. For example, in family households participants suggested that there is 'too much laundry to do', which could not wait for the appropriate time. In the case of working professionals, people suggested that: "Active professional life that doesn't allow to come home early or plan the return time. Possible is programmablemachine". Furthermore, the challenges become more intense when practices require coordination and planning, such as washing and then hanging clothes out to dry. One participant elaborated on this by suggesting that:

'Laundry that stays in for 3 hours smells bad' (and then, a grandma's advice: 'you should run a machine withvinegar to remove the odours'.

Others supported this argument saying that (It is) 'Ok if at home, but not if the clothes stay for 6 hours in the machine'. Suggestions to overcome this coordination issues were the ownership of appropriate appliances, such as programmable washing machines.

There are therefore **time-related coordination issues, which can restrict the flexibility of everyday practices, especially in family and working professionals' households.** Efficiency or flexibility therefore suffers these consequences, especially as household negotiations need to take place when a change needs to happen. In most cases, people are busy in their everyday life, and they have created routines and schedules to keep things

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together. This is particularly the case with family households in which time management is challenging. Forexample, in the case of cooking 'with the sun', one participant suggested that 'For most meals, there is a need forcooking on the spot', while another highlighted the challenge of coordination by saying that '3 children + lots ofwork. It is difficult to combine sun and available time'. Therefore alterations in routines, need to involve allmembers of the household and also take into account the wider social commitments and demands of its members.

The practices that were identified as more challenging to time-shift are cooking & dining and doing the laundry. However, these limitations were relevant to the particular participant group, which consisted primarily out offamilies and working professionals, who have more demanding needs for coordination and less space for negotiation between members.

Perceived opportunities

On the other hand, positive aspects, or opportunities were presented by participants living in similar households (families with children and working professionals). They suggested that during the pandemic, and with the**introduction of remote working**, they have been able to flex their consumption through the day and therefore spread the daily household practices across a wider range of times, while sharing it with their partners. This was also the case of working professionals, one of which suggested that:

Participant 1:'On teleworking days, we shower a bit when we want. We can take a shower later in the morning but we don't take a shower in the evening'.

Participant 2: 'Don't you have children?'
Participant 1: 'No'

Comfort and convenience are usually prioritised over efficiency or when considering the time-shifting of everydaypractices. However, small changes or amendments in practices such as heating could be more easily adopted by households than more drastic or complete change of practices, since participants suggest that they do not alter their (thermal)comfort that much or interfere with other ongoing household practices. Participants for example, referring to the suggestion of lowering their thermostat by 1°C have suggested that:

'Easy with help of the thermostat and doesn't impact comfort that much', or that:

'We have a good thermal insulation, and can put on a pullover if needed'

While most participants are protective of their comfort at home, such as, for example, maintaining a steady temperature throughout the house or using certain settings in appliances, they do not seem to be burdened by small amendments, if they know that these contribute to a more efficient community. However, it is important to help participants build appropriate meanings for the reasons why they need to amend or they could amend their practices. This is particularly important in order to avoid issues such as the participant below reports:

'I'm less productive when I'm cold'

Therefore, meanings of comfort and well being at home need to co-develop with recommendations of energy systems and community strategies for efficiency.

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Furthermore, interconnected practices such as cooking and dining could begin to be considered as ways to reducehouseholds' energy-intensive practices during peak hours. While these two practices are found to be some of theleast flexible ones in households, participants were quite positive towards experimenting with new ways ofdealing with food. For example, participants in Camille Claudel were excited and willing to try new ways of cookingor preparing food, such as using a solar oven or eating raw food for some of their meals. However, there is still along way to go in terms of coordinating these new suggestions with existing practices (of shopping, planning themeals etc.) in order to start to normalise them into regular routines.

It is important to mention that often, during the workshops, particularly in Camille Claudel, when recommendations were given for the change of practices, such as preparing meals; there was a particular genderdifference in the responses from participants belonging to the same household. In specific, the women in the group, who in their majority were more aware of the coordination of household routines and the physical housekeeping, were more reserved in accepting a practice change, such as cooking or preparing meals outside oftheir usual hours. The men on other hand accepted or perceived these suggestions as easy to perform, either by relying on cognitive household labour of their partners (such as the planning and organisation of meals), or by notbeing fully aware of the limitations of multi-tasking required. This is exemplified in the quote below:

'If there is sun!' 'Madam has an appropriate planning' (Participant in Camille Claudel, referring to the possibilityof cooking when the sun is out, and suggesting that probably it would be possible, since his wife usually takes care of it)

Therefore existing gender roles in the household, and set ways of performing everyday practices, need to beconsidered when new ideas or suggestions are given as alternatives to what is currently performed. Householders need to have the opportunity to adopt these within the existing structures of their everyday lives, and also it is important to have the chance to negotiate amendments between members of the household before these are taken on board.

4.4.2 Household and community dynamics

Learning (with others and tacit knowledge)

Social interactions and common experiences are important in the process of learning, as ways for people to createmeaning and make sense of the world (Vygotsky, 1962). This is a perspective presented by constructivist theoriesof learning, which suggest that people construct their learning with others, according to their prior experiences and knowledge.

Learning is an important aspect of the conduct of participatory workshops, which takes place more easily when people feel comfortable with their environment and their co-participants. During the workshops, the atmospherewas generally friendly and cheerful across all pilots. Initially some people were sceptical about the effectiveness and general concept of participatory workshops (for example in Voorhout and in Berchidda, where some participants had never experienced a similar event before). However, by getting into the activities and through the collaborative (and overall friendly and cordial) interactions with other participants, people suggested that they felt confident that this process can help them and their community progress their (energy) goals.

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In Voorhout, for example, participants felt that the event was a good idea as way to bring the community togetheragain, especially after a long period of disconnection because of COVID. Furthermore, they suggested that this was a good opportunity to share their experiences and even their frustrations of their energy systems, which frequently presented technical problems. One participant expressed the hope that Hestia's planned interactions, such as the pilot workshops, present an opportunity for citizens in the local community to come together in orderto get organised collectively for action regarding their energy community.

The workshops also presented a great opportunity to share and visualise the tacit know-how of everyday practices, i.e. the different ways in which participants perform different tasks at home, which are sometimesdifficult to put in words and is not explicit or specific. For example, citizens in Berchidda have been learning to operate their PVs by doing, e.g by time-shifting practices when the sun shines. Their way of finding out if that worked was by monitoring their energy bills, however, that took time and also was not shared between all members of the household. Learning how to live with technologies, taming their technical operation but also theirsymbolic meaning and therefore domesticating them, in all three pilots is still in process. However, there aredifferent points of departure for each community, with some, such as Voorhout, having already done the transition to a smart home and experienced living with smart systems, to others, such as Berchidda, in which thecitizens are on the way of acquiring smart technologies and slowly integrating them in their everyday life. Therefore, making sense of the energy technologies, as wells the practices that are involved in them, involvecontinuous negotiations between household members, and also require continuous support from others who are familiar with these in order to help their long-term integration and sustainment.

Moreover, learning does not only involve the technical operation of technologies, but also learning how to adapt everyday practices in line with these technologies. For example, in Camille Claudel the workshop facilitatorsintroduced several ideas of how to change some everyday practices in order to become more energy efficient. These included the practices of cooking and dining, by introducing them to concepts that were 'outside the box'.These included for example how to eat raw foods that does not need cooking (or fermented food), using solar ovens, batch cooking and the use of a 'Norwegian oven' (Marmite Norvegienne).



Fig.13 Examples of cards showing alternative ways of preparing food in Camille Claudel (ownership of User Studio).

Participants were challenged and intrigued by some of the suggestions. Some of them were already using such practices,

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such as fermented foods or solar ovens. However, it is challenging to make changes to practices such

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as cooking and dining, because of the complexity of their coordination but also because of the cultural meaning that these hold. For example, participants from a household with children suggested that they like the idea of reducing their energy, however, most of these suggestions involved not cooking or cooking less. This was problematic for them, since they want to encourage their children to learn to cook. Therefore, the cultural significance of cooking is preventing some of these 'new' ways of dining to establish.

There are also gender implications in these suggestions. Women in the group of participants were more interested and involved in the conversations about the different cooking and dining suggestions. One of them suggested that:

'My husband won't switch to fermented food. I'm already trying to introduce him to other kinds of [fermented]foods and he doesn't want to try'.

This statement indicated once more the prevalence of women, as housekeepers or coordinators of everydayroutines, who are therefore the gatekeepers for introducing new ways of performing everyday practices. Furthermore, it indicate the need to bring on board all members of a household in order to introduce and stabilisea new practice. Additionally, in regard to the orchestration of everyday routines, another participant suggested that:

'Batch cooking, I was only thinking about it only for the organisation and I wasn't tempted. But for changing myenergy consumption, I'm interested'.

In this case, the (female) housekeeper suggested that the coordination and time efficiency of certain practices (such as batch cooking to save time), could also work for energy efficiency purposes. However, in such cases, thecoordination of tasks and negotiation between household members is essential in order to make them sustainable. Learning how to adapt existing practices to different purposes is an ongoing learning process, whichrequires the appropriate skills and meanings to develop in households in order for them to get implemented.

Another important suggestion that came through some of the participants in terms of this learning process, is that **there is a need for appropriate intermediaries, who can support their learning** in terms of understandingwhat is required for energy efficiency and in what ways they can contribute. These **can be either known or trustedpeople with similar experiences or professionals who can share tips and suggestions informally**. Finally, participants in Camille Claudel suggested that **they would like more obvious and visible reference to issues such as energy efficiency, flexibility and ways of performing everyday practices,** through the use of community boards and posters, or through residents' Facebook groups, in order to **raise awareness and encourage collective action.**

Ownership of energy systems energy communities and the role of households

Community energy is a concept which has been discussed in the context of renewable energy in the recent years and is usually associated with decentralised forms of energy production(Brummer, 2018).In the process of developing smart grids, communities around the world are being called to contribute to energy production (Koirala et al., 2016). Furthermore, the ownership of energy systems is considered as one of the most advanced levels of participation that citizens could undertake in the transition to energy efficiency (Gorroño-Albizu et al., 2019). However, community energy or community ownership models are still largely undefined (ibid), while the role of the household is also unclear(Koirala et al., 2016).

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Workshops in two out of the three pilots addressed the concept of energy communities, exploring issues such aslocal perception of what energy communities are, what is the role of individual households and what characteristics citizens would like to see in their local community. Through serious games and small groupdiscussions, these issues highlighted some interesting insights of the participating communities.

First, in Berchidda, the energy community concept was used as a 'warming up' exercise. Participants were asked to address the following questions:

1. What is an energy community?
2. What is the role of each individual household?

The participants were given post-it notes and were requested to individually write their answers to each question. The facilitators collated their answers to different boards used for discussion.

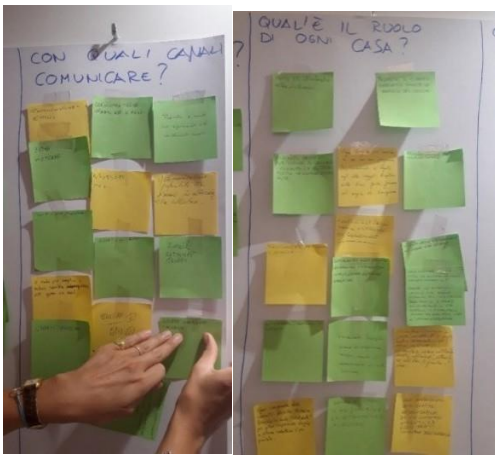


Fig.14 Collated suggestions from participants in Berchidda stating what they consider as an energy community and what isthe role of households

Some of the definitions that participants gave for the energy community were:

'A virtuous community which aims at energy efficiency through practices functional to consumption optimisation'(Participant, Berchidda)

And

'[…]a community that produces energy from renewables and exchanges/utilises it within the same community' (Participant, Berchidda)

A large number of the participants displayed a good understanding of the components of their energy systems at home, in particular their PVs and batteries and the different ways in which the energy distribution works. Theyalso, in their majority seem to appreciate and prioritise the importance of energy efficient household appliances, which they suggest should be used outside peak hours.

However, an observation, which was also pointed in section 4.4.1, is that **the digital housekeepers of homes,**which

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held that knowledge and competences in regard to what a smart energy system consists of, **were not the main housekeepers of the ordinary everyday routines**. There was a clear gender characterisation of the two sides (male participants overrepresented in the digital housekeeping and female participants overrepresented in the ordinary housekeeping), with **distinct gender roles in households**. **This discrepancy can be problematic in the collective organisation and participation in energy communities**. Decisions made in the public/ community, need to first be negotiated within the household, and therefore energy systems and their components need to bemanaged by the household as a unit rather than by isolated members. In addition, **it is important to have a diversity of perspectives woven into the development of new energy communities**, which represent different ages, genders and roles within the household and the community.

In regard to the role of individual households, participants in Berchidda suggested several points, **primarilyfocused around their responsibility of using energy-rated appliances and using them outside peak times**. A fewparticipants suggested that their contribution to reducing energy consumption happens through the running of appliances during the day, outside of peak times. **To support this flexibility, participants highlighted the importance of programmable appliances**, for those that are not at home outside peak times and the need fortraining or community education about the most efficient ways of performing different practices. Furthermore, they consider that it is important that **households act collectively for saving energy for their community, tryingto actively input energy in the local grid. Therefore ownership of the energy systems is not only a community**

responsibility, but it is also shaped by each individual household. A community-owned energy system, according to participants, also predetermines that each household acts fairly, and feels responsible for a balanced everyday consumption.

On the other hand, in Voorhout, the participants engaged with different kinds of energy communities through a serious game, which has been adapted by a previous project of the facilitating partner DW. The game, (discussedin section 4.3.4), titled 'The road to the energy community De Wals', used a near future scenario to put participants in the picture, in which their homes produce more energy than they consume. In this context, householders can store energy collectively, along with their neighbours, but also choose when they want to feed energy into the grid. The overarching question was what householders want to use the smart energy technology of their energy community for.

The participants had seven different ways to build their preferred energy community, each of which represented some selection or mix of three values: financial, environmental or social/community. Two themes emerged from the running of this game (over 4 different groups). First, the **issue of the energy technologies not working properly or being reliable and second, a collectivist approach to community**. Focusing on the second issue, it was clear that **most participants were keen to contribute to the 'common good'**. In this context, a limitation of this perception of doing 'common good' could be that there is more intense pressure to admit this, because of the presence of other community members and the need to appear positive towards the collective of the neighbourhood. While financial gains remain important considerations for participants, a common point of agreement was doing something good for the environment and the community, on top of or while also enjoying some financial gains. For example, one participant suggested that:

'If you choose financial return, you choose for your own future. I'd rather choose the future of my grandchildren'

However, doing something good for the environment needs to manifest itself in some way as a benefit to local community. One such suggestion was made by the participant quoted below, which suggests that if people were to join forces, they needed to contribute to clean energy for their community:

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Participant: 'The environment is too big. I would go prefer doing something for the community. Keep it closer toyourself. To have an overview.

Facilitator: 'To see more clearly what you're doing?'

Participant: 'Exactly. That's more stimulating. If you see that you're doing good for the neighbourhood, I thinkyou get more [encouragement]- I think [the environment] is too big, then you don't see it'.

Facilitator: 'And if you get feedback from the app, in terms of CO2 savings for example'?
Participant: 'No, nor how many trees I've grown. That doesn't mean anything to me'.
(Participant in Voorhout in conversation with Hestia researcher)
However, there was also scepticism about the contribution of each household, as quoted below:

'I think for me it's very important that you use your energy as efficiently as possible, and if I were to do that through a community, and I have all these lovely neighbours, but maybe some keep the windows open and theheating on high...Then I'm not in control...You know, find that very tricky. [on creating a 'business model' whereyou depend on each other to create value]' (Participant in Voorhout)

It is important to build trust and confidence within the community, in order to strengthen energy communities. The need for trust was highlighted by the participant below:

Participant: 'You have to be very much a unit, a community so to speak, to be able to say I agree, I will allowsomeone to drain my battery if he needs energy. You need solidarity as the basis'

Facilitator: 'Do you think you have that solidarity?' Participant:'Yes, I guess so. We'll have to try and see, carefully' (Participant in Voorhout in conversation with Hestia researcher)

Furthermore, with the introduction of new, energy-demanding technologies, such as electric cars, participants feel a little more uncertain or hesitant about how equal the contribution of energy will be from each household, as the quotes below suggest:

Participant 1: 'About those electric cars ... Let's suppose you're very frugal and someone else is not. Then you can get annoyed about that too. And that seems to be a problem to me, actually'

Participant 2: 'Yes but if my neighbour has a Tesla.....Then I'm filling up his car(...) but someone else says...I livefrugally and I don't do much and I keep my heating low. Yes, why should you have to pay for your neighbours?'

This could mean that the meanings of energy communities need to be collectively defined and regularly shared or updated according to the needs of each community. Furthermore, in order to do this strengthening of what it means to be an energy community and why, it is important to support people's technical and embodied know- how, in terms of dealing with the energy technologies and infrastructure.

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4.4.3 Living with technologies

Interacting with energy systems

How people live and interact with smart energy systems (whether these are already installed or if they are in the process of getting installed) are important to consider when designing DR solutions for households. The workshops helped uncover some aspects of people's experience so far with the interfaces of their energy system,or introduced them to some possibilities and track their reactions.

In Berchidda, where smart energy systems are not common, participants were introduced to the concept by a presentation of the Prosume interface as a prototype of an App which would help them control and manage theirhousehold energy. Participants showed an interest for services such as the digital wallet and also for the potential of combining the Hestia interface with existing Italian incentives, such as the selling of individual households' PVproduction surplus to GSE (Gestore Servizi Energetici). However, participants were not given the opportunity to have a more collaborative input into the development of this prototype but rather acted as reviewers of theservices that were presented. Overall, the app was received positively, however, since it was only presented on the screen, without the ability to test and play with its functions, the participants did not have the opportunity to experience the actual function themselves. Furthermore, the application, although designed for the Italian context, was a little removed from the current experience of everyday life of participants, who do not frequently use such an interface to operate their energy. Also, there was more interest and questions by men (often younger) in the workshop who mostly perform the digital housekeeping in their households. Therefore, there is still an ongoing learning process for both the technology developers and the community, in order to find ways of developing an application which responds and is sensitive to different age groups, genders and everyday lifepractices of the certain community.

In Voorhout on the other hand, where all the participants live in homes with smart energy systems, the workshopexplored initially the participants' existing experiences with the Swycs application they were using as an existinginterface. There was a prevailing suggestion that the existing App (Swycs) is not working properly and is not welldesigned, in terms of its graphic interface. A participant called it "a Mickey mouse App", suggesting that it isunclear and unattractive. In addition, other participants added that it was particularly difficult to find the data representing their consumption, suggesting that 'With this app you can't see the forest for the trees'. **The most well received functionalities of energy interfaces for participants in Voorhout, so far are PV production, source of** electricity for consumption and a 'stoplight' (green, yellow, red) to indicate favourableness for energy **consumption.**

However, some of the **challenging issues** in terms of interaction of participants with the energy interface included the **lack of communication or indication of how to get malfunctioning devices to function, a general uncertaintyand ongoing concern about devices not working, and a loss of faith in the stakeholders involved in the process**.Furthermore, the frequency and mode of communication between the app and the participants was challenging as a participant below suggests:

'I just find using apps difficult. You know, that's not what keeps me busy all day, chat messaging about this andstuff'.

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Overall, **there was a unanimous agreement of participants on the positive effect of insights received for their consumption**, both in household and at neighborhood level within the community. For example, a participant suggested that:

'Yes, we don't have any insight at the moment, but we can feel that it [the battery] is warm and that it switcheson. We both [partners] like the fact that you have insight into what your energy consumption is and also what you are generating. That insight is not there now. You want that trigger, to be efficient ["zuinig" = economical]'. People need to learn how to interact with their energy technologies and to be able to know if things are working as they should. At the moment, there seems to be some confusion about the functioning and efficiency of the system as the participant below suggests:

'How do we know if that system [in Voorhout] works? We basically have to assume that it works'.

Furthermore, it is important to contextualise the consumption insights, or recommendations sent by the system, within the immediate community, as suggested below:

'It is also nice to see the whole group, I don't need personal information of others, but the percentage of the group and that you see what the average is and see: 'I am below or above or ok'. And then also with informationabout [whether it's a] corner house or middle house etc. So that you can compare'.

While participants appreciate the comparison with their neighbours, they also do not want to compete with them.This is an important aspect to consider, especially if gamification is to be adopted as a way of household engagement in demand response initiatives.

There was no particular exploration or discussion about experiences with energy interfaces in the French pilot. However, within the different parts of the workshop, participants expressed their need to know more of how the local energy system works, as well as more detailed information about how their individual homes are heated (especially those living in apartment buildings). IN relation to this last point, participants expressed their views onissues of control (of energy systems and technologies), which are discussed below.

4.4.4. Control (of energy technologies)

Controlling and managing existing energy technologies were issues that concern participants to a great extent, particularly those who already live in smart homes (e.g in Voorhout) or those who have recently installed new energy technologies (Pvs for example), such as in Berchidda.

First, in Voorhout there was a decreased sense of involvement with energy management because participants suggested that their energy monitoring simply does not always work. Furthermore, issues of control influenced participants everyday comfort. For example, in Voorhout participants explained that the App they currently use, is able to control the Velux window, which would also open or close automatically according to the indoor temperature and levels of CO2. Participants suggested that you can also open the window manually and challenged the need for automation through the App. Some participants even complained about the windowopening at inappropriate times, such as in the middle of the night) responding to CO2 levels, which they did not think it was necessary to happen, rather than respond to temperature changes. Therefore, since the existing Appdoes not have sufficient automation customisation options, it does not meet householders' expectations. On the other hand, householders' expectations are not yet fully shaped, as they have limited exposure and experience in automated systems. Therefore, the meanings and expectations of householders' in relation to

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the need and ways
of control of energy systems need to co-evolve with the technologies (and their interfaces) involved.

Another important aspect of control came by participants in Camille Claudel and it regards the centrally controlledenergy systems in apartments or multi-dwelling buildings. For those living there, it is often impossible to controltheir heating. When, for example, participants were challenged about the possibility of reducing their temperature by 1°C, one participant suggested that:

'Seems possible as it's only 1°C. I'd do it if I had the possibility to control my heating'

Other participants also added to this by suggesting that "I don't control the thermostat in my flat" or that their heating *is* "Not controllable". This issue of non-individual control of energy systems, was also present in Berchidda, in some of the participants intergenerational homes. These dwellings often consisted of separateapartments within a building, which share their energy system, but that could not control the settings individuallyin each home. Some of the participants in those homes suggested that it is difficult to monitor their consumptionor to have customised indoor climate choices because of that. This lack of being in control, often results inparticipants' disengagement from a systematic and conscious understanding of consumption and could present a barrier in the adoption of proposed interventions.

Furthermore, the control of energy technologies is a gendered practice. Whether though the checking of smart meters or PV systems, it was obvious that more men than women undertook the task. In Berchidda for example, particularly some of the more senior women in the group, expressed their concern about the operation or management of technologies, as they feel they don't know much since their partners take care of it. In one case, one of the senior participants who turned up had lost her husband a few days before the workshop and came to the event in order to get some guidance and confidence about how to move forward in terms of their energysystem. While the younger generations are more digital literate than seniors, the gendered division of household labour still means that women take less responsibility in the monitoring and control of their energy than men. Furthermore, it was observed that there was an increased interest on smart technologiesfrom the men in theworkshops compared to the women. Additionaly, several of these technologies that enter the home are gendered(Kennedy et al., 2015), while most of the practices relating to this upkeeping of the digital (energy) systems (digital housekeeping) would be performed by men in most households. This can be problematic and contributes to the widening of the gap between the ordinary and the digital housekeeping of homes. It is an issue that needs to be addressed by both the designers/developers of technologies, as well as by the intermediaries that handle the installation and maintenance of the systems. For Hestia, this would mean that the interfaces develop need to address gender and age appropriately, considering the level of digital literacy and providing options for those householders that are not yet familiar withsmarttechnologies.

4.4.5. Privacy

The transition to smart energy systems requires the monitoring and collection of data from individual householdsand communities. Participants in the workshops expressed their reservations for the extent to which this collection interferes with their privacy and online safety.

In Camille Claudel for example, participants express their concern about the degree to which the sharing of their data satisfies an understanding of their consumption, while also respecting privacy and security. They suggested that they need to be fully aware of the process and the kind of data that are being collected from their homes and they would be satisfied if this information was available to them before they install a system.

Despite their reservations, participants agreed that it was important to have precise measurements (for things such as actual temperature, boiler consumption and hot water). They understand the need to also provide this accurate information to the

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technology interface, as a way to receive more accurate feedback.

4.5 Conclusions

This chapter presented the findings of the participatory workshops in line with an analysis and evaluation of these findings. It has initially shown that workshops are great engagement tools for uncovering collective insights into people's everyday life. However, it is important to design and facilitate them according to the communities' needsand context. It is equally important to follow them up with further interactions, such as interviews or small focus groups, in order to sustain the participants' engagement and in order to get more in-depth insights from individual householders.

Participants showed that they are, indeed, the experts of their experiences, confirming the point made by Sanders and Stappers (2008). Once recruited, communities were overall well engaged in the participatory process and were positive about continuing their involvement in the project. However, it was also observed that communities need customised approaches in order to allow for inclusive and socio-culturally specific demand respond solutions. It has become clear that before developing these solutions it is important to understand and map the current everyday practices and challenges that people face in terms of coordination and negotiation of their household routines. Furthermore, it has been highlighted that individuals need to be 'seen' as part of the community, rather than isolated units who makes rational decisions. Household decision-making and action are set within strong community norms and expectations.

The workshop findings revealed that the time-shifting and flexibility of everyday energy-related practices, rely on several societal and household-level rhythms, negotiations and dynamics, as well as existing infrastructures in the home and in the immediate community. It is important to address recommendations (for flexibility or energy / grid efficiency) with suugestions that address the coordination and association of everyday practices in each given context. Gender, age and digital literacy considerations should be 'designed in' the technologies that enter the homes, as well as the ability to customise and personalise these in each household or community. The materials involved in the performance of these everyday practices (including technologies and appliances) need to be available and familiar to householders before they are able to 'tame' or domesticate them in their everyday routines.

In addition, newly developed working and living conditions, deriving out of the COVID-19 pandemic, such as remote working/home working, are important considerations for the energy flexibility of households. Communities are currently in transition to this new way of being at home, while working and caring for others, and it will take time for these amended routines to be coordinated with new technologies, new understandings and newly developed know-how and skills of householders.

Householders are motivated to engage with new energy systems and there are various aspects that are attractive to them, such as financial benefits, collective ownership of energy systems, as well as 'doing good' for the environment and their local community. It has been found that most households do not want to compete with their neighbours; however a more general comparison with similar household types is beneficial and welcomed. It has been difficult to untangle or be specific about which priorities for adopting demand response solutions are most important for households, as these are not equally relevant to the different homes in each community. Several issues, such as the stage of life that householders are in, the house type that they live in as well as the expectations and social norms of their community are intersecting with their individual household motivations and intentions. Therefore, we need to be careful how we apply demand response solutions, avoiding 'blanket' or 'one-fits all' propositions across the pilots.

Moreover, we have established that living with energy technologies at home requires an ongoing learning and interaction between households in the community. Participants have different starting points, in relevance to their digital literacy and

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understandings, and therefore develop different 'speeds' and ways in which they respond and domesticate a technology. Gender and age should important considerations when designing and domesticating energy technologies. The workshops have also indicated that the designers/developers and distributors of technologies and interfaces need to provide ongoing assistance by familiar and reliable people and services. Finally, we have seen the importance of privacy and security of data for households. While accurate data measurements are important and useful, developers and distributors of technologies need to be clear about the data ownership and use.

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5. RECOMMENDATIONS AND IMPLICATIONS FOR PARTICIPATORY DR SOLUTIONS

5.1 Overview

This chapter presents some recommendation for the development of the Hestia platform, based on the findings of the household interactions and the participatory workshops that took place in the pilots.

It presents these recommendations in three sections:

4. As part of scenarios based on empirically identified household typologies
5. As recommendations for the design and the technical development of the platform
6. As recommendations for community engagement in relation to the development and maintenance of the platform

The chapter does not present new data, but it elaborates on the findings presented on the previous chapters, in order to generate more applicable and targeted propositions for action, based on the rich qualitative data we have collected so far. However, these recommendation should be read and applied together with the evaluations and analysis presented in the previous chapters in order to avoid inappropriate generalisations and inaccurate conclusions.

5.2 Recommendations based on household typologies

In the following section four generic household scenarios are presented, based on the composition of households and the stage of life of their inhabitants. The scenarios have been developed based on the qualitative findings of the household interactions and participatory workshops, presented in Chapters 3 and 4. The household scenarios are not generalisable to contexts outside the scope of Hestia. Instead, they represent a current picture of therather small sample of householders and of the ways in which they perform everyday practices. Within the target population of the three pilot sites in Hestia, the household scenarios should also not be regarded as representative. However, the household scenarios provide qualitative insights into 'typical' households typologies, found in the data, and present challenges and opportunities for enabling a flexibility of energy demand using a DR solution. The household scenarios aim to inform the development of quantifiable behaviour models e.g. the digital twin, which will be designed later in the project This framework should, however, be used with caution, considering the small sample size. Quantifiable models could therefore be inspired and developed on thebasis of these, but should not be a direct transfer or interpretation Quantifiable models must therefore be supplemented by further investigations with a generalisable setup.

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5.2.1 Families with children

Fig.15 Family with kids icon (adapted from Colourbox)

The first household scenario represents families with children (under 18 years of age). These household constellations were present in all of the three pilot sites.

Comfort and convenience

Among families with children, comfort and convenience is regarded as an important and often non-negotiable aspect of everyday life. They differ from other households, as comfort is related, most often, to the caring for others. In the findings, several families expressed how comfort was essential when caring for smaller children. This related to different spheres of energy consumption. Care and comfort were related to taking a shower, cooking a hot meal or maintaining a comfortable indoor environment. Comfort was also related to certain temporal rhythms of everyday life, and especially at peaks hours, comfort was regarded as important. Whenpreparing for work and school/kindergarten in the morning or when returning in the afternoon, the familiesunderlined comfort as important. This was ensured by e.g. having a bath in the early evening, cooking breakfast in the morning or keeping stable temperatures at these time periods.

Convenience was also regarded as important in the family with children scenario. Convenience was not directly related to the caring for others (as in the case of comfort), but instead perceived as a basic condition in a time- pressured everyday life. To the families with children, convenience was preferred in the coordination of activities. Convenience was often materialised, and washing machines and dishwashers were perceived as convenientmaterial objects, securing an easier coordination of everyday life.

In consideration of proposed DR solutions, it remains important to ensure and maintain notions of comfort and convenience for families with children. DR solutions need to consider caring needs, and how DR solutions might disrupt those. Here, coordination of everyday activities is central, and an opportunity is to exploit technological objects which can increase the convenience. Families with children may not perceive decreasing levels of comfort and convenience so positively, so recommendations should instead offer alternative ways of providing care and convenience to a similar degree. DR solutions, such as the Hestia platform, might be used, by families with children, to increase their comfort and convenience in everyday life. By making energy consumption and usevisible, new notions of comfort may develop, which could potentially risk the strengthening of inflexible practices.

Coordination of ordinary and digital housekeeping

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Due to many routinised activities, coordination between different rhythms is necessary for a successful DR solution. Families with children therefore need the necessary knowledge on how to perform alternative practices. New technologies also hold a possibility in ensuring the time-shifting of practices. Some of the participatinghouseholders, expressed that time-shifting of laundry and dishwashing was a possibility, but that they lacked the technological features to do so (e.g. programmable machines). An opportunity is therefore to focus on time- shifting practices, in which technologies already play a big part of the performance.

For families with children, coordination represents a temporal challenge. They regard coordination without using technological features inconvenient and something that risks taking time away for other activities. It is therefore important to secure an overlap between the ordinary housekeeping and the digital housekeeping. A challenge here is the gendered divisions of labour, which also exists inside these household, i.e. one partner takes care of the ordinary household chores and another of the digital. An opportunity for bridging ordinary and digital housekeeping, is that families with children often have a high digital literacy. This could be exploited, so that a broad engagement from many household members is ensure. A practical recommendation is to allow for remote monitoring and control using a app.

Another, relatively new, opportunity for enabling flexible practices is the newly introduced work-related rhythms. With the COVID-19 situation, many of the participating householders have had experience with working from home, and they saw this as an opportunity to ease the coordination between tasks e.g. making it possible to cook before the peak hours in the evening. It might be a good opportunity to utilise this availability of family members to be at home during non-peak times and present with alternative scenarios of how to coordinate and negotiate tasks outside their usual time-slots and within non-peak times.

Flexibility potential

Families with children have quite a low flexibility potential. Everyday life is structured along very 'sticky' practices, and ensuring coordination such context is regarded as a key priority. However, technologies may bring opportunities as they can ease some of the everyday doings (perceived as convenient ways of doing so), while shifting energy consumption in time. Doing so, it is important to ensure engagement from all household members, as ordinary and digital housekeeping is often split. However, high levels of digital literacy may providean opportunity and they may have a less steep learning curve when using DR solutions.

5.2.2 Individuals or couples living together without kids

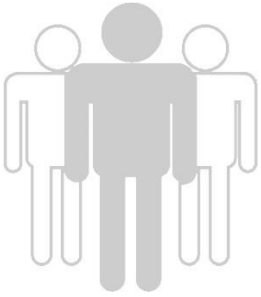


Fig.16 Individuals living together icon (adapted from Colourbox)

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A second household scenario represents a broad range of households, consisting of single person households orhouseholders living together without children (or adult children). This category does not include retirees, who areincluded in a latter scenario. Householders living together without kids and single-person households are often working professionals or university students. The household scenario was present in all of the pilot sites.

Comfort and convenience

Comfort and convenience are perceived as less important by this household scenario compared to families. This is, among others, due to easier negotiation between the few household members (if more than one). They might perceive comfort and convenience as important, but for other reasons than necessarily coordination between everyday activities. In this way, comfort can be regarded as an add-on. Notions of comfort are therefore more variable in this household scenario, and might be important to some while not so much to others.

Achieving comfort is also not necessarily connected to peak hours. As everyday life is more easily negotiated, practices involving high notions of comfort may be shifted in time. This offers an opportunity for modulatingenergy demand and decreasing the comfort standards at some times during the day.

Convenience is also regarded as less important in this household scenario, due to less negotiations between householders and hereby coordination of everyday tasks. An opportunity is therefore to time-shift performance of practices to off-peaks times, while maintaining householders' notions of convenience. A challenge for this household typology is the limited number of household members, which can potentially be involved in conducting this time-shifting of practices. If nobody is within the household when e.g. a recommendation is received, itbecomes difficult to time-shift a certain appliance. New technologies might be helpful in this turn. Remote control of domestic appliances could be a convenient way of enabling energy demand flexibility, by shifting appliances from outside of the home.

Coordination of ordinary and digital housekeeping

A detailed above, coordination of ordinary housekeeping may be regarded as easier for this household scenario, due to fewer negotiations and less routinised rhythms. However, it may still be the case that couples livingtogether have a gendered division of housekeeping chores, reflecting the gap between digital and ordinaryhousekeeping. As with the previous household scenario, securing an overlap between digital and ordinary housekeeping is essential for a successful DR solution, and should therefore be promoted in these cases. In the case of single-person households, the gap between ordinary and digital housekeeping disappears, and the useof new digital tools for management of energy consumption might be easier. Digital literacy might also be high(er)in the case of this household scenario. Householders are typically younger and might already have experience with energy technologies and therefore the learning curve for DR solutions might be less steep.

Flexibility potential

The flexibility potential for this household scenario is high. This is mainly due to fewer negotiations and more loose everyday rhythms, making coordination easier. The temporal performance of practices is therefore more flexible, and can potentially be shifted to other times of the day, without affected comfort or convenience of the householders. New modes for control, such as remote control from an app, might increase the flexibility potential, allowing householder to time-shift appliances in their everyday life that is characterised by spontaneous and ad- hoc activities.

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5.2.3 Intergenerational families

Fig.17 Intergenerational families icon (adapted from Colourbox)

A third, and more culturally dependent household constellation, are intergenerational families. These were especially present in the case of Berchidda. Intergenerational families typically consist of grandparents, their children and possibly grandchildren. They live together 'under the same roof', but might not share the same home or entrance, but have arrangements such as independent apartments or terraced homes under the same structure. Also they might share the same energy meter and energy system. Furthermore, intergenerational families are often closely connected socially, taking an active part in the performance in each-others' everyday practices, such as senior members regularly undertaking practices for their children or grandchildren.

Comfort and convenience

A central, yet different, aspect of intergenerational families is that there are multiple notions of comfort and convenience. Grandparents or more senior adults, may have one notion of what it means to be comfortable, while their children (and grandchildren) might have another. Comfort is important in this household scenario, and is often performed in a myriad of different practices that spread spatio-temporally throughout the day. While comfort remains important, the presence of multiple household members, with different everyday rhythms, may result in new ways of organising everyday life. Because of the diversity of this household type, several members (particularly the more senior ones), are present at home for longer periods of time, and therefore during non- peak hours. This presents an opportunity to be utilised, especially in terms of the coordination of everyday housekeeping. This could relate, for example, to cooking, cleaning or showering, which can be performed by other members of the household outside peak hours. Examples for the findings include grandparents taking care of their grandchildren, while the parents are at work. On the other hand, the challenge is that because of the longer presence at home at all times, energy consumption, especially heating or cooling, is more difficult to shift in time due to the required comfort across the day.

Convenience, which might be an important aspect due to the many different rhythms and negotiations, might also be easier to 're-arrange' due to the availability of the different household members. What is convenient can be negotiated and also achieved without the use of digital technologies. Everyday chores can be performed by several of the household members, and thereby increasing their convenience.

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The opportunities for intergenerational families are thus different from the other household scenarios. Whilemany different comfort standards are at play, notions of comfort related to care and cleanliness may be easierto shift outside of peak hours.

Coordination of ordinary and digital housekeeping

Coordination of everyday life is both a challenge and an opportunity in the intergenerational household scenario. It remains challenging, as multiple everyday rhythms have to be coordinated at once. If there is no shared rhythm
e.g. in relation to dining, cooking or doing the laundry, peak hours are not as important, because of the presenceof different household members in the house at all times or during non-peak hours. On the other hand, coordination in the intergenerational household scenario is often ensured by sharing the performance of everyday chores, such as cooking, taking care of the children etc. This provides an opportunity for shifting ordinary housekeeping chores outside of peak times and thereby enabling some energy demand flexibility. Communication and creating shared meanings are generally important, and to ensure that e.g. recommendations from the Hestia platform are followed by all members of the household, it is important that they are visible, easy to understand for all and to negotiate.

Digital housekeeping might be more challenging in the everyday life of the intergenerational households. Multiple household members with different levels of digital literacy might challenge the performance of digital housekeeping, which might be left in the hands of a few householders. It is therefore important to establish shared information on energy visibility and how the coordination can ensure a more flexibility energy demand. The solutions for enabling energy demand flexibility might be more analogue (rather than digital), but could be reached by exploiting the presence of household members within the home.

Flexibility potential/flexibility capital

The flexibility potential in the intergenerational household scenario is medium to high. The presence of many household members provides a challenge in the coordination of everyday rhythms and routinised activities. Adding to this, household members might possess very different skills, deviating the performance of practices. Creating shared meanings and skills is therefore essential. The opportunity is then to shift the performance of everyday practices outside of peaks time, by exploiting the idea that several household members can perform the same practice.

5.2.4 Retirees

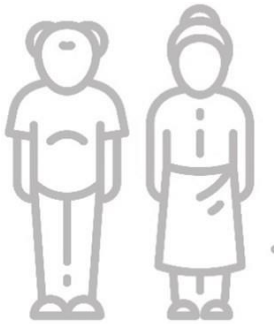


Fig.18 Retirees icon (adapted from Colourbox)

The last household scenario represents retirees. These were present in all of the pilot sites, but were more widespread in the Voorhout pilot site (compared to the population size). Retirees are considered as non-working people (individuals or

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couples), and characterised by everyday activities which are not structured or directly restricted by institutional rhythms.

Comfort and convenience

Retirees may already have well-established notions of comfort, which they have developed throughout life and experience with certain material objects. In the findings, many retirees expressed ways of performing everyday practices, which were not evident in the other household scenarios. Perceptions of comfort also differed, and comfort was not necessarily related to energy technologies. However, the expectations of comfort when e.g. moving into a new house, could spark frustrations. Among the retirees, it is therefore important to exploit that householders may not have 'normal' or culturally-dominant comfort standards. This is due to many reason, one of which is the concept that the generation of currently retired people, grew up and lived the majority of their lifein homes that did not offer the same level of thermal (or other kind)of comfort that contemporary buildingstandards are currently providing to many homes. Many senior adults have therefore developed different ways of achieving comfort at home, which was not based on the home's heating system. Suggesting, for example, to use a sweater or socks when cold, might therefore be possible alternative possibilities for time-shifting e.g.heating outside of peak-hours.

Convenience was also perceived differently by retirees, when compared to the other household scenarios. Engagement with technology was generally not regarded as a convenient way to ease the coordination of everyday activities. This household scenario may express frustration when not knowing how to engage properly with technologies, and feel out of control in their own home. Furthermore, they might prefer to use more manual,and less automated, ways of operating and maintaining their homes, such as controlling the indoor temperature manually or run household appliances themselves rather than program them or operate them remotely through apps. The implementation of DR solutions should therefore be aware that digital literacy is lower in this household scenario, and that retirees might have less desire or ability to engage with digital or automated systems. Therefore, a focus on more analogue practices might be a possibility.

Coordination of ordinary and digital housekeeping

As retirees are usually non-working professionals, coordination of everyday life may be less challenging (when compared to the other scenarios). Everyday activities may therefore be performed during off-peaks times e.g. eating breakfast late, being home early in the afternoon or being out late in the evening. The more loose rhythms in everyday life, may increase the possibility for shifting everyday chores, so that they will fit more with new and flexible ways of consuming energy. On the other hand, old routines, e.g. eating at certain times of the day, may be very established and 'sticky' and therefore more difficult to shift. Division of housekeeping chores may also bemore gendered among retirees, which could complicate the coordination of engaging in new and more flexible practices and closing the gap between digital and ordinary housekeeping chores.

In relation to the digital housekeeping, as already mentioned, generally digital literacy is lower among retirees. This might be a challenge for engaging householders in the coordination of digital housekeeping with ordinary housekeeping. It therefore remains important that the digital literacy is increased or encouraged. However, despite digital literacy challenges, most retirees showcase a relatively high interest in new energy technologies. One limitation of this though is the highly gendered interest in energy systems and technologies (over- represented by male participants). Therefore, a recommendation is to relate the DR suggestions to specific practices or certain skills, which the householders already possess and can take advantage of, while being mindful of the gendering of technologies that enter the home.

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Flexibility potential/flexibility capital

The flexibility potential in the retirees household scenario is medium to high. There are good possibilities among retirees to exploit the loosely routinised everyday life, which provides an opportunity to shift practices in time. Adding to this, negotiations may be easier in this scenario, due to the limited number of household members.

Another opportunity is to take advantage of the skills (often tacit) that are already in place. Householders in the retired household scenario are familiar and experienced with alternative ways of performing practices, which other practitioners may find inconvenient.

Challenges remain in relation to technology and the use of such for enabling energy demand flexibility. First, retirees might have certain expectations for what new technologies can deliver, and as these are not fulfilled, they may become frustrated. Second, the level of digital literacy is a challenge, and engaging occupant with the DR solution is regarded as more difficult.

5.3 Recommendations for the design and technical development of the Hestia platform

The findings presented in chapter 3 and 4, can be summarized in a series of recommendations for the design and technical development of the Hestia platform. For a further elaboration and background for these recommendations, see chapter 3 and 4.

5.3.1 Design of devices and technology interfaces

Fig.19 Devices and interfaces icon (adapted from Colourbox)

- Regarding the design or the consideration of digital devices that (potentially) are to be included in the DR solution, it is important to think about their appearance (aesthetics) as well as their visibility and safety in the homes. **Devices need to be aesthetically compelling and perceived as safe and easy to use** so thathouseholders will perceive and domesticate them as everyday devices within the home. To increase(long-term) engagement it is **important that householders perceive devices as 'natural' and non- disturbing within their home**. If devices are not properly domesticated, the consequence might be less engagement or householders simply disregarding the technology by e.g. putting it away/out of sight. The findings does not provide any clear recommendations for how to design devices that are properly domesticated into households but it important to involve the target population in the design process of these devices. By doing so, the familiarity with these devices will be secured and the design will reflect contextual circumstances.

- **Mobile phones** (particularly smart phones) **are preferred as the primary devices for engaging with theHestia**

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platform. Householders preferred to have the Hestia platform incorporated into well-known (and already domesticated) devices, and for most of the householders the mobile phone were a device which they felt confident and comfortable using. An app was preferred (contrary to a browser-based solution),as this was regarded as a more user-friendly solution. The app was also preferred due to the possibilities for performing remote control of appliances (in a future perspective). Using an App was therefore relatedto notions of control, which was highlighted as important by the householders

- A third design recommendation drawn from the findings, is the **importance of interoperability between various energy technologies**. As residential buildings become increasingly digitalized, various energytechnologies 'move in'. These are often design and distributed by different companies and also relates to different spheres of the home (e.g. lightning, heating, PV's etc). In the design of the Hestia platform, it is therefore important to consider how to integrate and create synergies between the different technologies and platforms at home. For householders, the possibility of monitoring and controlling for the same platform/entry point was preferred and should be pursued in the design of the Hestia platform.

- Many of the participating households consisted of several household members. While some everyday rhythms extend to most (if not all) household members, such as breakfast and dinner, other everyday rhythms differentiate between the household members e.g. in some families, one occupant may have breakfast before the others. An implication of this is that '**one -size'-initiatives does not fit all** — even within one household. **What is regarded as a 'normal' routine for some member of the household, mightnot be the same for others**. The reasons for this are multiple, and extends beyond the home, including institutional rhythms. The Hestia platform should allow households to setup different accounts, reflecting the different everyday rhythms and preferences that householders perform in everyday life.

- Related to the above recommendation, it also remains **important that the Hestia platform is responsiveto the material context of the household**. With the increasing use of AI, energy technologies are designedto detect and learn certain behavioural patterns and possible able to develop feedback and recommendation based on this. Within the three different Hestia pilot sites, **the spatial and materialsetup of the houses was very different, and it therefore is important that the Hestia platform reflect this in the setup**. Features and functions which do not relate do the specific context should not be visible to the householders, as this may create more confusion and uncertainty.

- Going further in detail, the findings suggest that there was no clear preferences for how feedback should be presented. While some householders preferred visualizations of their energy consumption, others preferred feedback in numerical values. A recommendation is therefore that householders **should beable to choose between different forms of feedback and how they would like to receive** it. Referring to the above recommendation, it is **also important that different household members are able to select inwhich form they prefer to get feedback**. **The Hestia platform should, in other words, support customization of feedback and suggestions**.

- In the data, it was clear that **most householders did not prefer a fully-automated DR solution**. This wasoften perceived as a feature that could risk putting them 'out of control' in their own home. While several of the householders expressed interest in automation and the benefits that this could bring them, they also expressed that an increased use of automation could risk putting the out of control. The householders therefore preferred to have the possibility for a DR solution that would allow them to override the system, and by doing so

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staying in control.

5.3.2 Frequency of interaction with users and content of interactions

Fig.20 Frequency of interactions icon (adapted from Colourbox)

- While notifications providing householders with insights on their energy consumption were generally were well received, there was not a clear indication in what ways they should be presented. The participating **householders generally preferred information which was context-dependent and meaningful to their everyday life and routinised activities**. To some householders, feedback on their consumption levels in e.g. Kwh was meaningful, but to others it was not. This related to different levels of 'energy knowledge' that the householders had, and thus what they would find meaningful. Some were able to understand feedback in e.g. KWh as meaningful and would know how to interpret such in a meaningful way. Others would prefer simple graphical representations, which they found easier to understand. **The Hestia platform should therefore allow for several ways of receiving feedback**.

- Suggestions providing by the Hestia platform on how, and when, to time-shift energy consumption were generally well received by the householders. They found **recommendations about performance of practices or alternatives more meaningful than merely getting energy feedback.** This underlines how suggestions could improve the everyday perspective of DR solution, as they **connect energy consumption to the task that householders perform** (and thus use energy). It was highlighted by householders, that **recommendations should be context-dependent and meaningful to their everyday life**. **The recommendations** presented to householders **should** therefore **reflect the different routinised activities that** householders **perform in everyday life** and a design implication is to **allow for different profiles in the platform, which reflect different everyday rhythms.**

- **The option of choosing how often to receive recommendations** (e.g. with suggestions) was expressed as important. Some householders preferred to get notifications on a daily basis, while others stressed that **notifications should not become too overwhelming**. A design solution could be **allowing for different modes in the platform, with the possibility of differentiating in relation to the frequency of notifications**. This could also be done at a household level, allowing some household members to receive frequent notifications with suggestions, and others not.

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5.3.3. Digital literacy and accessibility of users and ways to be inclusive (such as recommendations for voiceactivation or other accessibility issues)

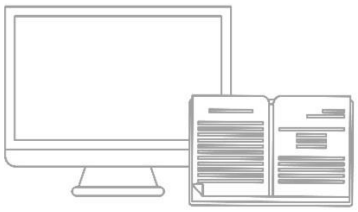


Fig.21 Digital literacy, accessibility & inclusivity icon (adapted from Colourbox)

- Given the quite different socio-demographic profiles of the pilot sites, a general recommendation is to support an inclusive design which pays attention to different 'user profiles'. Variables such as gender and age are related to digital literacy, but also to accessibility of a DR solution. The platform should support different levels of digital literacy and make it as accessible to as many household members as possible. This includes the possibility for customization of different profiles, from which the householders can choose. Furthermore, making the platform accessible on smart phones as an app, is supporting a wider accessibility among household members. However, householders should also have the possibility of accessing the platform through other devices(e.g. PC or tablet) which they prefer and use at home.

- As noted above, digital literacy differs among household members and while some find it easy to setup and understand energy feedback, others might not do so. It is therefore **important to design DR solutions that are user-friendly and age and gender appropriate in design and setup**. A recommendation is to allowfor learning opportunities of householders in regard to the use and domestication of energy technologies. Learning can involve peer interactions, 'warm' experts' interactions (familiar people who can provide assistance with technologies) and formal professional intermediation at regular intervals.

- When engaging with DR solutions (and energy technologies generally speaking), people will often find bugs and problems. **Reliable technologies are,** not surprisingly, preferred by householders. While breakdowns and bugs cannot be avoided, it remains important to **limit the area for bugs as much aspossible**. As householders experience technical challenges when engaging with the technology, thisexperience might increase their opposition towards and perception of technology in general. **Increasing the reliability of a technology thus remains important, also ensuring that it has been tested in an**

everyday life setting outside of the lab.

- Furthermore, it is important to **ensure technical guidance and service support to the householders**. **Guidance on how the technology works and how to operate it, should be given at an early point** (perhapswhen installing the technology). This will both increase householders' technical understanding, but also increase their perspectives for learning through doing. Technologies do break down, and when they do so, it is important that there is a clear route for support and that householders are well-aware of this. Inthe data, it was not clear to all householders how to obtain support for technical support, and several householders had experienced problems with getting the necessary help. For guidance and servicesupport, intermediaries such as installers and buildings managers, become vital. These are often the first point of entry for the householders, and establishing trust between these front-row professional and the householders is important.

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- Similar, 'warm experts' and familiar faces within the community, can act as important intermediaries for guidance and support. Householders often rely on communal or family support when they experience problems with their technologies. **A recommendation is thus to educate 'warm' experts, who within acommunity will act as transmitters of guidance for how to engage with the DR solution**. Besides learningthe necessary skills, helping out less technological literate householders, 'warm' experts can also helpto create a more meaningful notion of what DR is and how it can be used. Contrary to installers or similar intermediary professionals, **trust and familiarity is typically already established within a community, sowarm experts can thus provide a more easy to reach support.**

- Our data suggest that **not everyone in the household engages in the same way (and for the same time)with existing DR solutions**, such as energy apps. In addition, **the ones engaging with the technologiesmay not be the same ones that perform ordinary housekeeping at home**, meaning that the coordinationbetween everyday practices might be complicated. In order to mitigate this, **the Hestia platform should strive to make energy consumption visible to all members of the household** (e.g. laundry, cooking)**, creating a broader base for awareness of energy consumption.** Once again, **allowing for different profilesmay help in this turn,** letting householders install an app of their own phones. **A common and shareddevice** (e.g. IHD) **may also be an opportunity, as this can be accessed easily by multiple householdmembers in a shared space**.

- In addition to the above, it is important to highlight that **the Hestia platform should protect privacy,**through its design of DR solutions. It is **important to acknowledge the new possibilities for surveillance that this allows, while making sure that householders are aware of what data are collected and in what** ways they are processed, so that they can give informed consent.

5.4 Recommendations for community engagement regarding the Hestia platform

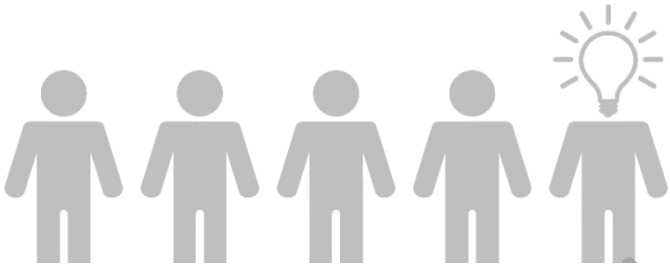


Fig.22 Community engagement icon (adapted from Colourbox)

The next recommendations relate to the broader community engagement relevant to the context of the Hestia platform. This concerns how communities can increase engagement through material and technical setups, but also how learning can be exploited across household boundaries.

5.4.1 Collective vs individual incentives and engagement

- The findings indicate that **peer networking can act as a way of mitigating uncertainty within the household**. When householders 'meet' new technologies in their homes, there may be differences ofdigital literacy or differences in the meanings they ascribe to this technology. This can often decrease engagement with the platform. However, the participating **householders perceived their local peers as important in sparking the necessary meanings and ideas of why it was important to shed or lower energy consumption**. **The community**

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thus acted as an incentive for the individual households. Furthermore, thecommunity acted as an informal way of introducing and learning about the new technology. **Exploiting peer communication in the Hestia platform is therefore an opportunity that can hinder some of the barriers of engagement**. Opportunities for increasing the communication between householders – both on the technical platform and in real-life settings should therefore be promoted.

- Contrary, the findings do not suggest a clear preference for specific kind of incentives. While some householders perceived financial benefits to be the greatest incentive for engaging with the Hestia platform, **others referred to sustainability, status, or community aspects as being the biggest incentive.** While individual **incentives** matter for householders, they **do not exist in a vacuum**. **Householders prefera mix of different incentives, and a recommendation is to combine both individual and collective** incentives in the Hestia platform.

- Adding to the above, **incentives should be sensitive to the routinised ways of performing practices athome**. What is perceived as easy or difficult to shift and at what times of the day this is possible, differs across households. Some everyday chores are more interlocked, or 'sticky' than others and they may therefore be very difficult to shift for the householders. **Despite an interest in a given benefit (e.g. financial savings), it can therefore not be expected that** householders **will perform shifts of certain practices, because they play a too important/meaningful role to them in their everyday life**. The**incentives** therefore **need to be** sensitive and **combined with suggestions for alternative, more efficientways of performing a practice**

5.4.2 (Generation and strengthening of) Energy communities

- **Communities play an important role in engaging householders in the Hestia platform**. This finding indicates that householders have an interest in their own energy consumption but also that of their community. They would like to know more on how they perform themselves, but also how they compareto their neighbours. However, **most householders did not express an interest in competing against theircommunity in relation to their energy consumption**. Increasing measures of competing could risk demotivating some householders instead of increasing their engagement. Instead **the Hestia platform should promote the role of the household and how their engagement with the Hestia platform can contribute to wider community efforts**. It seemed that householders perceived the idea of energy communities as a valuable consideration, and one in which they wanted to contribute. **The Hestia** platform should therefore promote the role of households within the community instead of increasingcompetition.

5.4.3. Ownership of the energy systems and implications for the design of DR solutions (GRD, DW, AAU)

- In the data, it seemed that householders who had experience with other energy technologies and systems, generally expressed an interest in new modes for shifting or shedding energy consumption levels. Householders who e.g. had PV's, were interested in ways of exploiting their own production and in new ways of shifting and storing energy. **The synergies that exist between experiences with otherenergy technologies could be regarded as an opportunity and a challenge to time-shifting energy loads**.On the one hand, it seemed that **ownership of energy technologies increased the occupant's interest in DR solutions. The Hestia platform could therefore exploit the presence of existing energy technologies,allowing householders to couple and exploit the synergies for this**. On the other hand, ownershipprovides a challenge for those who do not yet

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possess the technologies, as e.g. in the case of Voorhout, where householders moved into a pre-installed setup. In the latter, ownership was characterised by certain expectations to the technologies and the installers/developers. **When expectations are not met, householders become frustrated and decrease their engagement generally. The Hestia platform should** therefore promote knowledge and ownership of existing energy technologies as well.

- **Notions of ownership of energy systems should be promoted through the Hestia platform**, **as well as new business models to support this local collective ownership** At the household level, the platform should exploit that some householders already have energy technologies in place and would like tobenefit from the synergies between these and the new Hestia platform. Also, it is important to broaden the notion of ownership to other members of the household. A previously presented, there is often a difference between the householders who are engaged in digital housekeeping and those involve in everyday housekeeping. It is therefore important to increase the notion of ownership among the latter, as this will secure an increased coordination. Secondly, ownership of energy technologies can be promoted on a community level. **The data indicate that ownership could be promoted on a communitylevel,** and e.g. that **sharing certain energy technologies can increase engage and interest with the Hestiaplatform at home.**

5.4.4. Learning perspective – material vs social learning

- The process of learning about energy technologies and how to live with them (householders know-how) is important for the ways in which householders engage with the Hestia platform, besides their interest and reliance on technologies.. Furthermore, when breakdowns do happen, it remains important thathouseholders know who to contact or how they can troubleshoot the technology themselves. In the data, several householders expressed uncertainty in how energy technologies worked, and asked for more guidance. Going beyond guidance, the findings suggest that tacit knowledge plays an important role in a learning perspective. This knowledge is obtained through 'doing' and/or exposure. Know-how is therefore not 'easily' learned by e.g. reading a manual, and the learning curve will look different for different householder. It is therefore important for the success of the Hestia platform that tacit knowledge on how to use it is externalised, and thereby can be shared between householders. Learning-by-doing should be promoted as householders can obtain an experience with the platform. Showing **householders how the platform works and in what ways they can engage with it remains important and adds to the tacit knowledge.**

- **Intermediation and peer support is essential in the process of learning to live with energy technologies**.It should be acknowledged that learning takes time. Learning obtained through doing is a continuous process, and householders will add to this pool of knowledge every time they engage. **It is therefore important to contextualize the learning,** so that it becomes meaningful to the householders themselves. Contextualizing can happen using peer-to-peer communication or warm experts or peers in the same position as the householders, which can explain to householders how to use the Hestia platform and **carry out support in need of such.** Furthermore, **the role of intermediaries is important**. When installing the technology, intermediaries should include householders as much as possible, increasing the guidance and exposure on how to use and familiarise themselves with the technologies.

  Allow regular opportunities for community events, which can support the sharing of tecit knowledge (practical know-how) of the energy tehcnologies. These events could be in real life or virtual or a mix of the two. While explicit knowledge, such as that which can be found in manuals or online FAQ sections also remains important for guidance

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and support, the tacit knowledge aspect needs to be levelled, in order to achieve a successful engagement with the Hestia platform. The 'two types' of knowledge could go hand-in-hand and support each other, so that explicit knowledge is followed by practical experience and exposure to the platform and how to use it.

6. CONCLUSIONS

This chapter has presented some recommendations for the development of the Hestia platform based on the findings of the household interactions and participatory workshops that have taken place in the three pilots of Hestia.

The recommendations were presented under three categories:

1. As scenarios for action based on empirically identified household typologies and participants' stage of life
2. As recommendations for the design and technical development of the Hestia platform
3. A recommendations for community engagement

Initially the scenarios developed, which are not generalisable to contexts outside the scope of Hestia, represent a current picture of the rather small sample of householders and of the ways in which they perform everydaypractices. By developing these scenarios, we aim to contribute to the overall profiling of households, which will be done in further stages of the project (for use I the Hestia dashboard or the digital twin for example), attemptingto move further than the individual, toward a more collective perspective of household action. The scenarios haveidentified possible areas for energy flexibility, as well as challenges for this, based on the everyday practices of household members but also in line with collective societal rhythms, such as work, education and leisure engagements.

Then, the recommendations for the design and technical development of the platform derive from suggestionsof participants, either through their current experience with technologies and devices, or through visions developed in the interactions (workshops) which enabled them to simulate an experience and provide their views.Issues such as the prevailing preference for mobile phone applications for the operation of energy systems, as well as the need for interoperability of technologies and their domestication capacity were presented, as considerations for the technology developers in the Hestia project, as well as other relevant contexts in which these might be relevant.

Finally, the recommendations for community engagement in regard to the development of DR solutions, focused on the collective appreciation of communities in their current transition to more efficient energy systems. Ithighlighted that individuals need to be seen as part of the communities in which they belong in, and that theconstruction and ownership of (newly generated) energy communities is a complex issue which needs to behandled carefully. In specific, issues of privacy, digital literacy as well as gender and age parameters needs to be 'designed into' the platform, rather than applied as extras after the development of the platform. Also, that the ongoing learning that these communities are undergoing

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during this transition, needs to be supported by appropriate intermediaries and peer networks in order to enable a sustainable and long-term result.

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8. APPENDICES

Appendix 1: Template for reporting on virtual interactions

Template – Summary of interviews/virtual interactions

For each of the interviews, conducted as part of WP2, the relevant partners should prepare an English summary and send it to AAU. The aim of this summary reporting, is to inform the analysis of D 2.2 with the insights generated during the interviews. As the interviews has been conducted in different languages and in settings where AAU has not been present (and able to ask questions), it remains very valuable to gather these summaries and use them as secondary data points. The summaries should follow the same basic structure, as outlined in the sections below. If you have any question please don't hesitate to contact us.

In the template, the first two sections concern the basic/descriptive information of the interviews (e.g. number of participants present during the interview, age, gender representation), making it possible to draw out methodological limitations and evaluate the validity. Next, we ask for you methodological reflections, suggesting you to outline how the interview went, describing the engagement, 'mood' and potential deviations from the planned setup. Third, we ask you to highlight some of the 'most interesting' insights from the interview. Here we ask you to summaries a number of themes (you decide how many). We would both like to get 'raw data' i.e. summaries of what happened and what was being said, but we also ask for you analytical observations from the interviews.

We would furthermore like you to attach the interview-guide/ planned questions, which was used for the interview. We know that, as the interview was conducted semi-structured, there may have been deviations from the plan. If you have notes related to the interview, please also attach these if you think they are relevant.

Last, please feel free to attach photos or videos that you might have obtained during the interview, and which you think illustrate a certain point –methodological or content related.

The summaries should be written on basis of the audio/video recordings and analytical observations.

Please ensure that the participants are anonymised in the summaries. I.e. do not use their real names and take also care not to include personal information that would make it possible for others to identify them.

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Interview organisation: Time, place etc.

Interview day of the week and date	
Time the interview was conducted	
Duration of the interview	
Virtual or physical interview	
If physical: location / setting	
Any other practical information worthmentioning	

1. Participants and recruitment

Number of participantspresent in interview (if relevant)		
Gender distribution	**Female**	**Male**
Facilitator(s) (who ran the interview. If more than one person, please describe therole that each partner had)		
Participants' recruitment (how were participants recruited)		
Incentives / reciprocity offers for participating (if any)		
General qualitative comments on the participants' sample in regard to age and compositionof household		

Process

Introductions (please give us a brief statement about how you introduced the interview, what were participants told about the interview?)	

Did you introduce the Hestia project before starting or was the participants informed beforehand?	
Describe the **overall 'mood'** of the interview process. (E.g was it nice and 'cheerful'? Was it relaxed? Did the participant(s) feel comfortable about the setting and did they appear to trustyou? Or were the participants not so relaxed? Were there any tensions detected at any point? Please give us your reflection on what, how and why.	
Did you use **an interview plan/ guide/** determined questions?	(Please answer with yes or no, but also (if you used a guide)give us these questions as an appendix to this template)
Did the interview run according to plan? Were there any deviations from the original planning or interview guide?	
Comments/ reflections on the interview process and the (interpersonal) dynamics among participants(if more than one)	

Summary of topics

While the first three sections focus on the process and setting of the interview (including considerations of methodological validity), this section focuses on the content of the interview. Please divide your text into a number of sub-sections (4.1, 4.2 etc.), one for each topic or theme that you identify. You can for example follow themes from your interview guide, if used or empirically identified themes during the process of the interview.

Each sub-section (or topic summary) should contain 2 parts:

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1. Summary
2. Analytical observations/ reflections

Things you could consider when writing the **summary** of each theme:

- Please make a relevant and critical selection of themes, rather than a complete description ofwhat happened.
- Please include quotes where possible and relevant (you could give us both the original quote andits translation, using quotation marks to indicate the exact words

Things you could consider when writing the **analytical observations /reflections** of each theme:

- Please give us your personal interpretation and comments of the process and content of the theme identified. It is important to understand not only the objective encounter of what took place but also your opinion and explanation of why you think that was said or happened.

4.1 Topic 1: xxxxxxxxx (Please title)

Summary	
Analytical observations/ reflections	

4.1 Topic 2: xxxxxxxxx

Summary	
Analytical observations/ reflections	

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4.1 Topic 3: xxxxxxxxx

Summary	
Analytical observations/ reflections	

4.1 Topic 4: xxxxxxxxx

Summary	
Analytical observations/ reflections	

Photos and notes

Please add any material which you might find relevant for the reporting/summary of the workshop e.g. photos, screenshots or field-notes.

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Appendix 2: Template for reporting on workshops

Template – workshop summary

For each of the WP2 participatory workshops, the local site partners/facilitators should prepare a(short) English summary and send this to AAU. The summaries should follow the same basic structure in order to inform the final analysis. The template (presented below) outlines different sections and briefly describes (and give examples of) the focus and intended content of each section. If you have any questions, please don't hesitate to contact.

The aim of the summaries is to document and report some preliminary observations and findingsfrom the conducted workshops. As each workshop differs contextually and has been carried out in three different languages, we would like to gather some insights that can be comparable, thus opening up for a comparative analysis of the three different workshops. Taken together, the summaries will also be an input for the overall analysis (reported in D 2.2.) which aim to inform the design of the HESTIA interface (T 2.3 and T 2.4).

In the template, the first two sections will concern the basic/descriptive information of the workshops (e.g. number of participants, number of households, gender representation), makingit possible to draw out methodological limitations and evaluate the validity. Next, we focus on your methodological reflections, asking you to outline how the workshop went, describing the engagement, 'mood' and potential deviations from the planned setup. Third, we ask you to
highlight some of the 'most interesting' discussions from the workshop discussions. Here we suggest that you summarise each 'stage'/planned activity of the workshop. We would both liketo get 'raw data' i.e. summaries of what happened and what was being said, but we also hear your analytical observations from the workshop.

As D 2.2. is expected to inform the design of the HESTIA interface/dashboard (developed by i.LECO), please also consider including feedback from the occupants that directly concern the design of such (e.g. what features was liked, how occupants prefer to get energy feedback etc).

Finally, please feel free to attach in photos or videos that you might have obtained during theworkshop, and which you think illustrate a certain point – that being both methodological or content related.

The summaries should be written on basis of the audio/video recordings and analytical observations taken during the workshop.

Ensure that all participants are anonymised in the summaries. I.e. do not use their real names and take also care not to include personal information that would make it possible for others toidentify them.

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Hestia
Holistic demand response services
for European residential communities

Summary and analysis of workshop

(Insert photo)

Workshop organisation

Workshop day of the week and date	
Workshop time	
Workshop duration	
Venue	
Incentives/reciprocity offers for participating (ifany, e.g dinner)	
Hestia partner in charge	
Other Hestia partners present and their roles	

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Comments/reflections about the overall organisation of the workshop (positive comments about organisation and/or challenges about organisation)

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Participants and recruitment

Number of participants invited		
Number of participants that turned up		
Recruitment of participants (how they were invited to the workshop e.g personal invitation,letter, email etc.)		
Gender distribution:	Female	Male
Number of households present		
Comments/ reflections on participants' recruitment and turn out (e.g representation of age, gender, overallturnout of the specific community, practical issues that affected the running of the workshop and the participants' engagement etc.)		

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Workshop process and group dynamics

Welcoming process (what happened when participants arrived, e.g welcoming, names tags, introductions to othersetc.)	
Was there an introduction to the Hestia project in the beginning of the workshop? What did that involve? What were participants told?	
Were the participants split in smaller groups? If so how many and what were the specifics (e.g members of the same household split or stayed together, how many per group, purpose of eachgroup etc.)	
Did the workshop run according to plan? Were there any deviations from the original scheduleof activities?	
What was the overall 'mood' of the participating group during the workshop? (e.g. Was it good and "cheerful"? Was it relaxed? Did the participants seem comfortable about the settingand did they appear to trust each other? Or was the participant not relaxed or did any "bad atmosphere" occur? (If the latter, please includea few thoughts on why this might have happened)	
Please state 3 things that went really well in terms of engagement	1.

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	2. 3.

Please state 3 things that were challenging in terms of engagement	1. 2. 3.
Comments/ reflections on the overall dynamics of participants during the workshop (E.g In what ways participants engaged with the discussions, any challenges during the process, any positive remarks, did participants generally talk to each other or did they respond mostly to the facilitator? Did all participants participate in discussions? Or only a few? In the latter case: Indicate who?)	

Summary of each workshop stage/planned activity

While the first three sections focus on the process and setting of the workshop (including considerations on the methodological validity) this section focuses on *the content* of the workshop discussions. The section is divided into a number of sub-sections (4.1, 4.2, etc.), one for each of the different 'stages'/planned activities during the workshop. In the case that one 'stage'/planned activity was conducted in several small group, please include a summary of each discussion.

Each sub-section contains 3 parts:

1. Discussion starter and follow-up questions

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(Please give us the original discussion-starter summary and any possible questions or comments that wereintroduced to the participants)

2. Summary
(This is a summary of the discussions that took place. The following guidelines can be considered when writing this summary:
    - Please make a critical & relevant summary of what was said rather than everything thatwas discussed
    - Please include quotes where possible and relevant (you could give us both the original quote and its translation, using quotation marks to indicate the exact words

3. Conclusions/analytical observations: You conclude each summary with a brief sub-section with your own"Analytical observations". This is where you can add your personal interpretations of (or comments to) what the participants have said or discussed

4.1 Stage 1: xxxxxxxxx (please title the activity/stage)

Duration	
Any probes / tools etc. used	
Discussion starter	Please state how the session was initiated.
Questions asked to participants during the introductory part of theworkshop	Please give us the questions or comments that you have presented to theparticipants as part of the introductory phase of this activity
Summary of activity	
Facilitation	Please give us a brief statement about who facilitated this activity, what was planned (for example an activity can be partly presentation from facilitator and part co-creation activities), what worked well in regard to this method of facilitation you used, what were the challenges, what youwould do differently next time, etc.
Analytical observations/reflections	

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Highlights/ quotes	Please give us 2 or 3 points which highlight the process or content of theactivity. Also you can add relevant quotes or paraphrase some key sayings or findings during this activity.

4.2 **Stage 2**: xxxxxxxxx (please title the activity/stage)

Duration	
Any probes / tools etc. used	
Discussion starter	Please state how the session was initiated.
Questions asked to participants during the introductory part of theworkshop	Please give us the questions or comments that you have presented to theparticipants as part of the introductory phase of this activity
Summary of activity	
Facilitation	Please give us a brief statement about who facilitated this activity, what was planned (for example an activity can be partly presentation from facilitator and part co-creation activities), what worked well in regard to this method of facilitation you used, what were the challenges, what youwould do differently next time, etc.
Analytical observations/reflections	
Highlights/ quotes	Please give us 2 or 3 points which highlight the process or content of theactivity. Also you can add relevant quotes or paraphrase some key sayings or findings during this activity.

4.3 Stage 3: xxxxxxxxx (please title the activity/stage)

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Duration	
Any probes / tools etc. used	
Discussion starter	Please state how the session was initiated.

Questions asked to participants during the introductory part of theworkshop	Please give us the questions or comments that you have presented to theparticipants as part of the introductory phase of this activity
Summary of activity	
Facilitation	Please give us a brief statement about who facilitated this activity, what was planned (for example an activity can be partly presentation from facilitator and part co-creation activities), what worked well in regard to this method of facilitation you used, what were the challenges, what youwould do differently next time, etc.
Analytical observations/reflections	
Highlights/ quotes	Please give us 2 or 3 points which highlight the process or content of theactivity. Also you can add relevant quotes or paraphrase some key sayings or findings during this activity.

4.1 Stage 4: xxxxxxxxx (please title the activity/stage)

Duration	
Any probes / tools etc. used	
Discussion starter	Please state how the session was initiated.

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Questions asked to participants during the introductory part of theworkshop	Please give us the questions or comments that you have presented to theparticipants as part of the introductory phase of this activity
Summary of activity	
Facilitation	Please give us a brief statement about who facilitated this activity, what was planned (for example an activity can be partly presentation from facilitator and part co-creation activities), what worked well in regard to
	this method of facilitation you used, what were the challenges, what youwould do differently next time, etc.
Analytical observations/reflections	
Highlights/ quotes	Please give us 2 or 3 points which highlight the process or content of theactivity. Also you can add relevant quotes or paraphrase some key sayings or findings during this activity.

Photos

Please add any visual material which you might find relevant for the reporting/summary of the workshop. This canboth be pictures or videos.

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Appendix 3: Interview-guide used for doing virtual interactions in Berchidda

Interview-guide for use in short virtual interactions, HESTIA

Purpose/aim

- To develop an understanding of how, why and when occupants use energy in their homes
- To develop an understanding of how objects and technologies are entangled in energy intensive practices
- To develop understanding of how energy consumption is negotiated among occupants
- To develop an understanding on how occupant perceive DR technologies and identify

Presentation of HESTIA

The presentation is given by the facilitators of the interview. This should be a short introduction (5 min) The purpose of the presentation is to:

- Establish an open setting (making informants comfortable in participating)

- Making sure that the participants understand the purpose of HESTIA

- Making sure that the participants understand why their participation is important

- Making sure that the participants understand the format of the virtual interactions

- Making sure that the participants understand how their data is obtained, processed and stored and their rights to their data

- Making sure that the participants understand which kinds of personal data that the interview will collect

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Presentation of the interview-format – talking points

The format is open i.e. you as an participant can talk completely free (follow strain ofthoughts). There are no right or wrong answers.

The interview will take 45-60 min

The themes in the interview are; Everyday life activities/routines and in relation to the specific use-case [washing clothes, dishwashing, preference for visualization of consumptiondata]

The interview is confidential. Your data will be stored safely and are only processed bypeople directly involved in HESTIA.

Your data will be anonymized

The interview is recorded (voice and video) and undergo transcription after that.

Themes	Questions	Notes
Presen	Could you (all present at the interview) give a general presentation of yourself? Includingage, education, job and housing- career When and why did you move into your current house? - What was important is deciding on this?	Make sure to ask about all occupant who lives in the home. Cover details on: Age, gender, personal relations, ownership/housing type, length of living in the home, conditions of current and previous houses

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	In general, how is your current house different from previous places that you have lived)? Do you think about your energy consumption? - Do you know what your energy consumption is? (Kwh, price, generally andspecifically) - Do you know if you usemuch or little? - Do you know where yourenergy is produced/the source of energy?	Perception of energy consumption and provisionof such

New theme: After some introductory questions, the following will concern your everydaylife in general, how it unfolds and which role energy plays within		
Everyday life: Activities and routines	Please describe a typical weekday? Which activities occur routinely inyour home? - If several household members present: Who conducts these activities? How are household tasks distributed among members of the household? What are the biggest differences between activities and routines conducted on a weekday and in theweekend? Which activities are most time-consuming?	Morning/Afternoon/eveningroutines Negotiation of routinized activities Work/leisure Temporal organization of activities

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Perceptions of home and spatial use of the home	What do you think of your home? - What does home mean toyou? When and how do you use thedifferent rooms in your home? - Are there rooms where youspend more time than others? Why? - Has this changed over time (e.g. under the Covid- situation)? - Are there rooms in your home that you don't use?Why?	Perceptions of home and everyday life Spatial organization ofactivities

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Washing clothes	How often to you wash clothes? When do you wash clothes? - Time of the day of the week? - Why of these specific days? How is washing clothes related to other activities conducted in your home? - E.g. done when cleaning ingeneral - E.g. done when being at home(bored) - E.g. done before 'going out' … Do you always use the washingmachine? - Why/why not? Is the washing machine always fullwhen you run it? - Why/why not? Do you use different programs on you washing machine? For what?	

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	How to you decide on when to wash clothes ? – e.g. full laundry basket? – e.g. In need of certain clothes? – e.g. nice weather for drying outside? – e.g. routines (its Saturday, on wealways do it here) What makes you want to wash a certain clothing item? – Is that different among the household members? What do you do if you want to weara certain clothing item, but it is in the laundry basket? How is washing conducted? – How long does it take? – Is drying done inside or outside?And does that differ? – Who does most of the washing?Do others household members	

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	have certain tasks e.g. hang for drying, take clothes out of the machine etc.? – Do you wash when not being in the home e.g. starting the machineand then leaving for work? – Do you wash at night? Is washing done in the same way by all household member? – Can washing lead to "conflict" ? – an appropriate way of doing so? Would you consider washing at other times/washing less if: – You could save money? – You could use your own produced electricity? – It would benefit your community? – It would benefits the environment? … When would/is be the best time towash clothes ?	

	− Why is that? What would/isis be the worst timeto wash clothes ? − Why is that? How could technology assist/support you in shifting orlowering washing cycles? − Visualization of when it would be 'optimal to wash'? − Automation/pre-programming of the washing machine?	

Washing dishes	How often do you do dish washing? How do you do dish washing? Dishwasher, in hand? How do you dry dishes? dish drying rack/the dishwasher/dishtowel At what times do you dish wash? − Time of the day, days of theweek? − Why is that? How is dishwashing related to other activities in home? − e.g. having a dinner/eating breakfast − e.g. cleaning in general? − e.g. Having guests? − e.g. Cooking ? − e.g. routines (its evening and wealways do it here)	

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	Is the dishwasher also filled upbefore it is run? Are dishes always washed after ameal? – If so, which meal ? And why thatmeal? How to you decide when to washdishes ? – e.g. cleaning after a meal – e.g. in need of certain kitchenware – e.g. it is just a routines – e.g. it is an obligation – e.g. part of cleaning – e.g. In need of table space How is dish washing conducted? – How long does it take? – Who is involved – i.e. a task thatis done alone or nor? – Do you run different program on the dishwasher – which and why? – If no dishwasher, do you use a tubor running water?	

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	– Are plates rinsed before dishwashing? (why) Who's is doing most of the dishwashing? – Do anybody else do it? – Can washing lead to conflict ? – an appropriate way of doing so? Would you consider doing dishwashing at other times/ less if: – You could save money? – You could use your own produced electricity? – It would benefit your community? – It would benefits the environment? When would be the best time to dodish washing ? – Why is that What would be the worst time todo dish washing ? – Why is that?	

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	How could technology assist/support you in shifting the times you dish wash or the frequency? − Visualization of when it would be 'optimal'? − Automation/pre-programming of the dishwasher?	

Visualization of energy consumption	Do you look/monitor your energy consumption? – Why/why not? – In what way (e.g. meter, bill, online platform etc.) – How often do you look and in which circumstances? Have you made any changes to how energy is used, on the basis of looking at the energy consumption? – Have you ever changed a certain way of doing, because of receiving feedback on your energy consumption ? In what way and why? How do you receive information on your energy consumption? – Do you know what your monthly/annual energy consumption is? – Do you know what different appliances in your home use? – Do you know much different practices e.g. heating/washing/bathing use in [kWh, money]?	

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	If you have problem with yourbilling, who do you contact? − Have you had any visit from intermediaries? Which informationdid they give you? Who do you contact, if there is aproblem with your technical installations(e.g. boiler) or that there is a blackout? - What did you do the last time there was a problem?Did you try to solve it yourself? - Do you feel comfortable with sharing energy data with the utility? - Would you share data withothers (companies, public officials)? Do you currently receive feedback on energy consumption on any digital devices? - Why do you have these technologies? - By having these technologies, have you changed everyday life activities (how, why and when)?	

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	Would you like to receive information regarding your energyconsumption? − In which format? - In kwh - Price - Graphical/visual representation - Daily/monthly/week status? − On which platform/medium? − What would be your preferred device for received information andwhy? Who in the household would mainly look at/monitor energy consumption? − Is it the same person who would make changes based on the information? How could information on your energy consumption be used to shift or lower your energy demand? − What would be meaningful toyou?	

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	- Would you let your utilitycontrol your energy consumption? - Which role would economy, comfort, sustainability play inrelation to this? - Would you like increased automation of your home? - Are there situation where youbeing in control is more important.	
End of the interview	This was the end of the interview. Is there anything that you wouldlike to ask or do you have any questions? We will leave our contact information and if you have any questions or comments, don't hesitate to contact us.	

Appendix 4: Programme for participatory workshop in Berchidda

Warm up

The first warm up activity was to answer individually on a post-it to 3 questions:

- What is an energy community?
- What is the role of each individual household?
- Which mean of communication would you prefer us to use to communicate with citizens?

Energy community in action

A low-tech simulation of how DR works, working in groups.

Interface presentation and feedback

Projected demo of Prosume's interface. Plenary.

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Appendix 5: Summary and analysis of workshop in Berchidda

Template – workshop summary

For each of the WP2 participatory workshops, the local site partners/facilitators should prepare a(short) English summary and send this to AAU. The summaries should follow the same basic structure in order to inform the final analysis. The template (presented below) outlines different sections and briefly describes (and give examples of) the focus and intended content of each section. If you have any questions, please don't hesitate to contact.

The aim of the summaries is to document and report some preliminary observations and findingsfrom the conducted workshops. As each workshop differs contextually and has been carried out in three different languages, we would like to gather some insights that can be comparable, thus opening up for a comparative analysis of the three different workshops. Taken together, the summaries will also be an input for the overall analysis (reported in D 2.2.) which aim to inform the design of the HESTIA interface (T 2.3 and T 2.4).

In the template, the first two sections will concern the basic/descriptive information of the workshops (e.g. number of participants, number of households, gender representation), makingit possible to draw out methodological limitations and evaluate the validity. Next, we focus on your methodological reflections, asking you to outline how the workshop went, describing the engagement, 'mood' and potential deviations from the planned setup. Third, we ask you to
highlight some of the 'most interesting' discussions from the workshop discussions. Here we suggest that you summarise each 'stage'/planned activity of the workshop. We would both liketo get 'raw data' i.e. summaries of what happened and what was being said, but we also hear your analytical observations from the workshop.

As D 2.2. is expected to inform the design of the HESTIA interface/dashboard (developed by i.LECO), please also consider including feedback from the occupants that directly concern the design of such (e.g. what features was liked, how occupants prefer to get energy feedback etc).

Finally, please feel free to attach in photos or videos that you might have obtained during theworkshop, and which you think illustrate a certain point – that being both methodological or content related.

The summaries should be written on basis of the audio/video recordings and analytical observations taken during the workshop.

Ensure that all participants are anonymised in the summaries. I.e. do not use their real names and take also care not to include personal information that would make it possible for others toidentify them.

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Hestia
Holistic demand response services
for European residential communities

Summary and analysis of workshop in Berchidda, Sardinia



Workshop organisation

Workshop day of the week and date	Thursday 16th of September 2021
Workshop time	6:30 PM
Workshop duration	2h30
Venue	Berchidda Wine Museum
Incentives/reciprocity offers for participating (ifany, e.g dinner)	Dinner

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Hestia partner in charge	Gridability

Other Hestia partners present and their roles	AAU: support with workshop set up and materialsSinloc: observer CDLT: support with workshop set up and photographic documentation
Comments/reflections about the overall organisation of the workshop (positive comments about organisation and/or challenges about organisation) The workshop organisation was overall very smooth. The mayor pointed us to the Wine Museum director, who provided information and pictures about the space and available equipment ahead of the workshop; she also made available her staff for a preparatory visit the morning of the workshop. The formalisation of the content and structure of the workshop was a collaborative process between GridAbility as pilot leader and AAU as WP2 leader.	

1. **Participants and recruitment**

Number of participants invited	30 households	
Number of participants that turned up	24 participants	
Recruitment of participants (how they were invited to the workshop e.g personal invitation,letter, email etc.)	Interviews participants were pre-alerted in July about thedate; a round of invitations was done early September, followed by a reminder the day before the workshop	
Gender distribution:	Female 5	Male 19
Number of households present 21 households		

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Comments/ reflections on participants' recruitment and turn out (e.g representation of age, gender, overallturnout of the specific community, practical issues that affected the running of the workshop and the participants' engagement etc.)

We had high participation from the engaged households: most of those who could not attend were not available at that time/day. We had a very good diversity of age groups, ranging from participants in their early 30ties to participants over 70. In terms of gender, women were 20% of the participants, half coming with their partner and a half alone. Women's participation in Berchidda will deserve attention during the following phases of the project, since from interviews and workshop it emerged that housekeeping roles are quite gendered. We need to ensure that the project interfaces and devices are not only utilised in the households to "supervise" energy, but also that they become an integral part of energy habits, tools for adapting proactively everyday consumption. Although gender roles are out of the direct scope of the project, we will seek to utilise HESTIA technology components as a way to spark conversations and change around them.

Workshop process and group dynamics

Welcoming process (what happened when participants arrived, e.g welcoming, names tags, introductions to othersetc.)	Participants were given some time to mingle and then westarted the workshop by asking them to sit around the different tables.
Was there an introduction to the Hestia project in the beginning of the workshop? What did that involve? What were participants told?	A very brief HESTIA introduction only took place after the warm up activity, to not influence the participants' answers.

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Were the participants split in smaller groups? If so how many and what were the specifics (e.g members of the same household split or stayed together, how many per group, purpose of eachgroup etc.)	Yes, they were split into 6 tables, each representing a mini energy community. Besides a table of two of late-comers, tables had 4-5 participants. We asked members of the same household to stay together to include family dynamics in theactivity.
Did the workshop run according to plan? Were there any deviations from the original scheduleof activities?	Overall the workshop ran accordingly to plan. The first sessionof the core part took longer than expected and had rich discussions: we decided on the spot to reduce the second part and keep it informative. The final part was longer than expected because we took time to answer all the participants'
	questions and to provide them with a more in-depth overviewof how an energy community works and how it fits the Italian regulation.
What was the overall 'mood' of the participating group during the workshop? (e.g. Was it good and "cheerful"? Was it relaxed? Did the participants seem comfortable about the settingand did they appear to trust each other? Or was the participant not relaxed or did any "bad atmosphere" occur? (If the latter, please includea few thoughts on why this might have happened)	The overall mood was cheerful, people were curious. Considering that most participants had never attended an interactive workshop, we were positively impressed by their engagement. Some were hesitant at the start of the activity, but they all participated actively with extra explanations fromfacilitators and peer support. Most of the participants already knew each other: some had set an appointment at the workshop, others were surprised to see each other. Two tables included early adopters of PV inBerchidda: these people took a leading role in the table and plenary discussions. Some tensions were present at the end when two participantsquestioned whether they were "losing time" because their PVwas installed before 2020. This doubt is more related to Italian regulations and incentives than to HESTIA directly. We provided answers on the spot and we are working on a FAQ page in Italian on GridAbility's website.

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Please state 3 things that went really well interms of engagement	1. Leading role of PV early adopters: the first wave of PV adopters was present at the workshop, and without any indications from our side they took a leading role in sharing their experience with the others. By doing so, they made tangible the impact of behavioural adaptation on consumption. 2. Women voicing their opinion actively in the workshop: although women were a minority in the workshop, they werethe spoke-person in three out of the four tables they were present at. It was also clear from their exposition that they tried to channel the group discussion towards concrete outputs. 3. Everybody "played the game" although they are not used to participate in interactive workshops.
Please state 3 things that were challenging in terms of engagement	1. Having more women attendees: participating in the workshop meant for most of the households sending the malerepresentative. In this community gender roles are quite traditional, and this is a threat because we need to engage with the person carrying out everyday tasks and not only withthe "decision-makers" or "technicians".

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	2. On-boarding latecomers (they were also the oldestparticipants) 3. Closing on time

Comments/ reflections on the overall dynamics of participants during the workshop

(E.g In what ways participants engaged with the discussions, any challenges during the process, any positive remarks, did participants generally talk to each other or did they respond mostly to the facilitator? Did all participants participate in discussions? Or only a few? In the latter case: Indicate who?)

It was important for us to set a participatory mentality from the very start: this is why we opened with an individual activity forcing every attendee to express their view and feel on the same level as the others.

Some tables had more animated discussions than others, but overall there was collaboration and debate to fulfilthe requests. As a facilitation approach, we choose to intervene the least possible after explaining activities, only on demand. The workshop was useful for establishing an in-person connection with people and make them understand through a playful activity how consumption peaks and demand response work. Moreover, it allowedus to prepare the ground for future actions: the installations and the app. Last but not least, it created a first community moment around the topic of energy, which so far has been dealt with individually by households.

Each table had a spoke-person, but other people did intervene. Women were spoke-person of 2 tables. A table of late-comers was made by two women.

Summary of each workshop stage/planned activity

Stage 1: Warm up (please title the activity/stage)

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Duration	15’
Any probes / tools etc. used	Post-its and cardboards on the wall
Discussion starter	This activity was framed as a way to gather participants’ views on fundamental aspects of DR and activities to come, without influencingthem with our own.
Questions asked to participants during the introductory part of theworkshop	The first warm up activity was to answer individually on a post-it to 3questions: • What is an energy community? • What is the role of each individual household? • Which mean of communication would you prefer us to use tocommunicate with citizens?
Summary of activity	Participants have a good grasp of what an energy community is and are aware that technology plays a role in it. In their definitions of what an energy community is they show an understanding of acting in the common interest and of the overarching goal of sustainable consumption. They also showed knowledge that physical installations likemeters and mobile interfaces like apps are tools to reach this goal. For what concerns individual households’ role, many expressed the needto reduce energy waste to reduce overall community consumption, to betuned to PV production peak hours to exploit solar energy, and to use energy “wisely”, “consciously” and “responsibly”, meaning reducing overconsumption and tuning-in to community production . Emails and WhatsApp emerged as the favourite channels of communication with the project. However, some participants also highlighted the importance of having a physical person to address.

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Facilitation	The activity saw one facilitator explaining the questions and two collecting and hanging post-its. We gave a couple of minutes to answer each question. The post-its were collected by the facilitators and put on apaper board under the related question. We read some samples out loudas a general wrap up. The activity worked well, and it was an excellent way to get participants focused. The limited number of questions made it easy and approachableto everybody; it also allowed a wise use of time.
Analytical observations/reflections	We are in front of a group with good energy literacy. Owners of PV know how energy distribution works and how relying on renewables can positively impact bills; energy consumers have a good interpretation of the purpose of an energy community. As emerged from the answers to the second of the introductory questions, tThey seem aware of the fact
	that individuals' behaviour can contribute (e.g. reduce unnecessary sources of consumption, or consume at special times of the day): in the following phases of the project, we'll see if it is easy to turn this consciousness into action by monitoring whether DR suggestions are followed up by households, or whether it will request extra support (e.g. incentives, gamification, extra information).
Highlights/ quotes	Sample definitions of what is an energy community: "A virtuous community which aims at energy efficiency through practices functional to consumption optimisation" "It is a community that produces energy from renewables and exchanges/utilises it within the same community" "Energy production and exchange within the community"

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	"A set of meters that produce energy"

1.1 Stage 2: Energy community in action

Duration	1h30
Any probes / tools etc. used	Big cardboards; cards with appliances and everyday activities; markers;
Discussion starter	The session was introduced as a low-tech simulation of how DR works. People were incentivised to think of their real everyday habits, and to act for the community's best. In the introduction, we explained in basic terms what is a collective consumption curve.

Questions asked to participants during the introductory part of theworkshop	- Which are the most relevant energy practices in your daily routine - What is the consumption curve of your group?
Summary of activity	The activity was divided into two parts. /// PART 1 Individuals had to place their household daily routine on a 24h schedulein relation to when they use 3-4 recurring appliances. Looking at these data and with the help of a sheet with the kWh consumption of most common appliances, they were tasked with drawing the group (in our fiction, the community) daily curve. Finally, a spoke-person for each group had to report to the plenary explaining similarities among households and peak hours. The objective of this activity was to raise awareness in participants about the relationship between individual and collective consumption: we did not expect precise curves. Below, the main discussion points of each group. Group 1: Participants had all the PV. They commented how they already changed their habits from utilising appliances at night or during the weekend to exploit tariffs, and concentrating consumption (namely charging phones, clothes washing and dishwashing) during the day, especially in the morning. This switch seemed to be mainly a change of habits for these households as they either have one or more members of the family at home (housewives, or retired couples), or they are independent workers with some degree of flexibility in their time schedule). Group 2: In this group it emerged how housekeeping roles are gendered:3 male participants out of the 4 had difficulties in saying how energy is utilised during the day. They admitted they don't play an active role in housekeeping (but they were quite precise in detailing their energy- demanding leisure activities). The female participant said she concentrates most of the activities in the

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	morning.,g, before going to work. Group 3: the early adopters of PV were concentrated at this table. They shared their experience of adapting to PV production, a learn-by-doing process that they fine-tuned over time, by monitoring consumption at least monthly and learning more about the consumption of individual appliances. They also highlighted their joy in seeing the energy bills godown and how this was motivating. Otherwise, the content of their

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	experience is similar to what reported for group 1. They also stated theimportance of having a battery to increase their self-consumption. Group 4: In this group, the female spoke-person highlighted how she wasthe one "consuming more" because of her housekeeping tasks, concentrated mostly in the afternoon/evening. Other PV owners stated they shifted consumption during the day. Group 5: Spoke-person made a "historical excursus" on how once upon atime there were not many energy-demanding habits and people were doing economies, while today people have less consciousness and wastemore. He was the only retired, so he had a good grasp of everyday consumption, while the others tended to leave the task to women. Everyparticipant of this table had PV. Group 6: participants admitted not to have a particular eye for consciousconsumption. /// PART 2 In the second part, the plan was to give them real-life solar production curves, and they had to discuss how to adapt their energy-reliant habitsand practices to stay in the curve. In the process, with support from us, they had to discuss how to adapt habits, who does what, what is a no go… Because of time constraints and people's interest in getting more information, we decided to stick to the informative part and explain how PV production curves and DR work, leveraging in the narrative the previous activities. We explained how PV produces accordingly to irradiation, and how the challenge of the group will be to make the best out of this production. We also explained the

	importance of data to steerDR, highlighting how with more data in hand it is possible to understand where consumption can be optimised, and that sometimes the answer is less straightforward than one may thing (we made the concrete example of electric boilers, and how there are two schools, keeping it always on or turning it on when needed. The answer is that none of the two is the right behaviour per se, it depends from the model).
Facilitation	After the introduction, we let the groups work for 30 minutes. We were available for questions but preferred not to interfere with individual

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	groups dynamics. We then moderated the plenary moments with questions and kept the time. Next time we could take more time at the beginning to explain the activity, since some groups had a slow start: they weren't sure of thelevel of detail needed. Overall the gamified approach worked: it was a good way to increase participants' awareness by putting them into context and let them see how individual households consumptions cumulate, and the "behind the scene" of an energy community. It was instrumental in creating expectation and interest around the creation of an energy community, and in building the mindset for behavioural change.
Analytical observations/reflections	There is a literacy "divide" between PV early adopters and simple consumers that we will need to consider in further phases: PV owners know best how energy production and distribution work, the incentivesprovided by the government and the savings made possible replacing dependency from energy providers with self-consumption. As highlighted already, gender roles attributing to women most of the housekeeping and care work will need to be addressed when setting upactive DR.

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Highlights/ quotes	Recurring start phrase: “Maybe it’s because I am a woman, but my everyday consumption is much higher than the one of my table companions…” (implicit: because she is the main user of housekeepingappliances” “New generations are energy-demanding, they leave everything on all the time with no consciousness of its cost” “A battery would increase my capability to rely on self-consumption: despite the reimbursements I obtain from selling energy to Enel, I thinkit’s more relevant to make the best out of solar energy”.

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Stage 3: Interface

Duration	45’
Any probes / tools etc. used	Projected demo of Prosume’s interface
Discussion starter	The activity and the information delivered created the ground for showing people how the user interface of a DR app works. We utilisedProsume’s interface, which is in Italian and already tuned to Italian legislation.
Questions asked to participants during the introductory part of theworkshop	//
Summary of activity	We introduced the main parts of the app and what is possible to visualiseas consumers, producers and as a community.
Facilitation	This was a one-way presentation with a Q&A at the end.
Analytical observations/reflections	Despite being the last part of the workshop, people were very attentive,probably because they could see something tangible. They were particularly interested in the payment screens.
Highlights/ quotes	People were particularly interested in the conjunction between HESTIAand Italian incentives for renewables. People with PV reiterated their experience with adapting to solar production.

Photos

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Appendix 6: Workshop programme, Camille Claudel

Hestia



Projet HESTIA sur le Quartier Camille Claudel
Déroulé de l'atelier du 16 octobre 2021

Rappel du projet

Le projet européen **HESTIA** vise à étudier la manière dont il est possible et acceptable de modifier les pratiques quotidiennes de consommation à domicile pour diminuer et moduler les consommations d'énergies et pouvoir favoriser le déploiement d'énergies renouvelables.

19 partenaires européens, dont EDF et l'Agglomération Paris Saclay, avec le soutien de la Ville de Palaiseau, participent à ce projet.

Ce qu'on a fait le 16 octobre dernier

L'atelier s'est déroulé en plusieurs temps, à savoir :

- **L'accueil et l'introduction**, pour présenter les énergies renouvelables locales d'aujourd'hui et à venir et les enjeux de réduction, lissage et flexibilité (à l'aide d'affiches)
- **Un jeu d'introduction « Green Mirror »** pour comprendre les enjeux et les utilisations liées à l'énergie et pouvoir se projeter
- **Un atelier « actions »** : de questionnements de certaines actions de la vie quotidienne (baisser son chauffage, faire sa machine et préparer ses repas aux heures d'ensoleillement, chauffer en dehors des pics de consommation) et identification des freins et leviers
- **Idées pour changer**, afin de faire découvrir des pratiques alternatives et inspirantes permettant d'initier une réflexion, des débats et des changements
- **Parcours à trou** : pour interroger ce qui peut être mis en place et les motivations à long terme



Les participants

10 participants, dont des étudiants, accompagnés de 3 enfants. Des participants motivés par les enjeux de l'écologie et investis dans la vie de quartier.

EDF Palaiseau PARIS SACLAY Communauté d'agglomération

This project has received funding from the European Union's H2020 program under Grant Agreement No. 957823.

Appendix 7: Summary and analysis of workshop in Camille Claudel

Template – workshop summary

For each of the WP2 participatory workshops, the local site partners/facilitators should prepare a(short) English summary and send this to AAU. The summaries should follow the same basic structure in order to inform the final analysis. The template (presented below) outlines different sections and briefly describes (and give examples of) the focus and intended content of each section. If you have any questions, please don't hesitate to contact.

The aim of the summaries is to document and report some preliminary observations and findingsfrom the conducted workshops. As each workshop differs contextually and has been carried out in three different languages, we would like to gather some insights that can be comparable, thus opening up for a comparative analysis of the three different workshops. Taken together, the summaries will also be an input for the overall analysis (reported in D 2.2.) which aim to inform the design of the HESTIA interface (T 2.3 and T 2.4).

In the template, the first two sections will concern the basic/descriptive information of the workshops (e.g. number of participants, number of households, gender representation), makingit possible to draw out methodological limitations and evaluate the validity. Next, we focus on your methodological reflections, asking you to outline how the workshop went, describing the engagement, 'mood' and potential deviations from the planned setup. Third, we ask you to
highlight some of the 'most interesting' discussions from the workshop discussions. Here we suggest that you summarise each 'stage'/planned activity of the workshop. We would both liketo get 'raw data' i.e. summaries of what happened and what was being said, but we also hear your analytical observations from the workshop.

As D 2.2. is expected to inform the design of the HESTIA interface/dashboard (developed by i.LECO), please also consider including feedback from the occupants that directly concern the design of such (e.g. what features was liked, how occupants prefer to get energy feedback etc).

Finally, please feel free to attach in photos or videos that you might have obtained during theworkshop, and which you think illustrate a certain point – that being both methodological or content related.

The summaries should be written on basis of the audio/video recordings and analytical observations taken during the workshop.

Ensure that all participants are anonymised in the summaries. I.e. do not use their real names and take also care not to include personal information that would make it possible for others toidentify them.

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Hestia
Holistic demand response services
for European residential communities

Summary and analysis of workshop in Camille Claudel, Palaiseau, France



Workshop organisation

Workshop day of the week and date	Saturday 16th October 2021

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Workshop time	10:00 – 12:30
Workshop duration	2:30 h
Venue	City of Palaiseau "Salle de la Rigole" municipal hall
Incentives/reciprocity offers for participating (ifany, e.g dinner)	Buffet for lunch at the end of the workshop
Hestia partner in charge	EDF & CPS
Other Hestia partners present and their roles	User studio (Design company) in charge of workshopanimation for EDF City of Palaiseau
Comments/reflections about the overall organisation of the workshop (positive comments about organisation and/or challenges about organisation) In spite of some challenges to recruit, the workshop went very well, in a good atmosphere with very involved participants	

1. **Participants and recruitment**

Number of participants invited	20-30 directly contacted during the recruitmentprocess (described below) 13 people confirmed their intention to participate

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<table>
<tr><td>Number of participants that turned up</td><td colspan="2">11</td></tr>
<tr><td>Recruitment of participants (how they were invited to the workshop e.g personal invitation,
letter, email etc.)</td><td colspan="2">Several recruitment channels:

• Communication in the monthly journal of the Municipality of Palaiseau + CPS's journal
• CPS website
• posters in the district's buildings, shops and bus stops
• Direct contact on 4th September during the "Associations Day"
• Face-to-face recruitment (in front of school, market),
by the social landlords and direct contact with "building council"
• Phone contact with respondents from previous surveys
• inside CPS</td></tr>
<tr><td rowspan="2">Gender distribution:</td><td>Female</td><td>Male</td></tr>
<tr><td>6</td><td>5</td></tr>
<tr><td colspan="3">Number of households present 3</td></tr>
</table>

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Comments/ reflections on participants' recruitment and turn out (e.g representation of age, gender, overall turnout of the specific community, practical issues that affected the running of the workshop and the participants'engagement etc.)

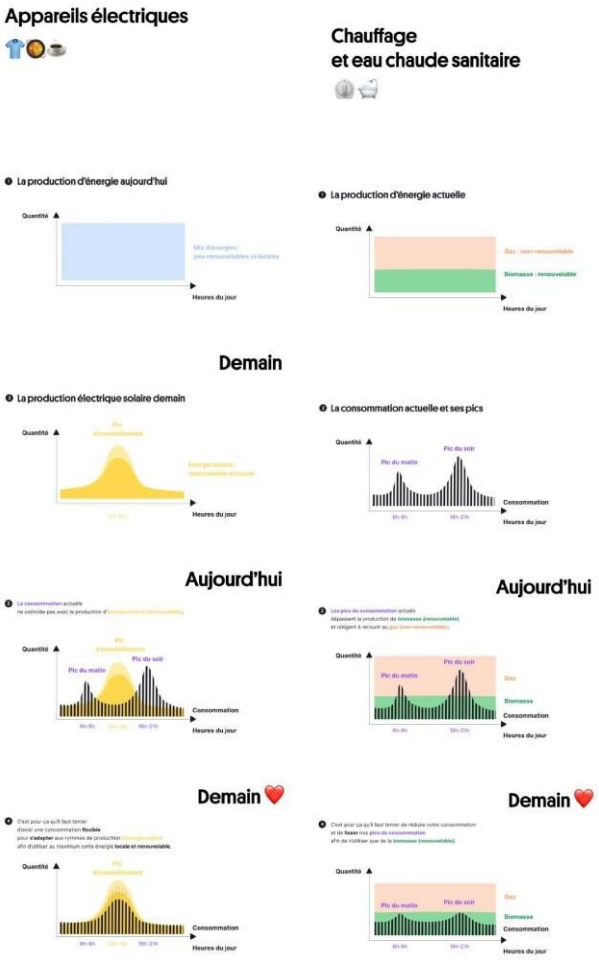
11 participants from 20 years old to around 50 years old. 3 students (physics studies) among the participants.

Several people who were available for the first workshop date (Sept 28) were unfortunately not available at thesecond date. Many solicited people were not available on weekends either. However in the end, we had a good turnout of motivated people with a good gender balance and representation of families with children: some parents came with their children, a table was dedicated to them and Anthony (CPS) took care of them.

The people present were motivated by ecology. They are also involved in the life of the neighbourhood (associations, co-ownerships,...). Some of them have been involved previously in the Hestia project through the questionnaire and some of them were interested in becoming ambassadors for the project and talk about it withtheir neighbours or acquaintances.

Workshop process and group dynamics

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Welcoming process (what happened when participants arrived, e.g welcoming, names tags, introductions to othersetc.)	Participants were welcomed over coffee/tea, and asked to sign the consent form for the use of their image on photos. The User Studio team in charge of facilitating the workshop asked them to choose a card indicating their household profile(single, family…), that would be used to split the participants in homogeneous groups. Informal discussions took place while waiting for all participants to arrive and the workshop to start. Someparticipants already knew each other. During this welcome time, participants were proposed to havea look at two posters pinned at the wall, aiming at presenting high-level principles of energy consumption and production, to highlight the issues at stake and the associated possibiles (peak shaving, load shifting…). 

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Was there an introduction to the Hestia projectin the beginning of the workshop? What did that involve? What were participants told?	Mathieu (EDF) introduced the workshop with a short presentation of the project (European project, duration, others countries involved, goals), a very first basic introduction on flexibility and the questions raised by the integration of renewable energies. He also mentioned the phases of the project (1/ Baseline: installing equipment for understanding consumption and practices, developing solutions and interfaces ; 2/ Experiment: asking people to act and record their impact). He also mentioned that during the whole project, we wanted to create a community of motivated participants, and would work on the best way to communicate during the experimentation. He also mentioned the fact that the project will use simulation, especially regarding PV models and it aims to bereplicable in other situations
Were the participants split in smaller groups? If so how many and what were the specifics (e.g members of the same household split or stayed together, how many per group, purpose of eachgroup etc.)	The participants were split into 2 groups of 5 for the Green Mirror introduction, and then 3 groups of 3-4 for the following phases. The groups were constituted so that each group had balance of gender and household situation (family with children, student…). A couple participated to the workshop together (the wife thought that she would had to leave earlier, but finally stayeduntil the end).
Did the workshop run according to plan? Were there any deviations from the original scheduleof activities?	The workshop ran according to plan, and a few people stayedat the end to continue the discussions. However, most had toleave for lunch or afternoon activities.

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What was the overall 'mood' of the participating group during the workshop? (e.g. Was it good and "cheerful"? Was it relaxed? Did the participants seem comfortable about the setting and did they appear to trust each other? Or was the participant not relaxed or did any "bad atmosphere" occur? (If the latter, please include a few thoughts on why this might have happened)	The workshop mood was very good overall, participants were relaxed and interested in the workshop's topics, and interested in talking about it. There were no "awkward silences". Some people knew each other. For many, it was their first time in the municipal room of the district.

Please state 3 things that went really well interms of engagement	1. All participants were involved, talked, asked questions for clarification, made propositions 2. They easily shared their point of view 3. One person had to leave before the end of the workshop and arranged with her partner to stay until the end. Anotherwould have had to work but arranged with her work to stay until the end as well, which is a sign of strong interest and engagement -> Was it the same person ?
Please state 3 things that were challenging in terms of engagement	1. Nothing in particular happened that made the workshop challenging in terms of engagement 2. 3.

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Comments/ reflections on the overall dynamics of participants during the workshop

(*E.g In what ways participants engaged with the discussions, any challenges during the process, any positive remarks, did participants generally talk to each other or did they respond mostly to the facilitator? Did all participants participate in discussions? Or only a few? In the latter case: Indicate who?)*

The poster presenting the use cases allowed participants to quickly and visually grasp the issues at stake (consumption reduction, flexibility), and the differences between local energies (solar, biomass). These posters triggered questions and discussions among the participants, some participants explaining the concepts to the other participants.

"I'm between two heated flats, so I don't turn my heating on." "Is biomass really ecological?"

The participants want to understand how the local energy system work at present (interest in a visit of the boiler/heating network control room), how their building/home is heated (exposure, insultation…) and have access to analyses of energy consumption (they want to know for example if their heat consumption and comfort comesfrom the insulation or from the neighbour's heating).

A "state of play" of their basic understanding of the situation is necessary to get used to the subject.

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Summary of each workshop stage/planned activity

Stage 1: Green Mirror game (Anticipate future experiences of renewable energy : a playful and participatory simulation tool for futures exploration)

Duration	30 min
Any probes / tools etc. used	"Green Mirror" Board Game displayed on 2 tables
Discussion starter	Session initiated by Zoe Bonnardot (EDF PhD position in design and ergonomics), designer of the game.
Questions asked to participants during the introductory part of theworkshop	No particular questions asked except to listen to the game rules and playthe game!
Summary of activity	**Setting the scene** using the Green Mirror board game designed around practices and appliances used during a day + sharing and storage of renewables energy. Making people project and express themselves, encouraging discussions by placing people in situation with the help ofa board game For the participants, the objective of this part of the workshop was: • To understand the issues at stake related to energy consumption, production and sharing • To remember a real day of practices and appliance use • To project oneself in an energy context including photovoltaic energy (and later biomass)

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Facilitation	One game leader per table. The objectives of the game were achieved: to put the participants in a situation, to make them project themselves in a future energy situation and to facilitate exchanges. It was only a "warming up", but with more time, another round could have been played, with different outputs as people would have had reflected on theenergy questions before and, from experience, would have proposed alternatives and new options for energy use and sharing. The lack of time to keep on playing was a bit frustrating for all (organizersand players).
Analytical observations/reflections	Although played for a relatively short time, the game showedits powerful capacity : • to figure energy consumption by reflecting on a real day (they day before the workshop, a specific Friday) • to think quickly essential questions, such as thevalue of energy, proportions, the gap between production and consumption, the effect of energy storage,... Participants talked to each other, participated in discussions, talked abouttheir current practices or took the initiative to propose some advice during the game, e.g. *"you could have sold them [the chips representing locally produced energy] for more".*

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Highlights/ quotes	Examples of participants' comments during the game: - *"they [the game designers] forgot some "smartphone" cards""*... a participant would have liked to have more “smartphone” cards to showall the moments when they charged their smartphone during the day. - A participant apologising for not being able to participate much in theuse of energy during the day: *"I didn't do much, I wasn't at home, I was at the office [that Friday]".* - At one table, of the two energy producers, only one had the idea of selling her energy ("*I'm going to sell it*"), the other was offering it for free (a more 'collective interest' oriented posture). The participants did not spontaneously think of negotiating the price proposed for the sale either (something that can appear more naturally in a second round, but there wasn’t enough time for a second round).

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	- Comment from a satisfied participant with a "business/nego/sales/individual interest" orientation: *"you buy, you have the money in your hand" (talking about the game), "it's concrete, you 'see' your electricity"*.

Stage 2: Action cards

Duration	20 minutes
Any probes / tools etc. used	Action cards, one by group: - "Lowering your heating setpoint by 1°C" - "Heating outside of peak consumption (6-8h/18-21h)", - "Doing your washing when it's sunny", - "Preparing meals when it's sunny"
Discussion starter	Each game leader presented the card action that the group had to fill in.For each situation, the game leader asked participants to express their opinions, current practices and possible ones, barriers, etc…
Questions asked to participants during the introductory part of theworkshop	
Summary of activity	In this part of the workshop, the participants were asked to fill in a card associated with a possible energy efficiency or flexibility action, and explain why it would be easy or not easy to perform (why, what would bethe condition, and detail their family profile (solo, couple, family with children, retired (solo or couple)). Example:

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	 The objective of this part was: • To study the feasibility of proposed energyefficiency or flexibility actions. • To classify the actions according to theiradoptability • To understand the reasons if these actions wereidentified as difficult to perform • To identify levers And to initiate a reflection on the next steps
Facilitation	Activity facilitated by User Studio
Analytical observations/reflections	The exchanges during the game showed that domestic habits/practicesvary: - during the week and at weekends: *"We don't consume the same at weekends*".

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	- with the development of teleworking (an element that came up severaltimes => new flexibility lever) - with the presence of children *Man: "On teleworking days, we shower a bit when we want. We can takea shower later in the morning but we don't take a shower in the evening.* *Woman: "Don't you have children?* *Man: "No"* Other findings: - Actions that seem simple for some will be more difficult forothers. - Differences in practices between weekdays and the weekend, especially for working people, students and parents. - Telework is an important change that will last and could be a newlever for flexibility - People's profile have an important impact on the organisation and practices (e.g. students who do laundry in a common laundryroom) - Within the same household, the agreement of the othermembers is a prerequisite - Couple/family life: Important that the whole household is involved. If only one person is experimenting this can slow downor even distort the experimentation. - Gendered division of labour

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Highlights/ quotes	• "Lowering heating setpoint by 1°C" group Easy (7) - Couple: *"Easy with help of the thermostat and doesn't impactcomfort that much"* - Couple: *"We prefer not to heat too much"* - Family with 4 children: *"We have a good thermal insultation, andcan put on a pullover if needed"* - Solo: *"1°C doesn't make a big difference, except when winter isreally cold"* - Solo: *"It's possible to adapt with thicker clothes"* - Solo with kids: *"Seems possible as it's only 1°C. I'd do it if I hadthe possibility to control my heating"* - Family with children: *"Our building is already quite hot"*

	Not easy (2) - Family with 2 children: *"I'm already at 19°C during the day and 17°C during the night"* - Solo: *"I'm less productive when I'm cold"* • "Heat outside of consumption peaks" group Easy (5) - Family with 4 children: *"Controllable thermostat and good thermal insulation"* - Family with 2 children: *"It's not too cold in the morning*" - Solo: *"It's colder later in the evening. I can move more or have a more active activity until 21h"* Couple: *"We almost never heat, and we are often at home during weekends, so it is possible to heat during the day"* Not easy (4) - Family with children: *"We have to heat the bathroom morning and evening for the children"* - Solo: *"It's during these [proposed] times that it's cold"* - Family with children: *"Not controllable"* - Solo with children: *"I don't control the thermostat in my flat"* • Preparing meals with the sun" group *"Preparing meals with the sun: weird!"* Easy (7) - Family with children: *"[Possible] to prepare meals in advance"* but *"Need to have time to manage both children and cooking"* - Family with 2 children: *"Depends on the children (2 & 5 yo)"* - Couple: "*Possible but not all meals: anticipate 5/6 meals for*

	the week". "*If the weather forecast is reliable (two days before) to be able to set up a plannin*g" - Solo: "*If I have time*" - Solo: "*Meal for lunch and sandwiches or salad in the evening*" "*if I'm home*" - Solo: "*Possible to prepare meals during the weekend for the rest of the week*" - Family with 2 children: "*If* there is sun!" "*Madam has an appropriate planning*"

	Not easy (3) - Family with 4 children: *"I don't like to cook"* - Couple: *"For most meals, there is a need for cooking on the spot.But we could prepare 1 or 2 in advance."* - Solo with children: *"3 children + lots of work. It is difficult tocombine sun and available time"* Other quote: *"I love to do everything at the last moment but I know I am using a lot of* *energy"* Considered possibility of "preparing meals with the sun" quite related tothe presence of children. Idem for "staggering meal times". • Group "Doing your washing when it's sunny". Easy (6): - Family with 4 children: *"Working from home (2-3 times per week)* *+ programmable washing machine"* - Family with children: *"We have enough clothes to wait"* - Family with 2 children: *"Programmable washing machine"* - Solo: *"Need to be available and have time"* - Solo w/ children: *"Using washing machines when working from home"* Not easy (4):

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	- Couple: *"Ok if at home, but not if the clothes stay for 6 hours in the machine. Need to have an automatic/programmable machine."* - Solo: *"Student busy all day"* - Couple: *"Active professional life that doesn't allow to come home early or plan the return time. Possible is programmable machine"* Family with 2 children: "*Too much laundry to do*"

	Staggered use between 12-4pm with the sun/ washing machine: *"with teleworking, I can spread the machines out all week, whereas I used to run all my machines on the weekends before "* - family with 4 children. 1to 2 machines per day. Drying in the sun? *"No, my balcony is on the top floor, so it's not protected, I don't take therisk of leaving the laundry outside."* Running the machines between 12 noon and 4pm, when you get homefrom work around 7pm: *"laundry that stays in for 3 hours smells bad"* (and then, a grandma'sadvice: *"you should run a machine with vinegar to remove the odours"*...the convivial side of the exchanges...). The participants think that they make the best use of peak/off peak electricity tariff. And preferring the sunny hours is penalizing for them (they don't have a ticket). They would perhaps like a financial incentive toshift their use according to the sun.

Stage 3 : Ideas for change

Duration	20 minutes
Any probes / tools etc. used	Cards presenting practices with a short description
Discussion starter	At each of the 3 tables, the principles of the practices were presented before asking for the participant's reaction.

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Questions asked to participants during the introductory part of theworkshop	
Summary of activity	In this part of the workshop, participants discussed practices that are realbut also challenging and possibly inspiring (for instance: doing batch cooking, investing in a solar oven, etc…). The objective of this part was: - To make people aware that practices that are not unrelated to energycan contribute to sobriety or flexibility - Initiate changes and debate on practices - Helping to get out of the box - Identify a way to accompany experimentation
Facilitation	Activity facilitated by User Studio
Analytical observations/reflections	*The participants were seduced by these first inspirations. There is a need for guidance and inspiration to change habits.* *The desires and needs are different within the same family.*

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Highlights/ quotes	*"Getting inspirations like that is good, the cards appeal to me."* *"I love to do everything at the last moment but I know I am using a lot of energy"* One card suggests: "Invest in a solar oven", one participant indicates thatshe already has one, made by her husband (with sheep's wool); idem for the "Lacto fermentation" card, two participants are already doing this => profile sensitive to environmental issues and Do It Yourself. *"My husband won't switch to fermented food. I'm already trying to introduce him to other kinds of [fermented] foods and he doesn't want to try."*

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	On the other hand, less "attraction" for "freeze-dried meals" and "drink meals": feeling of being disengaged because of the absence of cooking gestures: *"We don't like to cook and we're not going to get the children used to it".* Women more enthusiastic than men. The family who moved to Camille Claudel because of their ecological convictions were delighted to have new "tricks" to consume less (they will try lacto-fermentation and will find out about the Norwegian pot). *"Batch cooking, I was only thinking about it only for the organisation and I wasn't tempted. But for changing my energy consumption, I'm interested."*

Stage 4: Gap course of a usage scenario

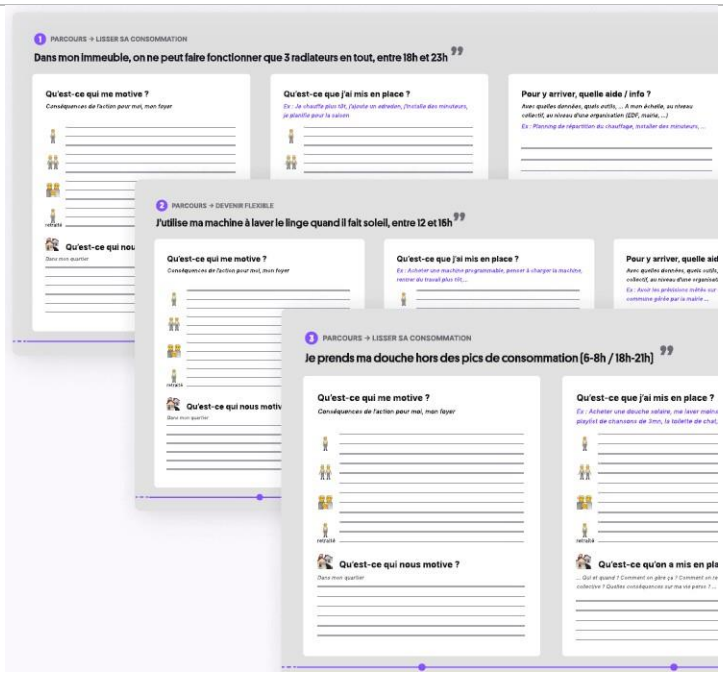
Duration	1 hour
Any probes / tools etc. used	3 main "roadmaps", each with a proposal: - Smoothing out consumption: "In my building we can only run 3 radiators in total between 6pm & 11pm" - Become flexible: "Use the washing machine between 12 & 16 hours withthe sun" - Becoming flexible: "Taking a shower outside of peak consumption times"
Discussion starter	*Please state how the session was initiated*.

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Questions asked to participants during the introductory part of theworkshop	(See summary of activity) Participants should indicate for the proposition they had to deal with : - what motivates them to respond to the request

	- what they will do - the useful aids/info to achieve this - their reactions in case of a "problem": e.g. neighbour who does not respect the radiator cut-off or cloudy weather to run the machines duringsolar production times
Summary of activity	"Extreme" usage scenario to be designed collectively. 3 main use case"roadmaps": - Smoothing out consumption: "In my building we can only run 3 radiatorsin total between 6pm & 11pm" - Become flexible: "Use the washing machine between 12 & 16 hours with the sun" - Becoming flexible: "Taking a shower outside of peak consumption times" The objective was to: -> Collectively question what can be put in place and with what help -> Understand the barriers and sources of long-term motivation -> Laying the foundations for real scenarios
Facilitation	*Facilitated by User Studio*

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Analytical observations/reflections	*[Mathieu: here I summarise the result of the activity, i.e. the content produced by the participants themselves]* <u>Use Case 1 - Smoothing out consumption: "In my building we can only run3 radiators in total between 6pm & 11pm"</u> The participants described a concept of building-level organisation formanaging heating at a collective level. Why we do it? -> *"We hardly ever heat up, so there is little impact [on us]."* Motivation toconsume less and to do something for the planet. -> No problem to share / alternate between residents. *"We* *heat little or not at all in some rooms." "There are no small actions."* -> To be in line with the values of the eco-neighbourhood. The future of children. Solidarity between neighbours. Financial aspect

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	How? -> Define priority persons for heating (elderly, sick, children, etc.)

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	sick, children...) -> Define who can turn on the heating according to the presence in thehouse -> At the collective level, union council with a global consent -> Co-owner general assembly to be triggered to define the rules to be putin place With which tools/information? -> App to collect everyone's timetable + remote control of heatingheating remotely -> Access to weather forecast -> Know the exact temperature in the dwellings -> Know the consumption of the dwellings (according to exposure / insulation) -> Raising awareness at the town hall level, via interventions (includingartistic interventions...) And if it doesn't work? -> Financial incentives -> Need for a A mediator Reactions: -> Sharing of data and information: *"This goes a bit far.""To what extent is the information visible to everyone?" "It's a bit intrusive to know my lifestyle habits."* *"You have to anonymise! Today, we don't know how to do that with individual level data."* <u>User Case 2 - Become flexible: "Use the washing machine between 12 &16 hours with the sun"</u>

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	Why we do it? Consuming green energy and avoiding waste of energy already produced How? -> Implementing financial incentives such as the principle of off-peakoff-peak hours -> *"We must motivate the public actors + the co-owners' general assembly to install solar equipment."* With which tools/information? -> Raise awareness among residents at the level of "public places" in thedistrict -> To have a view of their energy consumption and savings made(energy and euros). -> Alert/message on the appliance when you consume at the right time And if it doesn't work? -> Wash by hand -> Postpone the machine for one day Reactions: -> Put your laundry in the janitorial service, which works when there issunshine -> Difference between weekday and weekend time slots? Possibility of extending the time slots at weekends? <u>Use Case 3 - Becoming flexible: "Taking a shower outside of peak consumption times"</u>

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	Why we do it?
	-> Ecological motivation, need to be able to measure real (not theoretical) impact, to be able to monitor progress over the long term. -> Have a real impact on the climate, so have a quantitative return on the impact at the neighbourhood / group / individual level -> Financial impact How? -> Change the time of the shower -> Shift the time but also reduce the duration or frequency of showers With which tools/information? -> Access to boiler/heating network data. *"We don't know these curves as of today." "If it were in real time, I could say: OK, I can wait an hour."* -> The general consumption curve of the building -> Individual consumption of the neighbours (how much does the neighbour's shower cost) (!) -> information on peak of water use at the collective level. *"We are not aware of the influx of people, so we could take a shower because no one is taking one!" (a*nd therefore be able to shift the time of water use) -> Consumption rates (have a price associated with each hour?) -> CO2 impact And if it doesn't work? -> Present things as a voluntary action: each person at his or her own paceat their own pace and according to their own constraints. *"I don't want to make an effort that benefits to someone else."*

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	Reactions: Questions about the "buffer" type hot water tanks: *“Is this an Efficient system?” “Should we opt for larger equipment?”*

General findings:

Motivations to act/ perceived levers to encourage commitment :

- The “collective”, in this case the neighbourhood, is a stimulus to action but must not be exclusive and need for anonymised data [=> privacy issuecame up several times]

- Act with others "Residents are not the only ones who can act [alreadynoted in previous experiments]: "There are also shops and a swimmingpool in the neighbourhood".

- Communication, awareness-raising, financial incentives (such as off-peakrates) for the "most recalcitrant

o Awareness-raising through posters in the condominiums, shops & via theCamille Claudel Facebook group (receive incentives via this group), via awareness-raising moments, a known/trusted person who shares his or her experience, who acts as a relay for what is good to do

o Provide benchmarks: inform about consumption, inform aboutconsumption peaks in the neighbourhood

o Need to have reminders, alerts such as "Well done, you are within therecommended time frame".

o Need to have smart machines (which indicate that the machine has beenused at the right time "congratulations, it's the right time to use...")

o Have differentiated price ranges "Today we don't really know how theprice is calculated

⇨ [Need for feedback, benchmarks, solicitations].

The perceived benefits of the actions proposed during the workshop:

- For the environment

- For the district: to give a positive image of the district, to enhance itsspecificity as an eco-district

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	The perceived risks, fears/reservations expressed : - linked to privacy - that these practices of moderating consumption represent efforts for individuals and that Dalkia (heating company) recovers the benefits => signof mistrust of the economic actor [already noted in previous experiments]

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	Summary, as a conclusion: - The initial motivation is the planet, but if some people don't doanything, then the financial aspect can take over to allow the effort to be sustained over time and prevent those who don't make an effort from reaping the same benefits. - Volunteering is essential to recruit participants - The workshop participants felt that the neighbourhood lacked community life. They want more activities. The Hestia project could be a good relay. - Although it has been awarded the Eco-district label, the participants feel that the Camille Claudel district does not emphasise neither the ecology nor the commitment of the inhabitants or the elected representatives. Here too, the Hestiaproject could be a good relay. - For control/heating management and feedback, the participants seem to prefer a mobile app. - Telework is an important change that will last and could be anew lever for flexibility - Some participants are well aware of how their dwelling works, while others are not at all. The ways of controlling the heating also seem to vary according to the promoters. This information will have to be collected prior to the experiment. - Although willing to share data, participants will need safeguards to be confident. What is the limit to sharing that ensures understanding while respecting privacy and security?It will be necessary from the moment of recruitment communicate on the security of the data and reassure on the experimental aspect of the project. - All the participants said that it was important to have precise measurements (actual temperature, boiler room consumption, hot water). It will be necessary to give feedback to make the efforts made in terms of consumption visible.
Highlights/ quotes	*Included in the previous section*

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Photos



Figure 1 : Place where the workshop took place

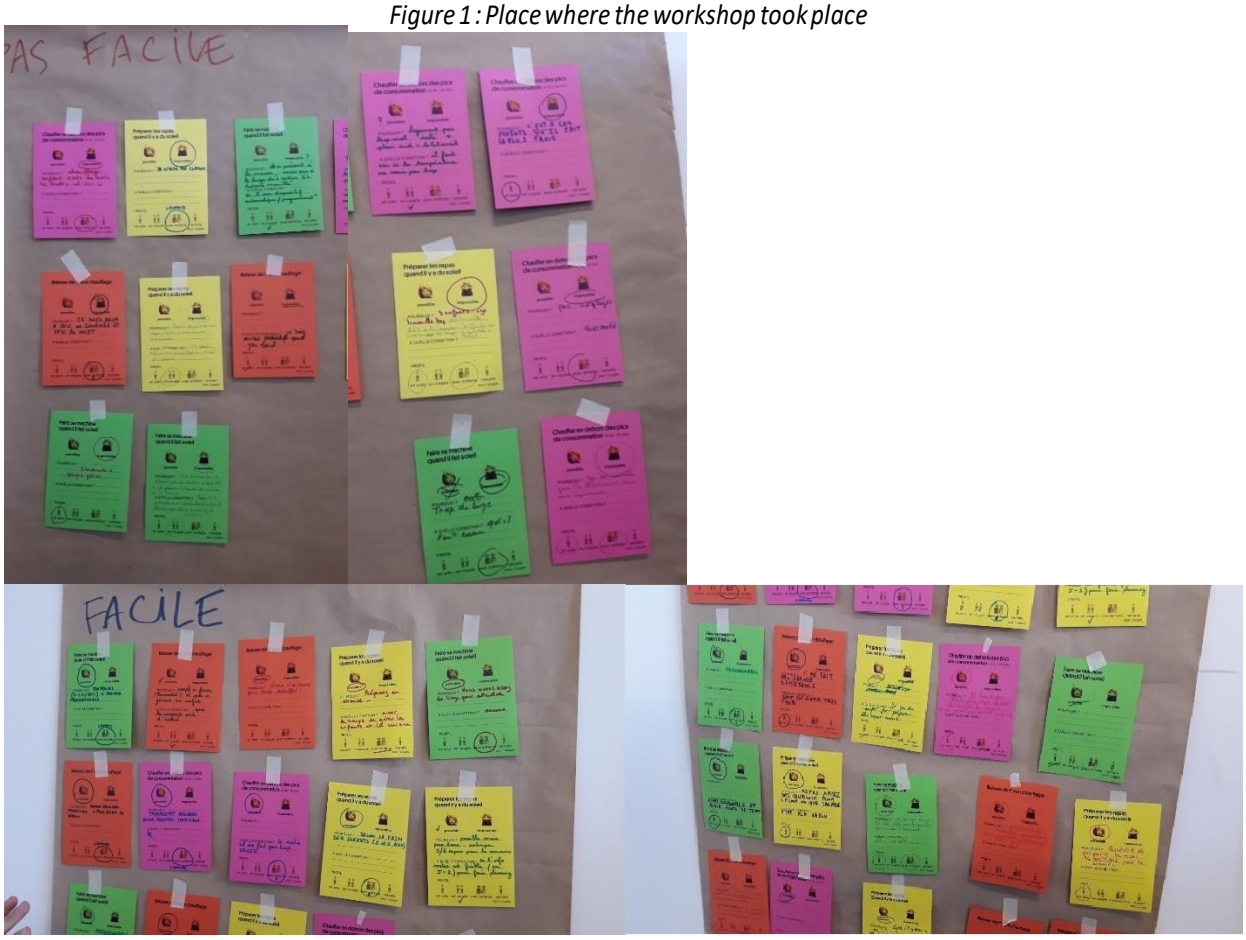


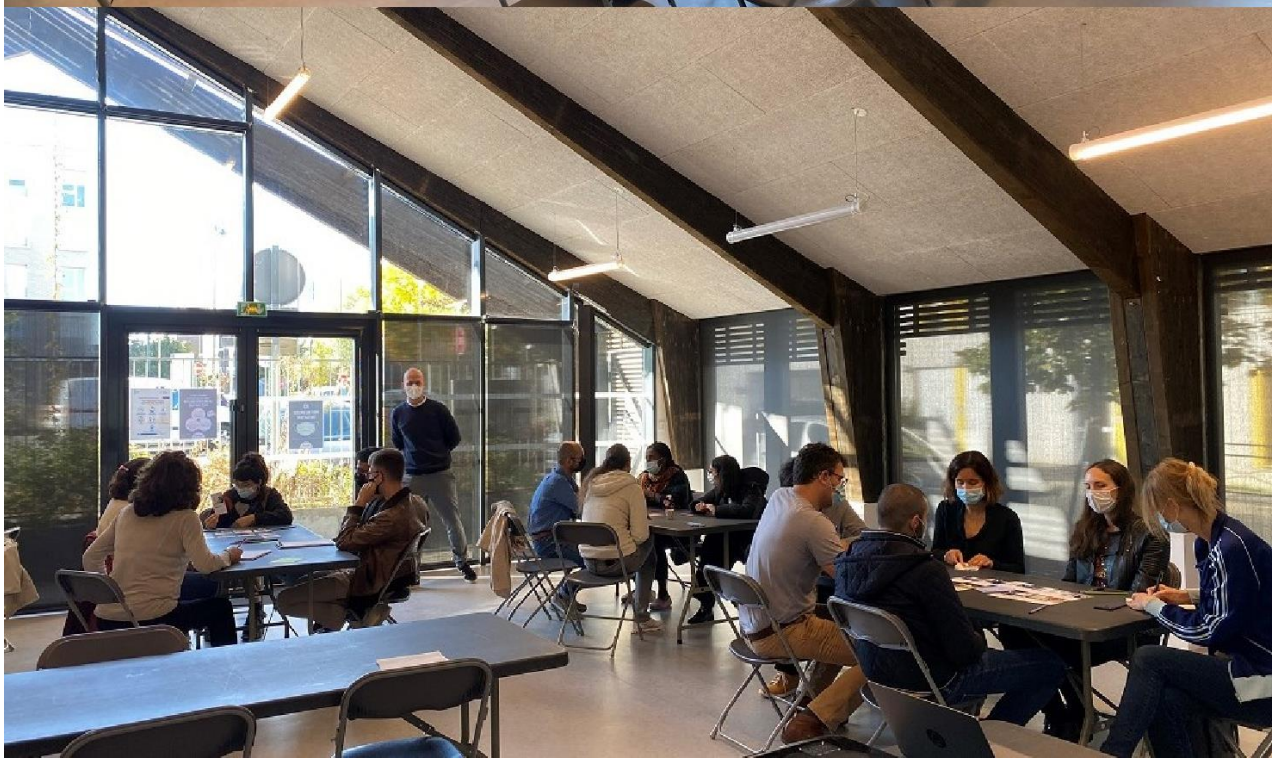
Figure 2 : different points of view on how hard habits are to change

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Appendix 8: Interview-guide used for doing virtual interactions in Voorhout

Introduction

The purpose of the interviews is to be a bridge between the survey and the workshops. They allow some deepening of the initial insights from the survey, but most of all, they should make meaningful workshops in September possible. Incidentally, this is also why they shouldn't take too long, because the engagement with the project should not be felt as a burden. The interviews are therefore simply part of a series of interactions. As a result, each interview doesn't need to be a universe contained in itself, with a holistic view of the participants. Nor do the interviews need to be comparable to another. There is no scientific ambition to compare how different people respond to similar stimulus or anything. Instead, they function more according toa design paradigm, intended to give a functional image of the users, their needs, experiences and skills. Functional in the sense that it allows the (co-)design process to move forward. In a word, then, they're an exercise in stock-taking.

Voorhout interviews

Concretely, in Voorhout, that means that we are doing two interview types: one with a focus on residents' experiences in Voorhout, and the other with a focus on possible "use cases". This division allows us to keep individual interviews to 45 minutes, while still covering all necessary topics globally. The first type of interviews will serve primarily to get a feeling for a) the neighbourhood and (the potential for) community, b) the influence of the (malfunctioning) net-positive homes on their (embodied) awareness of energy issues and flows as well as experience of comfort. The second type will serve to discuss the possible use cases for Hestia in Voorhout and what that could imply for automatic and behavioural DR. That discussion includes the values people want to see drive the platform and initial estimations of feasibility or desirability of certain DR actions. For now, we'renot planning on zooming in on one particular practice, because we simply don't know yet which practice they will be required to change, if any.

Note that while the focus of each of the interviews is different, there is space for fundamental themes and concerns of ours to surface in each type. Thus, while comfort will be explicitly on the agenda (a.k.a. topic guide)for the first type of interview, it's likely to come up in the second type as well, when we discuss behavioural DR. Similarly, the issue of community is important, since at the moment collective self-consumption is the only official use case that has been discussed for Voorhout. Hence, we include the experiences with the 'community'in De Wals explicitly in the topic guide for the first type. Finally, gender is an underlying point of attention in both interviews, with all questions. There are a number of dimensions in which gender has proven to be a significant (differentiating) factor: sensory experience, attitudes towards and familiarity with digital technology, and role division in the household. The role these dimensions will vary with age, which is something we'll pay attention to. (They will also vary with class and education, but we expect De Wals to be fairly homogenous in terms of class positions.) We have requested both partners to be present at the interview and we hope to suss out differences and similarities between them continuously, throughout the interview.

Type 1: living in Voorhout

Three parts: one about the community, one about comfort generally, and one about the smart home. The lattertwo parts will be closely interlinked. The questions should be quite open, allowing residents to narrate their experience and opinions. What's important is to have follow-up questions and topics ready, if necessary. The

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topic guide below is flexible – there are more topics than can be covered in 45 minutes, so the guide provides anchors wherever the conversation goes within the limits of time. We'll take at least 20 minutes to walk aroundthe house though.

- Community: what do you think/like about your street, public spaces; what do you like to do in your neighbourhood and can you do that here; contact with other residents, where; tell me about the visionand ambitions of De Wals at the beginning – how have their fared?;
- Comfort generally: compare advantages/disadvantages this and previous home (in terms of comfort/daily living); do you have any special needs (health, mobility, leisure);
- Smart home: see below

Show and tell

- Show me how you regulate the temperature
    - How often do you set/change it?
    - How you do like it?
    - Any differences in between you?
    - How does it compare with your previous home?
- What is nice about this home? What do you value in a home?
- Where do you spend most of your time in the house?
    - Is it also where you enjoy spending time the most?
    - What makes this an (un)comfortable space? How do you want to feel here?
    - When are you here? What do you before, after, when/where are your 'breaks' from this space?
- Show me the battery
    - When was the first time you saw this? (Before/after buying/getting the house) What did youthink?
    - *If there have been problems*: tell me about how it's been doing? Who did you call to fix the situation? What happened then? (Important links to safety here, as dimension of comfort.)
- Show me the where you can monitor energy production and consumption
    - *Important theme*: autonomy, control (real and perceived)
    - Managing energy consumption before and after Voorhout
    - Do you ever feel like you're consuming too much in this house?
    - Do you ever discuss this with other people (neighbours, friends from other places)
    - If they show that they do this often: do you know your monthly usage/production?

Type 2: uses cases for Hestia in De Wals

The interview could start with some 'warm-up questions' about whether they have had any first-handexperience with or second-hand knowledge about either:

- Smart energy
- Energy monitoring
- The challenges of the sustainable grid

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So, questions about EV, PV (in Voorhout and previous), smart meters, monitoring apps, what they've read about in the news, and follow-up questions about how they felt, what they understood or didn't etc.

Then move on to the use cases. While so far only collective self-consumption has been on the table, I'd like to propose the full range of possibilities that the batteries and flexible assets could afford. That includes: individual self-consumption, maximum market integration (arbitrage), collective self-consumption, and congestion management (i.e., at DSO level). One strategy here could be to show rudimentary visualizations of each strategy, explain each, get some feedback of initial reactions, explain some more (like breakdown in financial, environmental and social values), again get reactions.

When they give reactions pay attention to how they rate:

- Feasibility (am I or are 'they' ready for this, are institutional stakeholders ready for this, what will it mean practically)
- Desirability (what is most important: type of value, convenience)
- 'Excitement': does it energy, enthuse, or does it feel like a burden?

As with the first type of interview, in all answers I'll pay attention to and suss out differences and similarities between partners.

'Assignments' sent beforehand

At the moment, I'm not planning any assignments. Most of the interviews will be live, so we will be able to 'tour'the home with many of them. We will also send out little reminders ahead of time, which will include an announcement of the topics we will cover, as a way for them to mentally prepare (and practically, if that announcement includes a home tour).

What was the impact on daily life of malfunctioning devices?What dates would be good for the workshop?

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Appendix 9: Programme for participatory workshop in Voorhout

Voorhout Workshop outline

Es n of workshop: **2hrs 30 minutes**

Hestia



7.15pm: Walk-in, drink, snacks7.30pm: Introduction

- 5 minutes - introduction to decentralization of energy communities and new roles forprosumers
- 5 Purpose of the participatory trajectory in Hestia and purpose of the workshop, introduction of facilitators

7.45pm: **Round 1**: Participants split up into two groups

Marten & Sylvia: Group 1Razia &
Jordan: Group 2

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Group 1: Community Game

Two groups of 5-6 persons

Starting scenario for the game (not final):

It is the near future. Your new batteries are installed, the solar panels are running at full capacity, and your house generates more energy on an annual basis than it consumes. Together with your neighbours, you areone of the first to store self-generated energy using a home battery. But with the help of smart energy technology, you can do more than just store energy and feed into the grid what you don't use yourself. Ifyou join forces with your neighbours, you can realize more than if you stand alone. Therefore, together with your neighbours, you are now looking for your ideal energy community. What does 'De Wals' want to achieve?

- 5 minutes - introduction to types of energy communities + explanation of communitygame.
    - Purpose of the game is to allow participants to form and discuss opinions ontheir ideal energy community.
    - Therefore participants 'play' as one group. One member of the group reads outthe 'instructions' at each stage of the game.
    - Moderator guides discussions primarily by making sure everyone is heard andarguments are made explicit, while the assistant moderator records discussions.
- 20 minutes – walk through the stations
- 15 minutes – presentation of other final stations, reflection on the results /comparison with other potential results

Group 2: What should a Dashboard do?

- 5 minutes - Moderator explains aim of activity: discussion regarding experiences with the presentation of consumer data, and current preferences on which data you want to see, and how you want this information to be presented.
- Divide the group in smaller subgroups, 5-6 participants each. Groups are assigned todifferent tables.

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Two main topics:

- What do you need to understand energy flows in the house and neighbourhood?(Graphs, numbers, source of energy, impact of energy use.)
- How do you want the platform to communicate with you about the best choices forenergy consumption? F.i. traffic light visualization, notifications, tips .

1. Discussion of **experiences with Voorhout SWYCS app**. What worked, what didn't?
2. **Questions + discussion** of selection of other dashboards. What looks attractive,intelligible, which information is important?
3. **Voting for elements/features/functionality** that participants want to see in the digital environment.

- On a poster, participants vote with three (gender-colour-coded) dots, on what theywant to see most on the dashboard / app.
- This way we collect highest priorities for further development

8.25pm: BREAK (15 minutes for coffee / refreshments etc, groups switchrooms)

8.50pm Round 2: Participants switch activities

Same (sub) groups, switched activities.

9.30 Wrap – up:

thank you and bye. If people want to tell us about something they learned or they think is still important tomention that they walk up to one of us. Evaluation forms.

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Appendix 10: Summary and analysis of workshop in Voorhout

Template – workshop summary

For each of the WP2 participatory workshops, the local site partners/facilitators should prepare a(short) English summary and send this to AAU. The summaries should follow the same basic structure in order to inform the final analysis. The template (presented below) outlines different sections and briefly describes (and give examples of) the focus and intended content of each section. If you have any questions, please don't hesitate to contact.

The aim of the summaries is to document and report some preliminary observations and findingsfrom the conducted workshops. As each workshop differs contextually and has been carried out in three different languages, we would like to gather some insights that can be comparable, thus opening up for a comparative analysis of the three different workshops. Taken together, the summaries will also be an input for the overall analysis (reported in D 2.2.) which aim to inform the design of the HESTIA interface (T 2.3 and T 2.4).

In the template, the first two sections will concern the basic/descriptive information of the workshops (e.g. number of participants, number of households, gender representation), makingit possible to draw out methodological limitations and evaluate the validity. Next, we focus on your methodological reflections, asking you to outline how the workshop went, describing the engagement, 'mood' and potential deviations from the planned setup. Third, we ask you to highlight some of the 'most interesting' discussions from the workshop discussions. Here we
suggest that you summarise each 'stage'/planned activity of the workshop. We would both liketo get 'raw data' i.e. summaries of what happened and what was being said, but we also hear your analytical observations from the workshop.

As D 2.2. is expected to inform the design of the HESTIA interface/dashboard (developed by i.LECO), please also consider including feedback from the occupants that directly concern the design of such (e.g. what features was liked, how occupants prefer to get energy feedback etc).

Finally, please feel free to attach in photos or videos that you might have obtained during theworkshop, and which you think illustrate a certain point – that being both methodological or content related.

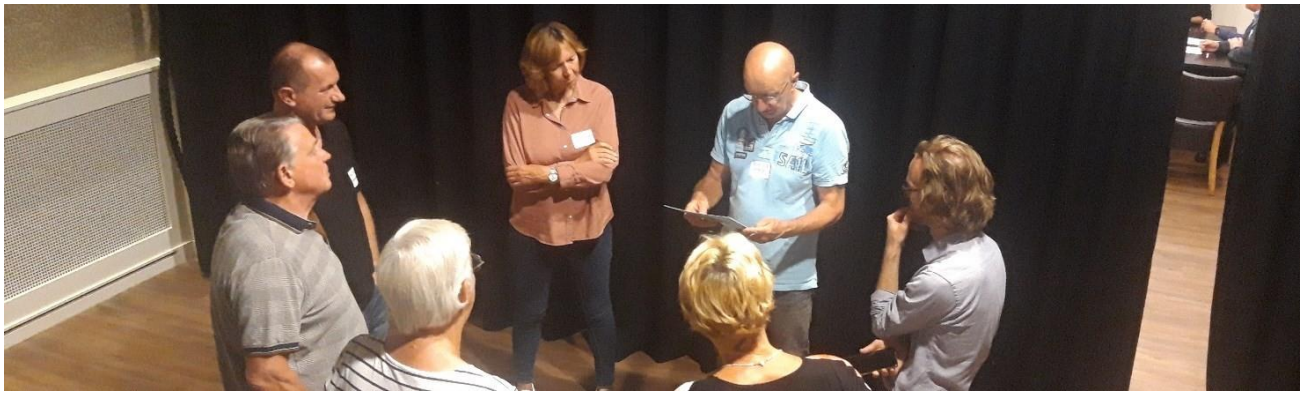
The summaries should be written on basis of the audio/video recordings and analytical observations taken during the workshop.

Ensure that all participants are anonymised in the summaries. I.e. do not use their real names and take also care not to include personal information that would make it possible for others toidentify them.

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Hestia
Holistic demand response services
for European residential communities

Summary and analysis of workshop in Voorhout, The Netherlands



Workshop organisation

Workshop day of the week and date	Wednesday 22nd of September 2021
Workshop time	19.15
Workshop duration	2,5 hours
Venue	Cheers café & restaurant, Voorhout
Incentives/reciprocity offers for participating (ifany, e.g dinner)	Booze
Hestia partner in charge	DW

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Other Hestia partners present and their roles	i.LECO, 4YEF (observation)

Participants and recruitment

Number of participants invited	32 households	
Number of participants that turned up	25	
Recruitment of participants (how they were invited to the workshop e.g personal invitation,letter, email etc.)	Personal e-mail; letter; postcard reminder	
Gender distribution:	Female	Male
	10	15
Number of households present	15	

Comments/ reflections on participants' recruitment and turn out (e.g representation of age, gender, overallturnout of the specific community, practical issues that affected the running of the workshop and the participants' engagement etc.)

A lot of couples attended, which helped with the gender distribution. Of those who came alone, I believe only one was a woman.

Our major recruitment challenge pertains to the age group: we recruited far more older people than younger ones. We had 4 households representing the younger population (and they were all represented by a single person, rather than couples.)

Undoubtedly there is a time factor here: if you are retired and you don't have to stay home for the kids, you are
more available for a workshop.

However, there are some additional dynamics as well. As a reminder, the neighbourhood is divided into two more or less equal groups – the senior citizen homes and the family homes. Each group consists of two streets. On the whole, residents are more closely connected with neighbours on their own street. However, the street- based bonds are stronger on the 'senior' side of the neighbourhood than on the 'family' side. Moreover, the uncertainties surrounding the smart home energy appliances have been the basis for community building on thesenior side, especially for one of the streets. On the family side of things, the connections are less strong and frequent and the energy infrastructure has played a smaller role in street communication.

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In other words, the topic of the energy community is much more 'alive' for the senior citizens than for the families.

Workshop process and group dynamics

Welcoming process (what happened when participants arrived, e.g welcoming, names tags, introductions to othersetc.)	There were about 15 minutes for people to get their nametag, get a drink and chat with their neighbours. This last bitwent quite natural, since people know each other well enough or even quite well.
Was there an introduction to the Hestia projectin the beginning of the workshop? What did that involve? What were participants told?	We decided not to emphasize Hestia as such, but instead focused the introduction on an explanation of the changing energy system and the possible new roles for end- users/citizens it opened up – individually as well as collectively. After that introduction, we introduced Hestia asthe occasion to find out what roles *residents of De Wals (Voorhout)* would want to take up ([active or smart] prosumers, individually or as part of a collective) – and how (what support would they need, to what end would they contribute). We presented this also also as a way to take thePlusLeven concept further.

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Were the participants split in smaller groups? If so how many and what were the specifics (e.g members of the same household split or stayed together, how many per group, purpose of eachgroup etc.)	The room we were in was split into two sides, separated by a curtain. The participants were divided into halves accordingly. Each half was then subdivided into two smaller groups that would actually be going through the activities together. This subdivision had been arranged beforehand and was indicatedon their name tags. Care was taken to evenly distribute gender and age (that is to say: to evenly spread out women and younger participants over the 4 groups). We also chose tosplit up couples, in order to allow each member of the couple to contribute autonomously, rather than one member taking the lead to represent the household, or for subtle negotiations about how to respond to queries to talk place, which couldn't be made explicit in the rushed setting of the workshop.
Did the workshop run according to plan? Were there any deviations from the original scheduleof activities?	No serious deviations from the schedule.

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What was the overall 'mood' of the participating group during the workshop? (e.g. Was it good and "cheerful"? Was it relaxed? Did the participants seem comfortable about the settingand did they appear to trust each other? Or was the participant not relaxed or did any "bad atmosphere" occur? (If the latter, please includea few thoughts on why this might have happened)	The mood was good: some people were a perhaps little sceptical at the beginning, after we explained the general set- up (the energy of the walk-in meet and chat had dissipated, some friendly raised eyebrows that suggested that perhaps they had expected something else), but once we got going, everyone participated actively, the interactions were cordial and friendly, and afterwards the mood was energized. Several people commented that they had found it enjoyable, useful or educational and expressed the hope that it would set the stage for a new phase of the PlusLeven concept.
Please state 3 things that went really well in terms of engagement	1. Small groups (5, 6) allowed for active participation by all(encouraged/monitored by moderators) 2. Starting the dashboard session off with something people knew (Swycs app) lowered barrier for participation. It got people involved right away. 3. The energy community values game seemed like it was a good tool for community-building – allowing residents to talk about who they were to each other and what role they wanted or felt ready to take up in the world (of energy). Thus,we talked about whether it was realistic that they could jointlyset investment goals for a shared energy fund, whether they'd be willing to 'compete' *or* to 'share' with each other, but also at which scale they felt they could make a meaningful contribution and/or take responsibility (do they take on the task of saving the future of their grandchildren, or 'the environment', or just make their own neighbourhood more cohesive and prosperous – in a 'clean' way)?

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Please state 3 things that were challenging in terms of engagement	1. Serious game is quite complex and intentionally confusing(sort of) – requires intimate familiarity with the case (Voorhout) and with game structure to guide well. 2. Gender dynamics: women were interrupted more often intheir explanations 3. The relative lack of younger participants made didn't allow us to explore more in detail the question of distribution (of burdens and benefits).

Comments/ reflections on the overall dynamics of participants during the workshop (E.g In what ways participants engaged with the discussions, any challenges during the process, any positive remarks, did participants generally talk to each other or did they respond mostly to the facilitator? Did all participants participate in discussions? Or only a few? In the latter case: Indicate who?) Interaction among the participants – only moderated by facilitators. This therefore allowed us and the residents to go beyond expressing their opinions. If one considers the residents to be in a (learning) process, where certain values or dispositions may be strong and relatively immutable, but what these values mean in practice to still in flux, with changing circumstances and changing understanding, then bringing people together and allowing them to exchange ideas (and – hopefully – bond in the process) accelerates that process. (Now, this workshop didn't really serve any other purpose than that – to be educational and to a certain extent, be a bonding experience. Once actual decisions have to be made – in negotiation with each other and other stakeholders – differences that can now be glossed over, because they don't really matter yet, will become more pointed.) One participant expressed the hope that the workshop (and Hestia's involvement more generally) would allow residents to come together and do something collectively, get organized. Preceding troubles were a significant factor in the discussions.

Summary of each workshop stage/planned activity

Stage 1: debating dashboards

Duration	40 min
Any probes / tools etc. used	Screenshots of various energy monitoring apps / digital environments;poster with dashboard features

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Discussion starter	- what were your experiences with [this or that part of] the Swycsapp? - Take a look at alternative dashboard/app A or B? What do youthink of feature x OR y? Would it be useful? Is it understandable? - Go to the poster and vote on your 3 most important/favourite functions of an energy monitoring app

Questions asked to participants during the introductory part of theworkshop	Nothing specific, we did mention the upcoming change to the contract, which might mean leaving behind the flat rate tariff (which might influence their interaction with energy monitoring apps).
Summary of activity	We tried to make the threshold as low as possible to participate, startingout with plenary y/n questions that people could answer by hand-raising,keeping the language simple and giving turns to people.
Facilitation	Two tables of 5 to 6 people. One facilitator to moderate the discussion,give turns, ask for explications. Other facilitators in background to keeptime. Some challenges: interfering noise from other group; social desirabilitywas difficult to counter when one or two participants were dominant/vocal. Images helped to focus discussion. Voting was seen as fun exercise. Groups small enough to be able to let everyone speak.

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Analytical observations/reflections	- People felt quite unanimously that more insights would allow themto adapt their energy consumption - The previous app they were given didn't work but was also badly designed – unclear and unattractive: A "mickey mouse app" (referring to graphic representation) (M). (The fact that some devicesdidn't work may have made the app feel worse: "I couldn't find out how much I had produced that day, but maybe it was because we didn't have the right smart meter" (W). "With this app you can't see the forest for the trees" (M). - Mobile devices were seen as primary interface with monitoring - Three most upvoted functionalities: PV production; source of electricity for consumption; a 'stoplight' (green, yellow, red) toindicate favourableness for energy consumption Obstacles to overcome as project proceeds: - Lack of communication about how get malfunctioning devices tofunction - Uncertainty and even fear about devices not working - Loss of faith in parties involved - Decreased sense of involvement with energy management because monitoring simply doesn't work
Highlights/ quotes	It is also nice to see the whole group, I don't need personal information ofothers, but the percentage of the group and that you see what the average is and see: 'I am below or above or ok'. And then also with information about [whether it's a] corner house or middle house etc. So that you can compare.

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	Yes, we don't have any insight at the moment, but we can feel that it [thebattery] is warm and that it switches on. We both [partners] like the fact that you have insight into what your energy consumption is and also whatyou are generating. That insight is not there now. You want that trigger, to be efficient [“zuinig” = economical]. (M) How do we know if that system [in Voorhout] works? We basically haveto assume that it works." (M) "I just find using apps difficult. You know, that's not what keeps me busyall day, chat messaging about this and stuff".

1.1 Stage 2: The road to energy community De Wals

Duration	40 min
Any probes / tools etc. used	Square thick-paper handouts with instructions, laid out as path to walk; A2 posters on the wall as final ‘destinations’

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Discussion starter	It is the near future. Your new batteries are installed, the solar panels arerunning at full capacity, and your house generates more energy on an annual basis than it consumes. Together with your neighbours, you are one of the first in the Netherlands to store self-generated energy using home and community batteries. But with the help of smart energy technology, you can do more than just store generated energy that you don't use or feed it into the grid. You can also intervene more actively in energy flows. And joining forces with your neighbours is even more possible than doing it on your own. Together with your neighbours, you are therefore now exploring the possibilities of an energy community called 'De Wals'. The most important question: what do you want to use the smart energy technology of your energy community for?

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Questions asked to participants during the introductory part of theworkshop	Series of value statements at each station from which participants had tochoose the one they felt most in tune with. Debate among each other.
Summary of activity	The goal for the group was to 'build' step by step their preferred energy community. There were 7 final, archetypal energy communities that theycould reach. Each of these represented some selection or mix of three values: financial, environmental or social/community. At each step, one member of the group would read out the scenario description, the statements to choose from and/or the result of their choice for the architecture of the energy community. It was made clear that this was purely a thought exercise, a conversationstarter and not a procedure to make actual choices for Voorhout.
Facilitation	Each group (again of about 6 persons) had one moderator. The moderator had to make sure everybody got a chance to speak and thatpeople explicated their preferences, so that they could be debated, laidout in the open for examination, and the group couldn't steamroll through the stations based on groupthink or dominance.
Analytical observations/reflections	Two themes played a central role in the discussions. One is the experience with malfunctioning technology that could not be easily repaired. This pushed many people to prefer the financial value over other values as a hedge against future trouble. The second is community and collectivity. Quite a few people expressed to desire to either do something with neighbourhood or do something *for* the neighbourhood.It was also a place where the financial hardliners could find some common ground with the people who emphasized the importance of doing something for the environment. If a collective could be created that would invest its revenue into more clean energy, a compromise could be found. That is not to say that community was idealized: most were sceptical ofthe possibility of make the collective work, if it meant that energy became a common pool resource (begging questions of equity among unequal participants) or that it would require joint decision-making (where to invest the revenue).

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Highlights/ quotes	I think for me it's very important that you use your energy as efficiently aspossible, and if I were to do that through a community, and I have all these lovely neighbours, but maybe some keep the windows open and the heating on high...then I'm not in control...you know, find that very tricky. [*on creating a 'business model' where you depend on each other to create value*] (W) You have to be very much a unit, a community so to speak, to be able tosay I agree, I will allow someone to drain my battery if he needs energy. You need solidarity as the basis…

	S: do you think you have that solidarity? Yes, I guess so. We'll have to try and see, carefully. (M) [There is] little contact [among the family homes]. Some isolate themselves from the neighbourhood anyway – but there's even little contact with the rest - an occasional chat message - for example, aboutwho would be coming [to the workshop] tonight. (M) We have expensive batteries, expensive heat pumps, all expensive things, and at some point they have to be replaced - that has to be paid for… The disadvantage of this game is that we know too much. You see all the worries, you see that *this* will break down and *that* will break down. You know too much. If I didn't know - if everything would have worked properly, then I would have easily gone for community and the environment, all those good things. (M) [*About whether to choose financial return as primary objective for the energy community or social orenvironmental values*] The environment is too big. I would go prefer doing something for thecommunity. Keep it closer to yourself. To have an overview. MB: *"To see more clearly what you're doing?"* "Exactly. That's more stimulating. If you see that you're doing good for the neighbourhood, I think you get more [encouragement]- I think [theenvironment] is too big, then you don't see it. MB: *"And if you get feedback from the app, in terms of CO2 savings for example?"* No, nor how many trees I've grown. That doesn't mean anything to me.(M) If you choose financial return, you choose for your own future. I'd ratherchoose the future of my grandchildren. (W)

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	I'd want to support the upgrading of the network to some extent - everything that goes with it. Which I might also benefit from myself. (M) W: Yes, then you are going to sit in the cold [with a jumper].W2: I'm really not going to do that, whatever.

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	MB: *You're afraid it will go too far?* W: Yes." W2: And that you will get annoyed with people who don't.W: You get nasty conversations. W2: We have such good relationships in De Wals, so we would like to keep it that way (haha). [*About gamification features like competitions orleaderboards*]

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